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Span-based Classifier for Short Answer Scoring and Beyond

Master's Thesis

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Acknowledgments

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My deepest gratitude goes to my family. I would especially like to thank my father and mother for always supporting and motivating me in the past. Without them this thesis/project would have never been finished.

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Abstract

Note *- In most cases it is easier to write the abstract at the end than trying to finish it directly at the beginning.*

An abstract is a brief summary of a research article, thesis, review, conference proceeding, or any in-depth analysis of a particular subject and is often used to help the reader quickly ascertain the paper's purpose [6]. When used, an abstract always appears at the beginning of a manuscript or typescript, acting as the point-of-entry for any given academic paper or patent application. Abstracting and indexing services for various academic disciplines are aimed at compiling a body of literature for that particular subject.

It may also contain brief references, although some publications' standard style omits references from the abstract, reserving them for the article body (which, by definition, treats the same topics but in more depth).

Abstract length varies by discipline and publisher requirements. Typical length ranges from 100 to 500 words, but very rarely more than a page and occasionally just a few words [4]. An abstract may or may not have the section title of "abstract" explicitly listed as an antecedent to content.

According to [52] an abstract should be:

- **Objective** - It should refrain from any personal evaluation,
- **Short** - It should be as short as possible,
- **Comprehensible** - It should have a clear, comprehensible language and structure,
- **Complete** - All essential facts should be included,
- **Accurate** - It should accurately reflect the content and the opinion of the original work

Contents

1 Introduction **1**
- 1.1 Motivation . . . 1
- 1.2 Problem Statement . . . 2
- 1.3 Research Questions . . . 2
- 1.4 Structure of the Thesis . . . 3

2 Background **5**
- 2.1 Automatic Short Answer Grading (ASAG): Task Definition . . . 5
- 2.2 Evaluation Metrics in ASAG . . . 8
- 2.3 Encoder-Based Approaches to ASAG . . . 13
- 2.4 LLM-Based Approaches to ASAG . . . 14
- 2.5 Rubric-Guided Scoring: Guided Alignment Architectures . . . 15
- 2.6 Fundamentals of the Transformer Architecture . . . 17
  - 2.6.1 Self-Attention and Multi-Head Attention . . . 19
  - 2.6.2 Causal vs. Bidirectional Attention Masks . . . 20
- 2.7 Parameter-Efficient Fine-Tuning (LoRA) . . . 22
- 2.8 Fusion Mechanisms in Rubric–Answer Alignment . . . 23

3 Datasets **25**
- 3.1 ASAP-SAS: Structure, Statistics, Label Scale . . . 25
- 3.2 Beetle: Domain and Test Splits . . . 26
- 3.3 SciEntsBank: Domain and Test Splits . . . 27
- 3.4 Origin of the Rubric Texts and Reference Answers . . . 29
- 3.5 Comparison of the Benchmarks and Rationale for Their Selection . . . 29

4 Methodology: The RUSPAN System **33**
- 4.1 System Architecture: Span-Alignment Model . . . 33
  - 4.1.1 Input Format . . . 34
  - 4.1.2 Span Extraction and Pooling . . . 35
- 4.2 Fusion Mechanisms . . . 37
  - 4.2.1 CONDIFF: Conditional Difference Fusion . . . 37
  - 4.2.2 CONCAT: Plain Concatenation as a Baseline . . . 38
  - 4.2.3 DIFF: The Difference as the Sole Input . . . 38
  - 4.2.4 GATE: Gated Fusion . . . 39

4.3 Bidirectional Attention in Decoder-Only Models . . . 40
4.3.1 Motivation . . . 40
4.3.2 Implementation: Partial Removal of the Causal Mask . . . 41
4.4 Backbone Models . . . 43
4.4.1 Llama-3.2-1B/3B . . . 43
4.4.2 Mistral-7B . . . 43
4.4.3 BERT/RoBERTa as Encoder Baselines . . . 43
4.5 Variants for the Ablation Experiments . . . 44
4.5.1 Fixed Classification Head as a Baseline . . . 44
4.5.2 Ablation of the Rubric Texts . . . 44
4.5.3 Variation of the Task Context . . . 45
4.5.4 Answer Span as the Anchor of the Fusion . . . 45
4.5.5 Transfer Between Benchmarks . . . 45
4.6 Training Configuration . . . 46

5 Experiments and Results **51**
5.1 Experimental Design . . . 51
5.2 Noise Threshold: Variation Between Runs . . . 53
5.3 RQ1: Span Alignment Versus the Baselines . . . 56
5.3.1 Span Alignment Versus a Fixed Classification Head . . . 56
5.3.2 The Contribution of Task-Related Information . . . 58
5.3.3 Encoder- Versus LLM-Based Approaches . . . 63
5.3.4 Transfer Between Benchmarks . . . 64
5.4 RQ2: Adapting Decoder-Only Models . . . 65
5.4.1 Fusion Mechanisms . . . 66
5.4.2 Anchor of the Fusion . . . 68
5.4.3 Attention Mask . . . 69
5.4.4 Effect of Model Size . . . 73
5.5 Supplementary Analyses . . . 75
5.5.1 Results per Task on ASAP-SAS . . . 75
5.5.2 Unstable Runs . . . 76
5.5.3 Robustness to the Choice of Metric . . . 77
5.6 Summary of Findings . . . 81

6 Discussion **83**
6.1 Answers to the Research Questions . . . 83
6.2 Relation to Prior Work . . . 85
6.3 Methodological Implications . . . 87
6.4 Limitations . . . 87
6.5 Practical Implications . . . 88

7 Conclusion and Outlook **89**

List of Tables **91**

List of Figures **95**

Bibliography **97**

Appendices **103**

1 Introduction

1.1 Motivation

Short, freely formulated answers are a widely used instrument for assessing subject-matter understanding. Unlike selected-response items, they require learners to recall or infer a fact themselves and to express it in their own words. This strength comes at a price: every answer has to be read individually and checked against the scoring criteria, and the effort grows with the number of learners. In large groups in particular, this makes timely feedback difficult. Automatic Short Answer Grading (ASAG) aims to support this assessment computationally and has long been an independent research field within educational technology [7, 2].

Methodologically, the field has evolved from manually engineered features via pre-trained encoder models such as BERT [15] to large language models that score an answer on the basis of a prompt [30, 9]. For use in educational practice, two requirements arise that research has so far treated separately.

The first concerns data protection. Student answers are personal data as soon as they can be attributed to an identifiable person, and children's data merit specific protection [20, Art. 4(1) and Recital 38]; guidelines on the use of generative AI in education likewise emphasise corresponding safeguarding duties [58]. Transmitting answers to external model providers is therefore problematic, and ASAG systems should be operable locally. Locally operated generative models of sufficient size, however, are computationally expensive. An alternative is opened up by a more recent line of research that uses decoder-only language models not generatively but as encoders, exploiting their internal representations for discriminative tasks [3, 32]. This makes it possible to use the knowledge that large language models acquire during pre-training in a single forward pass, without having to generate text at inference time.

The second requirement concerns the heterogeneity of scoring scales. A classical classifier maps the answer to the levels of a particular benchmark through an output head of fixed width. It is thus bound to the number and meaning of these levels; a task with a different scale requires a new head and renewed training [23]. Even within a single benchmark such as ASAP-SAS, the number of levels varies between tasks (Chapter 3). An obvious answer to this problem is rubric-guided scoring: each scoring level is represented not by an index but by its rubric, that is, the textual description of its criteria, and the model assesses how well the answer matches the criterion description. Such a model is not tied to a fixed label space and can in principle be transferred to unseen rubric sets. The topicality of this approach is also reflected in the fact that rubric-based scoring is the subject of a current shared task [24].

1.2 Problem Statement

Rubric-guided methods can be implemented in two ways. Pairwise methods encode the answer together with each rubric separately and therefore require as many forward passes per instance as there are scoring levels [51]. Span-alignment methods, by contrast, encode the answer and all rubrics jointly in a single sequence, extract the representation of each rubric as a span from the same hidden-state matrix and align it with a context representation [23]. They thus combine the flexibility of pairwise comparison with the cost of a single forward pass (Section 2.5).

Transferring this principle to decoder-only language models raises three questions that have not yet been investigated systematically.

The first concerns the *premise* of the approach itself. Rubric-guided scoring presupposes that the model actually uses the criterion descriptions for scoring and that extracting them explicitly as spans offers an advantage over a simple classifier with a fixed head. Neither is self-evident, especially since rubric texts are closely related in content to the question and the reference answer, and since no original scoring guidelines exist for two of the three benchmarks used (Section 3.4).

The second concerns the *design of the alignment*. It is open which context representation should be aligned with the rubric spans – the representation of the answer span or that of the entire sequence – and by which fusion function the two should be combined. Under causal attention this choice is not neutral: the answer span precedes the rubrics and has not seen them, whereas the last state of the sequence summarises the entire input including all rubrics.

The third concerns the *attention mask*. Decoder-only models are pre-trained under a causal mask that allows each position to attend only to preceding positions. For a pure scoring task this restriction has no substantive motivation: the answer cannot take the subsequent rubric texts into account, and the rubric representations depend asymmetrically on the respective preceding levels. Work on repurposing language models as encoders shows that the mask can be removed, but that this intervention is not without consequences relative to the pre-training condition [3, 35]. Whether a partial or complete removal benefits ASAG within parameter-efficient fine-tuning is an open question.

A methodological difficulty applies to all three questions. The expected differences between architectural variants are small, whereas the choice of random seed alone can substantially influence the outcome of fine-tuning [44, 16]. A comparison of single runs therefore cannot distinguish an effect of the architecture from mere training noise. Likewise, the significance of a result depends on whether a model merely scores new answers to known questions or is transferred to unseen questions and domains [18].

1.3 Research Questions

The aim of this thesis is to transfer a span-based classifier for rubric-guided scoring to decoder-only language models and to investigate the questions above in a controlled manner. It is guided by two research questions:

RQ1 *Does the span-based classifier outperform the baselines?* This is examined against a classifier with a fixed output head that receives the same input, against conditions without rubric text or without any task-related information, and against encoder models. In particular, it is examined what the rubric texts contribute relative to the question and the reference answer and whether a trained model can be transferred to another benchmark via the rubric texts.

RQ2 *Can decoder-only language models be adapted to the span-based classifier?* This covers the design of the alignment between rubric spans and context – the fusion function and the anchor representation, i.e. sequence representation or answer span – as well as the partial and complete removal of the causal attention mask.

These questions are investigated on three English-language benchmarks (ASAP-SAS, Beetle and SciEntsBank) with Llama-3.2-1B, Llama-3.2-3B and Mistral-7B as decoder backbones and BERT and RoBERTa as encoder baselines.

The thesis thereby makes four contributions. First, it compares span alignment with a fixed classification head on identical input and, in a full 2×2 design, separates the contribution of the rubric texts from that of the remaining task information; in addition, it tests the transfer of a model trained on ASAP-SAS to Beetle and SciEntsBank. Second, it transfers span alignment to decoder-only models and compares four fusion functions and two anchor representations within an otherwise identical architecture. Third, it varies the attention mask layer by layer from the purely causal to the fully bidirectional condition. Fourth, it evaluates all substantive comparisons over three random seeds against an empirically determined noise threshold, separately for each generalisation condition and under two metrics, and checks the reliability of this noise threshold on a group with eight seeds.

1.4 Structure of the Thesis

Chapter 2 introduces the ASAG task and its evaluation, presents encoder- and LLM-based approaches as well as rubric-guided scoring, and describes the fundamentals of the transformer architecture relevant to this thesis, in particular the attention mask, as well as parameter-efficient fine-tuning. Chapter 3 describes the three benchmarks used, their evaluation splits and the origin of the rubric texts. Chapter 4 develops the span-alignment model under investigation, the fusion mechanisms, the partial removal of the causal mask and the baseline conditions. Chapter 5 first determines the noise threshold and then reports the results along the research questions. Chapter 6 answers the research questions, relates the findings to the literature and identifies the limitations of the study. Chapter 7 summarises the thesis and gives an outlook on further research questions.

2 Background

This chapter lays the conceptual, methodological and technical foundations on which the RUSPAN system described in Chapter 4 is built. First, the task of Automatic Short Answer Grading is defined formally and distinguished from neighbouring tasks (Section 2.1). Next, the evaluation metrics used in this thesis are derived and discussed with respect to their measurement assumptions (Section 2.2). Sections 2.3 and 2.4 review the state of research on encoder- and LLM-based methods; Section 2.5 introduces rubric-guided scoring architectures, from which the approach of this thesis is directly derived. The technical prerequisites – transformer architecture, self-attention, attention masks and parameter-efficient fine-tuning – are developed in Sections 2.6 to 2.7. Section 2.8 situates the fusion mechanisms under investigation within the broader literature on combining vector representations.

2.1 Automatic Short Answer Grading (ASAG): Task Definition

Definition and Delimitation

Automatic Short Answer Grading (ASAG) refers to the automatic assignment of a score to a freely formulated, short student answer given in response to a question with a well-defined content. The seminal survey by Burrows, Gurevych, and Stein [7] delimits the task by five characteristics: the question asks for a recalled or inferred fact, the answer is formulated in natural language, it typically comprises one to a few sentences, the assessment concerns the content and not the linguistic form, and the answer space is open but narrowly bounded in content. ASAG thus differs systematically from Automated Essay Scoring (AES), in which writing skills such as style and structure enter the assessment in addition to content [2]. It is likewise distinct from multiple-choice formats, which require no language understanding. In recent literature, the term Automatic Short Answer Scoring (ASAS) is also common [2]. This thesis uses both terms synonymously and follows the more established abbreviation ASAG.

The practical relevance of the task stems from the mismatch between the didactic value of constructed responses and their marking cost: in contrast to closed formats, constructed responses require learners to formulate an answer themselves and to justify their reasoning [2]. Their manual assessment, however, is laborious, and large groups of learners require efficient solutions [7].

Formalisation

An ASAG dataset is viewed as a set of annotated instances

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}, \tag{2.1}$$

where y_i is the reference label assigned by human raters. Each instance x_i is a tuple of textual fields

$$x_i = \big(q_i,\ s_i,\ a_i,\ \mathcal{R}_i\big), \tag{2.2}$$

consisting of the question q_i, a sample solution or reference answer s_i, the student answer to be scored a_i, and the set

$$\mathcal{R}_i = \big(r_{i,0},\ r_{i,1},\ \ldots,\ r_{i,K_i-1}\big) \tag{2.3}$$

of K_i rubric descriptions, each of which characterises one performance level in words. The availability of the fields s_i and $\mathcal{R}_i$ depends on the benchmark; their origin for the datasets used in this thesis is described in Chapter 3.

The classical formulation of the task is a scoring function

$$f_\theta :\ (q_i, s_i, a_i) \ \longmapsto\ y_i, \qquad y_i \in \mathcal{Y}, \tag{2.4}$$

that maps to a *globally* fixed label space $\mathcal{Y}$. As a rule, $\mathcal{Y}$ is ordinal, i.e. there is a substantively justified ranking between the levels, without the distances between adjacent levels being interpretable as equidistant. The cardinality and semantics of $\mathcal{Y}$ vary between benchmarks and in some cases even between the individual tasks of one benchmark (cf. Chapter 3).

It is precisely this variability that motivates the alternative pursued in this thesis. Instead of a global label space, the task is formulated in a rubric-conditioned manner: for each instance there is an instance-specific candidate set $\mathcal{Y}_i = \{0, \ldots, K_i - 1\}$ whose elements are described semantically by the corresponding rubric texts $r_{i,k}$. The scoring function then assigns a score to each level,

$$\sigma_\theta :\ (q_i, s_i, a_i, r_{i,k}) \ \longmapsto\ s_{i,k} \in \mathbb{R}, \qquad \hat{y}_i = \operatorname*{arg\,max}_{k \in \mathcal{Y}_i} s_{i,k}. \tag{2.5}$$

Formulation (2.5) is strictly more general than (2.4): it is applicable even when K_i varies between instances, and it permits transfer to unseen rubric sets, because the model learns not an index assignment but the match between answer and criterion description. The concrete architectural implementation of (2.5) and the corresponding training objective are developed in Chapter 4; throughout, this thesis compares it with the classical formulation (2.4) as a baseline. Figure 2.1 contrasts the two formulations.

Generalisation Dimensions and Evaluation Splits

A single score on a single test set says little about an ASAG system as long as it is unclear *what kind of novelty* the system encounters. A system that scores the same exam question across several cohorts of a course faces an entirely different task from one that is applied to a newly formulated question or even to a different subject area. The split introduced in SemEval-2013 Task 7, which has been customary since then, reflects exactly this gradation by distinguishing

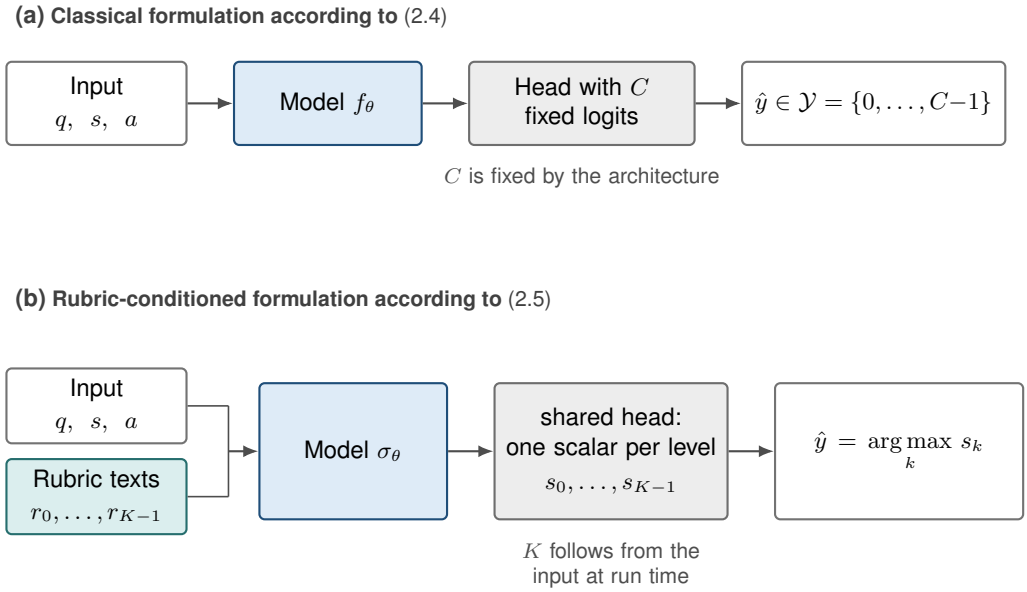


Figure 2.1: Classical **(a)** and rubric-conditioned **(b)** task formulation. In **(b)**, a shared head assigns one scalar per rubric, so that the number of outputs follows from the number of rubrics.

three test conditions according to which components of a test instance already occurred during training [18]:

- *Unseen Answers* (UA): only the student answer is new. The model has already seen the question, sample solution and rubric texts many times during training. What is measured is the ability to reproduce the scoring convention of a *known* task.
- *Unseen Questions* (UQ): question and answer are new, while the subject domain remains the same. The model must have abstracted the scoring principle from the individual task but can still rely on familiar content.
- *Unseen Domains* (UD): question, answer and subject area are new. This condition is the most demanding, since neither the task formulation nor the subject area occurred during fine-tuning.

Figure 2.2 depicts the three conditions as nested degrees of novelty. The split is not merely nominal but structurally realised in the datasets used: in Beetle, for example, the UA split uses exactly the same 47 questions as training, whereas the 9 questions of the UQ split have an empty intersection with the training questions. In SciEntsBank, the 46 questions of the UD split come exclusively from three topic blocks that do not occur in a single training question (Chapter 3).

For this thesis the distinction is more than a convention. Under UA, a model can already achieve high agreement through recurring phrasing patterns of known questions without processing the scoring criteria in substance; Section 5.3.2 shows that this shortcut actually occurs in the present data. Above all, however, the promise of the rubric-conditioned formulation (2.5) – transferability to unseen tasks and rubric sets – can only be tested under UQ and UD, since under UA the rubric texts are identical to those seen in training. All results in Chapter 5 are

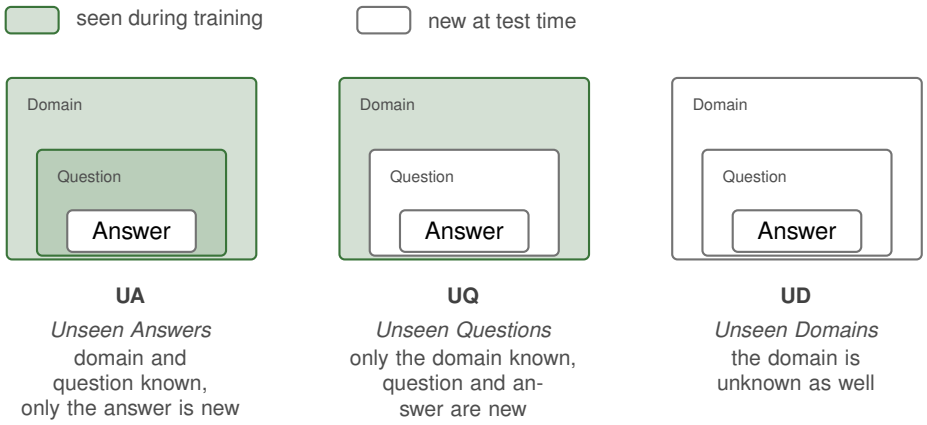


Figure 2.2: The three evaluation conditions according to [18]: new are the answer in UA, additionally the question in UQ, and also the domain in UD.

therefore reported separately by split, and statements about transferability rest exclusively on UQ and UD. As a limitation, only SciEntsBank provides a UD split (Chapter 3); all findings on domain transfer therefore rest on a single benchmark.

2.2 Evaluation Metrics in ASAG

In this thesis, ASAG results are reported using three measures: Quadratic Weighted Kappa (QWK), Macro-F_1 and accuracy. All three can be derived from the confusion matrix, but they assume different scale levels and react differently to class imbalance. This section first develops them formally and then discusses their assumptions.

Notation

Let $\mathcal{Y} = \{0, 1, \ldots, C-1\}$ be the label space under consideration with C levels, and let y_i denote the reference labels and $\hat{y}_i$ the model predictions over N test instances. The confusion matrix $O \in \mathbb{N}^{C \times C}$ is defined by

$$O_{jk} = \big|\{\, i \in \{1, \ldots, N\} \;:\; y_i = j \;\wedge\; \hat{y}_i = k \,\}\big|, \tag{2.6}$$

i.e. O_{jk} counts the instances with reference level j and prediction k. The marginal distributions are denoted by $n_j^{\text{ref}} = \sum_k O_{jk}$ and $n_k^{\text{pred}} = \sum_j O_{jk}$.

Accuracy

Accuracy is the proportion of instances predicted exactly correctly,

$$\text{Acc} = \frac{1}{N} \sum_{j=0}^{C-1} O_{jj}. \tag{2.7}$$

It is directly interpretable but has two serious drawbacks for ASAG. First, it weights all errors equally: confusing a maximum with a minimum score counts just as much as confusing adjacent levels, although the latter is far less serious from an assessment perspective. Second, it is not robust to imbalanced label distributions: a trivial model that always predicts the most frequent level already achieves the relative frequency of that level. Since the benchmarks used in this thesis are in part markedly imbalanced (cf. Chapter 3), accuracy is reported only as a supplementary measure.

Macro-F_1

For a level k, precision and recall are defined per class as

$$P_k = \frac{O_{kk}}{n_k^{\text{pred}}}, \qquad R_k = \frac{O_{kk}}{n_k^{\text{ref}}}, \qquad F_{1,k} = \frac{2\,P_k\,R_k}{P_k + R_k}, \tag{2.8}$$

where $F_{1,k} := 0$ if $P_k + R_k = 0$. Macro-F_1 is the unweighted arithmetic mean over all levels,

$$F_1^{\text{macro}} = \frac{1}{C} \sum_{k=0}^{C-1} F_{1,k}. \tag{2.9}$$

The unweighted averaging is the decisive point: each level enters with equal weight regardless of its frequency, so that weak performance on rare levels cannot be compensated by strong performance on the majority level. Macro-F_1 is thus the stricter measure under class imbalance. It is, however, blind to the ordinality of the label space, since it treats the levels as nominal categories.

Quadratic Weighted Kappa

Cohen's κ measures the agreement between two raters while correcting for the agreement expected by chance [12]. The weighted variant κ_w generalises this principle by weighting deviations of different severity differently [13]. Let $E \in \mathbb{R}^{C \times C}$ be the frequency matrix expected under independence of the marginal distributions,

$$E_{jk} = \frac{n_j^{\text{ref}} \cdot n_k^{\text{pred}}}{N}, \tag{2.10}$$

and let $W \in \mathbb{R}^{C \times C}$ be a weight matrix that assigns to each cell the severity of the corresponding deviation. Then

$$\kappa_w = 1 - \frac{\sum_{j,k} W_{jk}\, O_{jk}}{\sum_{j,k} W_{jk}\, E_{jk}}. \tag{2.11}$$

The quadratic weighting used in this thesis sets

$$W_{jk} = \frac{(j-k)^2}{(C-1)^2}, \tag{2.12}$$

so that $W_{jj} = 0$ on the diagonal and $W_{jk} = 1$ for the maximum possible deviation. Substituting (2.12) into (2.11) yields the Quadratic Weighted Kappa

$$\text{QWK} = 1 - \frac{\sum_{j,k} (j-k)^2\, O_{jk}}{\sum_{j,k} (j-k)^2\, E_{jk}}, \tag{2.13}$$

in which the normalisation factor $(C-1)^{-2}$ cancels out. The range is QWK ≤ 1; a value of 1 corresponds to perfect agreement, 0 to chance-level agreement, and negative values to systematically worse-than-chance agreement. The quadratic term penalises distant confusions disproportionately and thus reflects the ordinal structure of the scoring scheme. QWK is the established standard measure in automated scoring and was used, among others, as the official metric of the ASAP competition [56, 2]. Figure 2.3 contrasts the quadratic weighting with the $0/1$ weighting underlying accuracy.



(a) $0/1$ **weighting**
every error counts equally
(corresponds to accuracy)

(b) quadratic weighting
$W_{jk} = (j-k)^2/(C-1)^2$
adjacent levels hardly count

Figure 2.3: Weight matrices for $C = 4$: $0/1$ weighting of accuracy **(a)** and quadratic weighting of QWK according to (2.12) **(b)**.

Assumptions and Limitations of QWK

The use of QWK is tied to a substantive assumption that is not always made explicit in the ASAG literature: the quadratic weighting (2.12) presupposes that the label indices j and k lie on an ordered scale and that their numerical difference can be interpreted as a meaningful measure of the severity of a deviation. In effect, QWK thus assumes interval-scaled distances,

although scoring scales are, strictly speaking, only ordinal. For point scales such as that of ASAP-SAS this idealisation is defensible, since the levels reflect the number of correctly stated content elements and the distances are therefore approximately equivalent.

The assumption becomes problematic, by contrast, for nominal label sets. The original five-way scheme of SemEval-2013 Task 7 distinguishes the categories *correct*, *partially_correct_incomplete*, *contradictory*, *irrelevant* and *non_domain* [18]. These categories do not form a ranking: a *contradictory* answer is not in any well-defined sense “better” or “worse” than an *irrelevant* answer, and confusing *correct* with *non_domain* is not four times as serious as confusing *correct* with *partially_correct_incomplete* merely because the arbitrary indexing suggests so. In this five-way setting, QWK is consequently not a meaningful measure, and the results depend on the arbitrary order of the categories.

For Beetle and SciEntsBank, this thesis does not use the nominal five-way scheme but a three-level ordinal version with the levels *incorrect*, *partially correct* and *correct*, which represents a genuine ranking of the degree of performance (cf. Chapter 3). It is not to be confused with the official three-way variant of SemEval-2013 Task 7, which distinguishes *correct*, *contradictory* and *incorrect* and therefore likewise forms no ranking [18]. Under this choice, the ordinality assumption of QWK is met and the metric can be interpreted comparably across all three benchmarks. This decision is explicitly declared here as a methodological choice, because it limits comparability with earlier work reporting on the five-way or three-way scheme of SemEval-2013.

Irrespective of the scale level, QWK has further known weaknesses. Doewes, Kurdhi, and Saxena [17] show that its value depends considerably on the shape of the marginal distributions and on the length of the scale, that the paradoxes known from the kappa literature also occur in automated scoring, and that QWK should therefore not be used as the sole measure. For this reason, Macro-F_1 is used as a second metric in this thesis: QWK captures the ordinal proximity of misclassifications, Macro-F_1 the performance on rare levels. The result tables in Chapter 5 mainly report QWK; all comparisons with several random seeds are repeated under Macro-F_1 in Section 5.5.3, where the discrepancies between the two measures are also discussed.

Reliability of Measured Differences

The measures introduced so far describe *what* is measured. For the evaluation in Chapter 5 it must additionally be clarified *how precise* such a measurement is – that is, which difference between two configurations may count as a result at all.

Stochastic Components of Training

Training a neural model is not a deterministic procedure. Several components rely on random decisions:

- the initialisation of newly added modules – in this thesis the fusion and scoring parameters (Section 2.7) and the LoRA matrices,
- the order in which training instances are assembled into batches,

- the random deactivation of units by dropout,
- the split into training and validation sets if no official development split exists; in this thesis it is performed with a fixed random seed that is independent of the seed of the run (Section 4.6).

These sources can be controlled via a *random seed*: a fixed starting value of the random number generator determines the sequence of all random decisions and in principle makes a run repeatable. Full reproducibility is nevertheless not guaranteed. Concurrent computations on graphics processors sum floating-point values in a non-deterministic order; since floating-point addition is not associative, this alone can lead to diverging results. Parallel data loading by several processes can also affect batch composition.

Variation Between Runs

In practice, this means that two training runs with identical configuration but different random seeds do not yield the same value. Reimers and Gurevych [44] show for sequence labelling tasks that the choice of random seed alone can lead to statistically significant differences, and argue for reporting distributions over several runs instead of single values. Dodge et al. [16] demonstrate the same effect for fine-tuning pre-trained language models and attribute it in comparable proportions to weight initialisation and data order.

This gives rise to an evaluation rule that is applied throughout Chapter 5. A single measured value is not a point measurement but a sample from a distribution. The difference between two configurations can only be interpreted as an effect if it is *larger* than the variation that the same configuration exhibits when merely the random seed is changed. This variation is referred to below as the *noise threshold*; it is determined empirically in Section 5.2 rather than assumed.

If $\{m_1, \ldots, m_S\}$ denotes the measured values of a configuration over S random seeds, the range

$$\Delta = \max_s m_s - \min_s m_s \tag{2.14}$$

is used as the measure of variation. With the number of $S = 3$ runs per configuration feasible here, the range is a more robust measure than a standard deviation, whose estimate from three values would be very uncertain. For the same reason, inferential statistical tests are not used: their assumptions are not met with three observations per cell, and comparing the observed difference with the observed range is the more honest presentation. Figure 2.4 illustrates the resulting decision rule.

Ablation

A recurring evaluation pattern in this thesis is *ablation*: *one* component is deliberately removed from or replaced in a working configuration, while all other components remain unchanged. An observed performance difference can then be attributed to this one component. The informative value of an ablation depends directly on the noise threshold described above: it demonstrates a contribution of the removed component only if the difference exceeds the variation between runs.

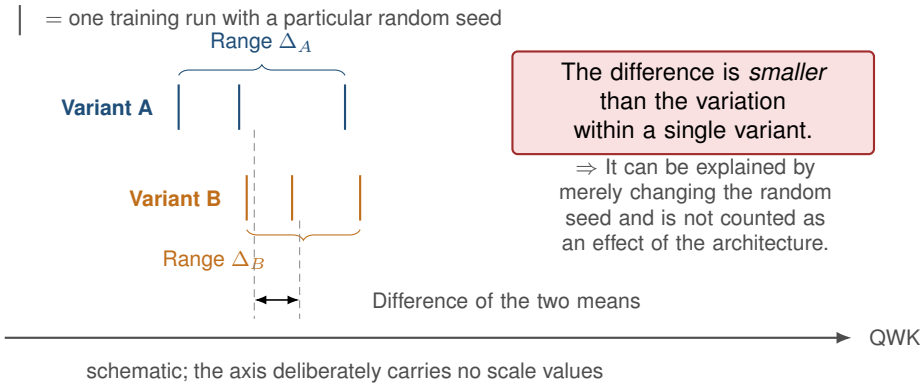


Figure 2.4: Schematic evaluation rule: ticks mark runs with different random seeds, braces the range (2.14). If the difference of the means lies below the range, it is not interpreted as an effect.

2.3 Encoder-Based Approaches to ASAG

Early ASAG systems scored answers using manually engineered features, such as semantic similarity measures to the reference answer [40, 54] or matching against pre-formulated scoring templates [39]; Burrows, Gurevych, and Stein [7] give an overview of this phase. Their common drawback lies in the task-specific effort of feature engineering and in the fact that paraphrases as well as answers that are correct in content but lexically divergent are captured only insufficiently.

Neural Encoders

With the availability of pre-trained language models, the field shifted to neural encoders. Riordan et al. [46] systematically investigated recurrent and convolutional architectures for short answer scoring and showed that neural models can outperform a strong feature-based baseline, with performance and optimal parameters varying between tasks. Kumar et al. [31] combined various feature types in AutoSAS, including lexical diversity, Word2Vec embeddings and overlap with the prompt and content, into a fast and scalable scoring system.

The approach that dominates today was established by transferring BERT [15] and related encoder models such as RoBERTa [37] to the task. Sung et al. [55] showed that domain-specific further pre-training of BERT on subject resources markedly improves scoring quality; Camus and Filighera [8] investigated the transfer of various transformer encoders to ASAG benchmarks, and Ormerod [42] demonstrated gains from ensembles of pre-trained models.

Architecturally, these works follow a uniform pattern. The textual fields are concatenated into one input sequence, separated by delimiter tokens such as `[SEP]`, and processed by the encoder. A single sequence representation is obtained from the resulting hidden-state matrix – usually the state at the `[CLS]` position – and mapped by a linear output layer to C logits, from which a softmax yields a distribution over the global label space $\mathcal{Y}$. This construction corresponds exactly to the classical formulation (2.4).

A related but architecturally different branch dispenses with the fixed classification head and scores via similarity. Bexte, Horbach, and Zesch [5] compare the student answer with anchor answers of different scoring levels and derive the level from the highest embedding similarity, using Sentence-BERT-style bi-encoders [45]. Li et al. [34] model the relations between answers, reference answers and scoring levels as a relational network. These methods already anticipate conceptually what is systematised as rubric-guided scoring in Section 2.5: the scoring level is represented not by an index but by a text.

Limitations

The encoder line is strong and computationally cheap within a benchmark but has two structural limitations. First, the fixed classification head binds the model to the cardinality and semantics of a concrete label space; a change of benchmark, or even a single task with a different point scale, requires a new output layer and thus renewed training [23]. Second, the model receives the scoring criteria at best implicitly through the training data; the rubric, as an explicit textual description of what constitutes a level, does not enter the prediction. Both limitations motivate the architectures presented in Section 2.5.

2.4 LLM-Based Approaches to ASAG

Generative Scoring

With the availability of large instruction-tuned language models, it is natural to formulate scoring as text generation: the model receives the question, reference answer, student answer and, where applicable, the rubrics as a prompt and generates the scoring level as a string. Kortemeyer [30] evaluates GPT-4 in this configuration on the SemEval datasets SciEntsBank and Beetle, Chamieh, Zesch, and Giebermann [9] systematically investigate zero- and few-shot prompting, Chang and Ginter [10] transfer the approach to Finnish, and Ferreira Mello et al. [21] compare prompt-optimised GPT-4 with classical models.

The approach is maximally flexible, since it does not presuppose a fixed label space, but it has three drawbacks. First, autoregressive decoding at inference time is considerably more expensive than a single discriminative forward pass. Second, the results are sensitive to the wording and format of the prompt, which impairs reproducibility. Third, generation requires post-processing, since the model is not guaranteed to output a parsable label.

Data Protection Context

An additional restriction that is central to education concerns the processing of student texts by commercial model providers. Student answers are personal data as soon as they can be attributed to an identifiable person, and under the General Data Protection Regulation children's data merit specific protection [20, Art. 4(1) and Recital 38]. When generative AI is used in education, duties to protect children and young people must also be observed [58]. Transmitting

such texts to external application programming interfaces therefore raises data protection and governance issues. This gives rise to the requirement that ASAG systems remain operable locally. Locally operable generative models of sufficient size, however, are computationally expensive, so a method is desirable that exploits the representational power of large language models without having to generate at inference time.

LLMs as Discriminative Encoders

Exactly this possibility is opened up by a more recent line of research that repurposes autoregressive language models as text encoders. BehnamGhader et al. [3] show with LLM2Vec that decoder-only models become powerful encoders through removal of the causal mask, masked further pre-training and contrastive post-training. Lee et al. [32] develop with NV-Embed training procedures that turn decoder models into generalist embedding models. Lin et al. [35] investigate training-free conversion and identify the attention sink – the tendency of the model to concentrate attention on the first token – as a central obstacle, while Qiao et al. [43] reinterpret decoder-only models as masked autoencoders. Further work demonstrates the effectiveness of the approach for classification [47, 36] and dense retrieval tasks [38]. Gekhman et al. [22] additionally show that language models represent factual knowledge in their hidden states even when they cannot reliably retrieve it generatively – a finding that further motivates encoder-based use for knowledge-intensive discrimination tasks.

For ASAG, this approach is attractive for two reasons. On the one hand, a language model can jointly represent the heterogeneous fields question, reference answer, student answer and rubrics in a single long sequence instead of comparing them pairwise. On the other hand, it brings along the world knowledge required for subject-matter assessment. At the same time, there is a research gap: previous work on repurposing LLMs as encoders evaluates predominantly on single-sequence or sentence-pair tasks such as those prevalent in retrieval and similarity benchmarks [3, 32, 36]. The joint processing of several heterogeneous input fields, as required by ASAG, has hardly been investigated so far.

2.5 Rubric-Guided Scoring: Guided Alignment Architectures

Rubrics as Label Semantics

In educational measurement, a rubric is a textual description that specifies which characteristics an answer exhibits at a particular performance level. From a modelling perspective, it is thus a *semantic representation of a label*: instead of treating level k as a meaningless element of an index space, it is characterised by the text $r_{i,k}$.

This idea is not limited to ASAG. In natural language processing, there is a broad line of work that improves classification tasks by incorporating label semantics. It includes Pattern-Exploiting Training, which maps labels to words via a verbaliser [48], its extension by a label encoder that replaces the verbalisation [26], and discriminative formulations in which input and

label option are jointly assessed as a statement [61, 19]. Such methods replace purely index-based classification and are particularly effective in zero- and few-shot scenarios. Rubric-guided ASAG can be understood as a domain-specific instance of this principle.

Rubric Retrieval and Pairwise Matching

A common implementation in ASAG treats the task as a retrieval or matching problem: each instance is expanded into K_i pairs $(a_i, r_{i,k})$, each pair is encoded separately, and a head decides whether the answer meets the criteria described in the respective rubric text [51]. A related variant uses anchor answers of different levels as implicit rubrics, either via similarity comparisons [5] or as nodes of a relational network [34].

The advantage of this formulation is its flexibility: it is independent of a global label inventory and thus directly applicable to unseen rubric sets. Its drawback is its cost. If c denotes the cost of one forward pass, scoring an instance requires $K_i \cdot c$ instead of c, both in training and at inference time. With large backbones and extensive datasets, this quickly becomes prohibitive. Rubric retrieval thus trades efficiency for flexibility. Figure 2.5 contrasts this method with classical classification and with the span alignment developed below.

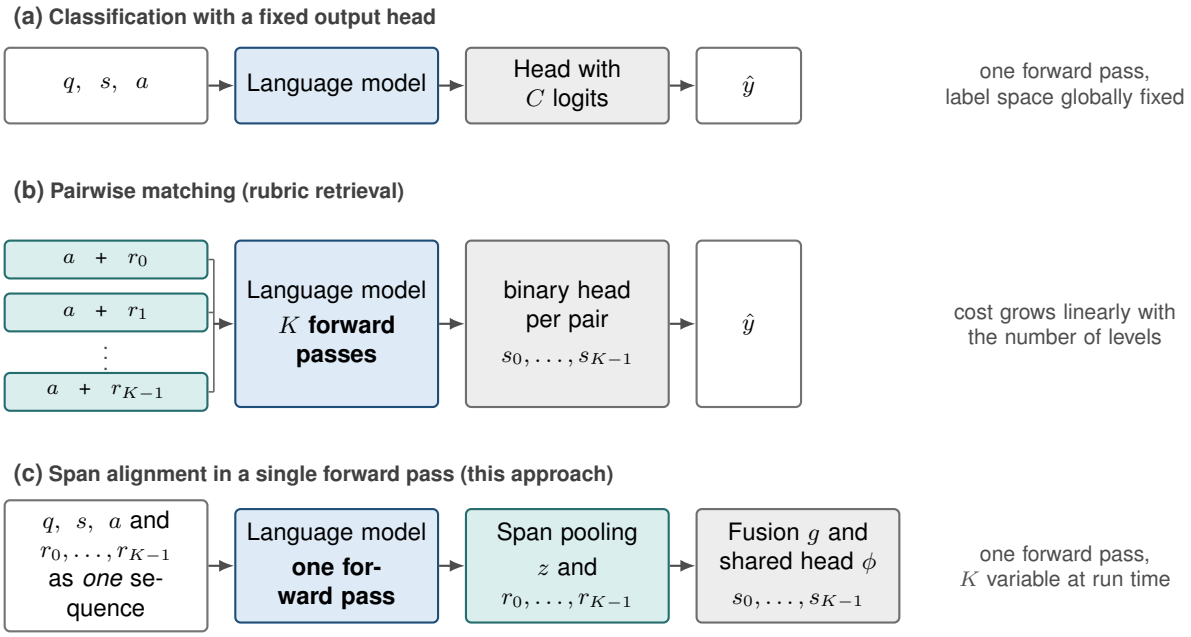


Figure 2.5: Fixed classification head **(a)**, pairwise answer–rubric encoding with K forward passes **(b)** and joint encoding with span pooling in one forward pass **(c)**.

Span Alignment in a Single Forward Pass

A way out of this trade-off is offered by encoding all fields jointly in a single sequence. Gombert et al. [23] encode the answer and rubrics together and align the representation of the answer

span with the representations of the individual rubric spans via a bilinear network. This removes both the fixed global label index and the pairwise expansion.

The general scheme of this architecture family can be summarised as follows. The fields of (2.2) are serialised into a single string, marking the field boundaries and keeping track of the character positions of the rubric spans. After encoding by the language model, a hidden-state matrix

$$H_i \in \mathbb{R}^{T \times d} \tag{2.15}$$

is obtained, where T denotes the sequence length and d the model dimension. Two kinds of representations are extracted from H_i:

$$r_{i,k} = H_i[e_{i,k}] \qquad \text{rubric representation of level } k, \tag{2.16}$$

$$z_i = H_i[T] \qquad \text{sequence representation.} \tag{2.17}$$

Here, $e_{i,k}$ denotes the position of the last token of the k-th rubric span. The choice of the last token is not arbitrary: under causal attention, the state at the end position of a span aggregates the information of the entire preceding context including that span (cf. Section 2.6.2). Analogously, z_i, as the state of the last sequence token, summarises the entire input and is accordingly used as the input of fine-tuned LLM classifiers [47]; for embedding models, by contrast, mean pooling sometimes proves superior [3].

Each rubric representation is then combined with a context representation via a fusion function g,

$$v_{i,k} = g\big(z_i,\, r_{i,k}\big), \tag{2.18}$$

and a shared scoring head ϕ projects the result onto a scalar,

$$s_{i,k} = \phi(v_{i,k}) \in \mathbb{R}. \tag{2.19}$$

The prediction is obtained as the argmax over the candidate levels according to (2.5). Since ϕ maps onto a scalar rather than onto C logits, the architecture is independent of the number of rubric levels and can process instances with varying K_i without modification.

This is precisely where the present thesis comes in. It investigates, first, which context representation should be used in (2.18) – the representation of the answer span, as in Gombert et al. [23], or the sequence representation z_i –, second, which fusion function g should be chosen (Section 2.8 and Chapter 4), and third, how decoder-only models can be adapted to this discriminative task by controlling the attention mask (Section 2.6.2).

2.6 Fundamentals of the Transformer Architecture

All models used in this thesis are based on the transformer architecture [59]. This section develops the components required to understand the methodology.

Overall Structure

An input sequence is first split by a tokenizer into a sequence of T tokens and mapped into a continuous space via an embedding matrix $W_E \in \mathbb{R}^{|V|\times d}$, yielding the initial matrix $H^{(0)} \in \mathbb{R}^{T\times d}$. This passes through L structurally identical blocks. Each block ℓ consists of a multi-head self-attention sublayer and a position-wise feed-forward sublayer, each embedded in a residual connection and stabilised by a normalisation layer. In the pre-norm variant that is common today,

$$\tilde{H}^{(\ell)} = H^{(\ell-1)} + \mathrm{MHA}\big(\mathrm{Norm}(H^{(\ell-1)})\big), \tag{2.20}$$

$$H^{(\ell)} = \tilde{H}^{(\ell)} + \mathrm{FFN}\big(\mathrm{Norm}(\tilde{H}^{(\ell)})\big). \tag{2.21}$$

As normalisation, the original model uses layer normalisation [1], whereas the Llama and Mistral models used in this thesis use the computationally cheaper root mean square normalisation [62] in a pre-norm arrangement. This arrangement was already used in Llama 2 [57], on whose architecture Llama 3 and Mistral build [25, 29]. In the original model, the feed-forward sublayer consists of two linear maps with a non-linearity in between; the models mentioned use a gated linear unit in the SwiGLU variant instead [49]. The output $H^{(L)}$ of the last layer forms the hidden-state matrix H_i used in (2.15). Figure 2.6 summarises the overall stack and the block structure.

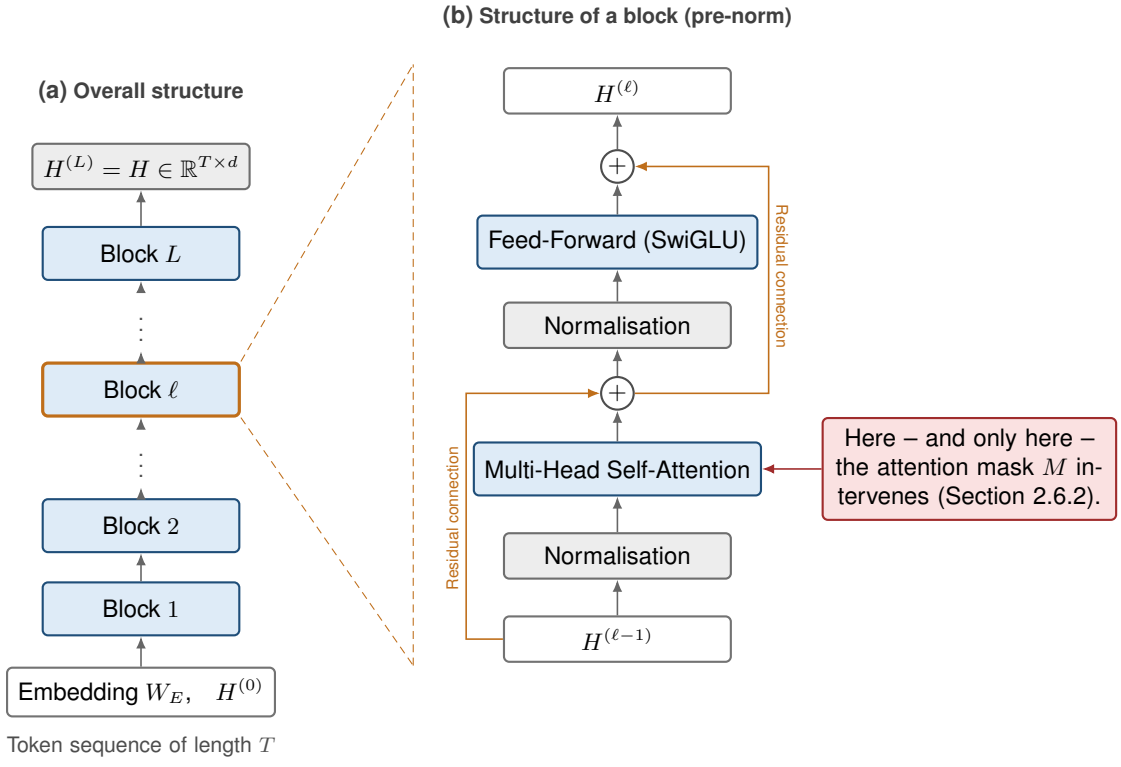


Figure 2.6: Overall stack of L blocks **(a)** and a single pre-norm block **(b)** according to (2.20) and (2.21).

The central architectural difference between encoder-only models such as BERT [15] and decoder-only models such as Llama [25] or Mistral [29] lies not in the block structure but exclu-

sively in the attention mask and the pre-training objective; this is elaborated in Section 2.6.2.

2.6.1 Self-Attention and Multi-Head Attention

Self-attention computes for each position a weighted sum over the representations of all positions, with the weights derived from the similarity of the positions to one another. Starting from $X \in \mathbb{R}^{T \times d}$, three linear projections are formed,

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V, \tag{2.22}$$

with $W_Q, W_K \in \mathbb{R}^{d \times d_k}$ and $W_V \in \mathbb{R}^{d \times d_v}$. The rows of Q, K and V are referred to as query, key and value vectors. The attention output is

$$\text{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V. \tag{2.23}$$

The matrix $S = QK^\top \in \mathbb{R}^{T \times T}$ contains in entry S_{ij} the dot product between the query at position i and the key at position j, i.e. an unnormalised measure of the relevance of position j for position i. The softmax is applied row-wise, so that for each i the attention weights α_{ij} form a probability distribution over j:

$$\alpha_{ij} = \frac{\exp\left(S_{ij}/\sqrt{d_k}\right)}{\sum_{j'=1}^{T} \exp\left(S_{ij'}/\sqrt{d_k}\right)}, \qquad \text{Attention}(Q, K, V)_i = \sum_{j=1}^{T} \alpha_{ij}\, V_j. \tag{2.24}$$

The scaling by $1/\sqrt{d_k}$ prevents the softmax from entering saturation regions with vanishing gradients for large dimensions [59].

Multi-head attention performs h such operations with their own projections in parallel, concatenates their outputs and projects them back onto the model dimension with W_O; the individual heads can thus capture different types of relations. The models used in this thesis employ the grouped-query attention variant, in which several query heads share one set of key and value heads [29, 25].

The computational and memory cost of self-attention grows with $\mathcal{O}(T^2 d)$, since the matrix S is quadratic in the sequence length. This is directly relevant to this thesis, because the joint serialisation of question, reference answer, answer and all rubrics leads to longer sequences than pairwise encoding – a cost that is, however, more than offset by the fact that only one instead of K_i forward passes is required (cf. Section 2.5).

Encoding Positional Information

Operation (2.23) is permutation-equivariant: reordering the input positions yields a correspondingly reordered output without the order itself entering the computation. Positional information must therefore be introduced explicitly. The original model adds sinusoidal positional encodings to the embeddings [59], and BERT uses learned absolute position embeddings [15].

The decoder models used in this thesis instead use Rotary Position Embedding (RoPE) [53]. RoPE does not encode position additively but rotates query and key vectors depending on their position, so that the attention logit of two positions depends only on their *relative* distance. Only this property matters for the present thesis: since RoPE contains no directional assumption, it remains applicable unchanged when the attention mask is modified, and the removal of causality described in Section 2.6.2 requires no adjustment of the positional encoding.

2.6.2 Causal vs. Bidirectional Attention Masks

Masking as an Additive Term

Which positions may see each other is controlled by an additive mask matrix $M \in (\{0\} \cup \{-\infty\})^{T\times T}$, which is added to the scaled logits before the softmax:

$$\text{Attention}(Q, K, V; M) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V. \tag{2.25}$$

An entry $M_{ij} = -\infty$ yields $\exp(\cdot) = 0$ and thus $\alpha_{ij} = 0$: position i cannot attend to position j. In the implementation, $-\infty$ is replaced by the smallest representable value of the floating-point format used.

Two types of mask must be distinguished. The *bidirectional* mask restricts attention only with respect to padding positions,

$$M_{ij}^{\text{bidir}} = \begin{cases} 0, & \text{if position } j \text{ is a valid token,} \\ -\infty, & \text{otherwise,} \end{cases} \tag{2.26}$$

so that every position sees every other. This is the mask of encoder-only models such as BERT and RoBERTa, whose pre-training via masked language modelling consistently presupposes a two-sided context [15, 37]. The *causal* mask adds a strict block on future positions to the padding condition,

$$M_{ij}^{\text{causal}} = \begin{cases} 0, & \text{if } j \le i \text{ and } j \text{ is valid,} \\ -\infty, & \text{otherwise,} \end{cases} \tag{2.27}$$

so that the attention weight matrix takes on a lower triangular form. This mask is constitutive for decoder-only models: it ensures that the prediction of the token at position $i+1$ is based exclusively on positions $1, \ldots, i$, and thereby makes training on all positions of a sequence possible in parallel. Figure 2.7 contrasts the two mask types.

Consequences for Discriminative Use

From (2.27) follows directly the asymmetry that is central to this thesis: in a decoder-only model, the state $H[i]$ is a function exclusively of the prefix $1, \ldots, i$. Two consequences follow.

First, the choice of pooling is determined. Since only the last position has seen the entire sequence, last-token pooling $z = H[T]$ is the natural sequence representation, whereas encoder

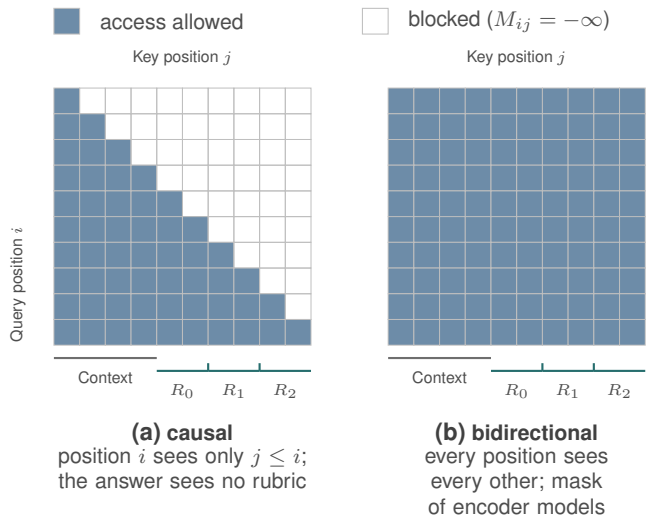


Figure 2.7: Causal **(a)** and bidirectional **(b)** attention mask for a sequence consisting of a context block and three rubric blocks; filled cells denote permitted access.

models use the `[CLS]` position or a mean over all positions. It likewise determines the choice of the *last* token of a rubric span as the pooling anchor in (2.16): only at this position has the entire span entered the representation.

Second, the causal mask is an unnecessary restriction for a pure comprehension task. For scoring a student answer, the causality condition has no substantive motivation; the model is meant to understand the input, not to continue it. Under a causal mask, the tokens of the answer cannot take the subsequent rubric texts into account, although precisely this information would be relevant for the assessment. It is exactly this observation that motivates the work on repurposing LLMs as encoders [3, 35, 43].

Removing Causality

The simplest intervention consists of replacing M^{causal} by M^{bidir}, thereby lifting the block on future positions. Since the mask is a purely additive term in (2.25), this intervention requires no change to the weights, the projections or the positional encoding: the scaled dot-product attention and RoPE remain unchanged, and only the set of admissible key positions is extended.

The intervention is, however, not without consequences. The model was pre-trained exclusively under a causal mask; the abrupt extension of the visible context represents a shift relative to the pre-training distribution. BehnamGhader et al. [3] address this with additional masked further pre-training that first teaches the model to use the two-sided context. Lin et al. [35] investigate the training-free case and identify the *attention sink* as the core problem: autoregressive models concentrate a considerable part of their attention mass on the first positions of the sequence, which serve as a store for surplus attention [60]. If the mask is removed, the normalisation of the softmax in (2.24) changes for all positions, which disrupts this mechanism [35].

Against this background, a *partial* removal of causality suggests itself, in which only the top n_{bidir} of the L layers compute bidirectionally while the lower layers remain causal. With layer

indices $\ell = 1, \ldots, L$,

$$M^{(\ell)} = \begin{cases} M^{\text{bidir}}, & \text{if } \ell > L - n_{\text{bidir}}, \\ M^{\text{causal}}, & \text{otherwise,} \end{cases} \qquad n_{\text{bidir}} \in \{0, \ldots, L\}. \tag{2.28}$$

The number n_{bidir} interpolates between the unchanged decoder ($n_{\text{bidir}} = 0$) and a fully bidirectional model ($n_{\text{bidir}} = L$). It is conveniently specified as a proportion ρ of the number of layers; the conversion used in this thesis, which rounds down and reads values of 1 and above as an absolute number of layers, is given in Section 4.3.2. The intuition behind this construction is that lower layers predominantly extract local, lexical and syntactic features and should therefore stay close to the pre-training behaviour, whereas upper layers perform task-specific aggregation and can benefit from a complete context there. The systematic investigation of ρ is a core contribution of this thesis; the concrete implementation is described in Chapter 4 and the results are reported in Chapter 5. Figure 2.8 shows the resulting layer split for the proportions investigated.

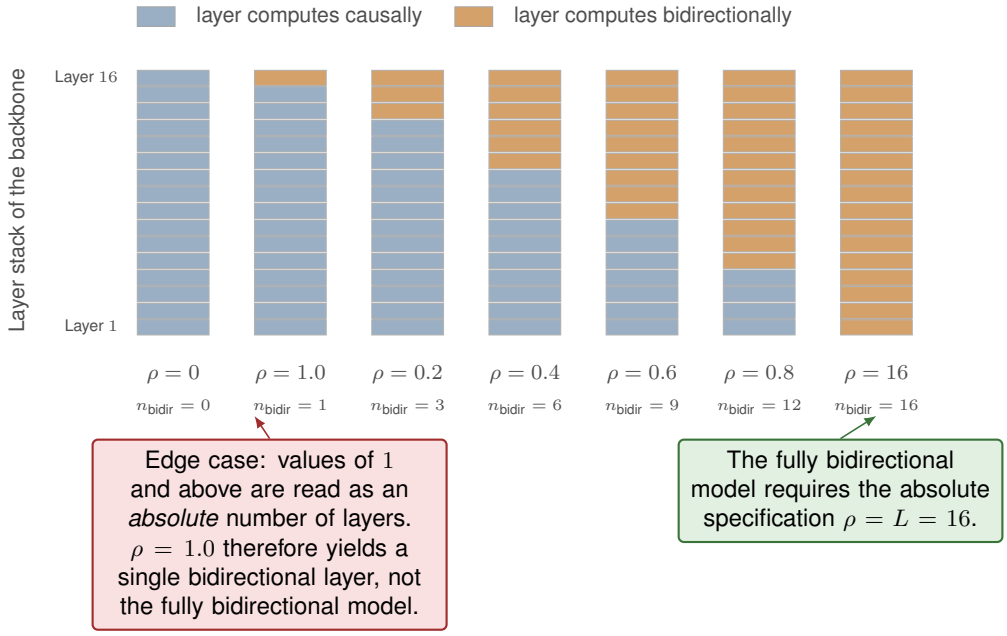


Figure 2.8: Partial removal of causality using Llama-3.2-1B ($L = 16$) as an example: the bidirectional layers always form the top group. $\rho = 1.0$ denotes exactly one layer (cf. (4.15)).

2.7 Parameter-Efficient Fine-Tuning (LoRA)

Full fine-tuning of all backbone parameters would be neither memory- nor compute-efficient given the number of configurations investigated in Chapter 5. Parameter-efficient methods therefore freeze most of the parameters and train only a small number of additional parameters [27, 33].

LoRA [28] parametrises the weight update of a frozen matrix $W_0 \in \mathbb{R}^{d \times k}$ as the product of two

low-rank matrices $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with $r \ll \min(d, k)$. The forward pass of the adapted layer is

$$h = W_0 x + \frac{\alpha}{r} BAx, \tag{2.29}$$

where the scaling factor α/r decouples the effective learning rate from the chosen rank. Since B is initialised with zeros, the adapted model corresponds exactly to the pre-trained one at the start of training. The configuration chosen in this thesis is documented in Section 4.6.

The fusion and scoring components introduced in Chapter 4 are not part of the pre-trained backbone. They are randomly initialised and trained in full, since the low-rank adaptation is applied exclusively to the language model.

2.8 Fusion Mechanisms in Rubric–Answer Alignment

The fusion function g from (2.18) introduced in Section 2.5 is the point at which rubric-guided architectures in ASAG differ from one another. This section situates the mechanisms compared in this thesis within this line of work.

Similarity-Based Methods

The simplest form of combination dispenses with learned parameters and scores via a similarity measure. Bexte, Horbach, and Zesch [5] encode the student answer and the anchor answers of the individual scoring levels separately and assign the level with the highest cosine similarity. The fusion here is thus a fixed function $g(u, v) = \cos(u, v)$ without a trainable component. The advantage lies in the direct transferability to new sets of levels, the drawback in the fact that the match between answer and criterion is reduced to a single scalar and no feature-wise weighting is possible.

Pairwise Classification

A second approach expands each instance into answer–rubric pairs and lets a model decide for each pair whether the answer meets the respective criterion [51]. Fusion here takes place not at the level of extracted vectors but implicitly within the encoder, which processes both texts jointly. This allows arbitrarily complex interactions but comes at the price of K forward passes per instance (Section 2.5) and prevents the model from seeing the criteria of a task in relation to one another.

Bilinear Alignment

Gombert et al. [23] encode the answer and rubrics jointly and align the representation of the *answer* span with the representations of the individual rubric spans via a bilinear form

$$g_{\mathsf{bl}}(u, v) = u^\top \mathcal{W} v + b, \qquad \mathcal{W} \in \mathbb{R}^{d\times d}. \tag{2.30}$$

The bilinear form captures all pairwise feature interactions between the two vectors and is thus more expressive than a linear map over the concatenation. The price is d^2 additional parameters, which carry weight for large model dimensions and increase the tendency to overfit. This work is the direct methodological reference point of the present study: it establishes the principle of treating rubrics as spans in a jointly encoded sequence, but uses the answer span rather than the sequence representation as the anchor.

Concatenative and Multiplicative Fusion

The mechanisms compared in this thesis start exactly here and vary the construction of v_k while keeping the rest of the architecture unchanged. CONCAT combines both representations purely additively in the subsequent linear head, DIFF uses only their element-wise difference, CONDIFF combines both, and GATE introduces a multiplicative conditioning of the context representation by the rubric. The formal specification is given in Section 4.2.

Comparing these four variants is informative in the ASAG context because they differ systematically in the number of additional parameters that have to be trained in full – from none (CONCAT, DIFF, CONDIFF) to d^2 (GATE) – and thus make it possible to separate the contribution of additional fusion capacity from that of the representational quality of the backbone. To the best of the author's knowledge, a systematic comparison of these variants in combination with the sequence representation as anchor and across several backbone sizes has not yet been carried out in ASAG; it constitutes a core contribution of this study.

3 Datasets

This thesis evaluates the RUSPAN system described in Chapter 4 on three publicly available ASAG benchmarks: ASAP-SAS, Beetle and SciEntsBank. This chapter describes the structure, statistics and label scale of each benchmark (Sections 3.1 to 3.3), documents the origin of the rubric texts and reference answers (Section 3.4) and finally justifies the selection of these three datasets for the comparative evaluation (Section 3.5). All reported figures were computed from the training and test files actually used in this thesis, not from the original competition or corpus descriptions, since the concrete splits may deviate slightly from the respective original publication.

3.1 ASAP-SAS: Structure, Statistics, Label Scale

Origin and Format

The Automated Student Assessment Prize – Short Answer Scoring (ASAP-SAS) is a competition on the automated scoring of short constructed student responses, organised by the Hewlett Foundation in cooperation with Kaggle in 2012 [56, 2]. Both public participants and commercial vendors of scoring systems took part; Shermis [50] provides an analysis of the results. The dataset comprises ten independent tasks (*items*) from the subjects *Science*, *Biology* and *English* / *English Language Arts*, answered by students in grades eight and ten. Each instance consists of the question q, the student answer a and an integer score y assigned by the task authors. Unlike Beetle and SciEntsBank, ASAP-SAS contains task-specific scoring guidelines written by the task authors (*general_scoring_rubrics*) but no reference answer s; the latter is supplemented with the help of a language model for this thesis (Section 3.4).

Variable Label Scale

A structural peculiarity of ASAP-SAS compared with the other two benchmarks is the task-dependent scoring scale: four of the ten tasks use a four-level scale $\{0, 1, 2, 3\}$, the remaining six a three-level scale $\{0, 1, 2\}$. Table 3.1 lists subject, grade, label scale and instance counts per task. This variability is the immediate empirical motivation for the rubric-conditioned task formulation (2.5) introduced in Section 2.1: a model with a fixed classification head according to (2.4) cannot easily be trained jointly across all ten tasks, since the output dimension alternates between 4 and 3.

The levels are unevenly distributed. Across all tasks of the training split, $39.5\,\%$ of the answers fall on level 0, $32.7\,\%$ on level 1, $23.4\,\%$ on level 2 and $4.3\,\%$ on level 3. Within individual tasks the imbalance is in part pronounced: for the Biology items 5 and 6, $77.5\,\%$ and $84.3\,\%$ of the answers receive level 0, respectively, and for item 4 only $7.7\,\%$ fall on the highest level. The Science items 1 and 2, by contrast, are comparatively balanced.

Table 3.1: ASAP-SAS: overview of tasks with subject, grade, label scale and instance counts (computed from `train.csv` / `test_ua.csv`).

Item	Subject	Grade	Label scale	Train n	Test n (UA)
1	Science	10	$\{0,1,2,3\}$	1 672	557
2	Science	10	$\{0,1,2,3\}$	1 278	426
3	English Language Arts	10	$\{0,1,2\}$	1 808	406
4	English Language Arts	10	$\{0,1,2\}$	1 657	295
5	Biology	10	$\{0,1,2,3\}$	1 795	598
6	Biology	10	$\{0,1,2,3\}$	1 797	599
7	English	10	$\{0,1,2\}$	1 799	599
8	English	10	$\{0,1,2\}$	1 799	599
9	English	10	$\{0,1,2\}$	1 798	599
10	Science	8	$\{0,1,2\}$	1 640	546
Total				**17 043**	**5 224**

Answer Length and Splits

The student answers are on average 42.3 words long (median 39, maximum 325 words in the training split) and thus considerably longer than the answers in Beetle and SciEntsBank (cf. Sections 3.2 and 3.3). This reflects the task type: several items require several aspects to be named (e.g. the description of several steps of a biological process), which explains the scoring scale with up to four levels – the level often encodes the number of correctly named aspects. For ASAP-SAS, only the Unseen Answers split (UA) is available in this thesis (cf. Section 2.1): the test answers belong to the same ten tasks as the training data, so ASAP-SAS does not permit any statement about generalisation to new questions, but it does test the scoring convention of a variable, task-specific scale.

3.2 Beetle: Domain and Test Splits

Origin and Domain

The Beetle corpus was created during the evaluation of the BEETLE II tutorial dialogue system and contains student answers to explanation and definition questions from basic electricity and electronics (circuits, voltage, short circuits). Together with SciEntsBank, it was released as a joint benchmark for the automated assessment of student answers within SemEval-2013

Task 7 [18]. Unlike ASAP-SAS, Beetle provides a native reference answer s; together with the generated rubric texts (Section 3.4), each instance thus comprises the complete tuple $(q, s, a, \mathcal{R})$ according to (2.2).

Task Structure and Label Scale

The dataset comprises 56 questions, which fall into three thematic groups with the prefixes `FF` (23 questions), `PC` (21 questions) and `SC` (12 questions). In terms of content, the `FF` questions mainly concern voltage measurements at individual terminals, whereas several of the `SC` questions explicitly ask about short circuits. Each question has a uniform three-level ordinal scoring scale whose levels correspond to *Incorrect*, *Partially Correct* and *Correct* and are explicitly named as such in the rubric texts. In the training split, $34.4\,\%$ of the answers are *Incorrect*, $23.3\,\%$ *Partially Correct* and $42.2\,\%$ *Correct*; the middle level is thus the rarest.

Test Splits

Beetle provides two test splits (Table 3.2). The Unseen Answers split (UA) uses the same 47 questions as training but disjoint student answers. The Unseen Questions split (UQ) contains 9 questions that do not occur in training; a check of the question IDs confirms that the intersection between training and UQ questions is empty. Unlike ASAP-SAS, Beetle thus permits a statement about generalisation to new questions within the same domain.

Table 3.2: Beetle: instance and question counts per split (computed from the provided CSV files).

Split	Instances n	Questions	Relation to training
Train	3 941	47	—
Test (UA)	439	47	same questions, new answers
Test (UQ)	819	9	new, disjoint questions

With an average of 9.8 words (median 9), the student answers are considerably shorter than in ASAP-SAS, which corresponds to the task design: the questions typically require a one- to two-sentence justification of an observed phenomenon (e.g. a voltage measurement) rather than an enumeration of several aspects.

3.3 SciEntsBank: Domain and Test Splits

Origin and Domain

SciEntsBank was released together with Beetle within SemEval-2013 Task 7 [18] and, unlike Beetle's narrow domain of electricity, covers a broad range of science topics. The version used in this thesis comprises 200 questions, which can be grouped into 15 topic blocks by their question ID prefixes (among others `ME`, `WA`, `MX`, `PS`, `SE`, `HB`, `EM`, `LF`, `VB`, `FN`, `MS`, `ST`, `EV`, `II`, `LP`, with

between 3 and 25 questions per block). As for Beetle, question, reference answer and rubric texts are available for each instance.

Label Scale

In its original SemEval version, SciEntsBank uses a five-way nominal scoring scheme (*correct*, *partially_correct_incomplete*, *contradictory*, *irrelevant*, *non_domain*) which – as discussed in Section 2.2 – has no ordinal structure. Following common practice in recent ASAG literature, this thesis instead uses a reduced version with the three ordinal levels *incorrect*, *partially correct* and *correct*. Of the 200 questions, 169 are annotated with this complete three-level scale; the remaining 31 questions are available only with two levels (*incorrect* vs. *correct*, without a middle level). This heterogeneity in the number of levels *within* a single benchmark is further empirical evidence for the need for the variable candidate set $\mathcal{Y}_i$ of (2.5). In the training split, $32.9\,\%$ of the answers are *incorrect*, $26.6\,\%$ *partially correct* and $40.4\,\%$ *correct*.

Test Splits

SciEntsBank is the only one of the three benchmarks that provides all three generalisation conditions introduced in Section 2.1 (Table 3.3): Unseen Answers (UA), Unseen Questions (UQ) and Unseen Domains (UD). A check of the question ID prefixes confirms the intended structure of the splits: the 46 questions of the UD split come exclusively from the three topic blocks `WA`, `HB` and `EV`, which are *completely* absent from the training split – the UD split thus genuinely tests transfer to entirely new topic areas and not merely to new individual questions. The training split covers the remaining twelve topic blocks. The UQ split comprises 15 questions from eleven of these twelve blocks, but without overlap in the concrete question IDs. Finally, the UA split uses additional, disjoint test answers for 131 of the 135 training questions; the small residual difference of four questions is due to the concrete split construction.

Table 3.3: SciEntsBank: instance and question counts per split (computed from the provided CSV files).

Split	Instances n	Questions	Relation to training
Train	4 969	135	—
Test (UA)	540	135	mostly the same questions, new answers
Test (UQ)	733	15	new questions, known topic blocks
Test (UD)	4 562	46	new questions from unseen topic blocks

With an average of 11.4 words (UD) to 12.6 words (training), the student answers are similarly short as in Beetle. Owing to the large number of questions and topic blocks and the availability of the UD split, SciEntsBank is the benchmark with the greatest thematic diversity and the most demanding generalisation condition among the three.

3.4 Origin of the Rubric Texts and Reference Answers

None of the three benchmarks natively provides all fields of the tuple (2.2). Since this thesis makes the rubric texts its central subject, their origin must be stated explicitly.

ASAP-SAS contains task-specific scoring guidelines written by the task authors but no reference answer. The latter was generated with the help of a language model from the task description alone.

Beetle and SciEntsBank, conversely, contain native reference answers but no scoring guidelines. The rubric texts used in this thesis were generated for both benchmarks with `gpt-4o-mini`. The input consisted of the task description, the available reference answers and up to ten example student answers per scoring level.

This origin is essential for interpreting the results. The generated texts are not an information source independent of the task description but a paraphrase derived from it that visibly incorporates its content. For the SciEntsBank question *What is the difference between a rock and a mineral?*, for example, the description of the highest level states that correct answers accurately distinguish between rock and mineral and explain that a rock consists of several minerals, whereas a mineral consists of a single substance – which is exactly the content of the corresponding reference answer. Section 5.3.2 shows that this proximity is reflected empirically: rubric texts and task context prove to be largely interchangeable there.

Two limitations must be noted. First, the workflows used for generation are not part of the provided code base; the prompts and the exact selection of example answers could therefore not be inspected. Second, and as a consequence, it cannot be verified from which split the example student answers were drawn. If only training answers were used, the procedure is unproblematic. If test answers had also been included, however, the rubric texts would contain information about the test set, and the rubric-conditioned results would be optimistically biased, in particular on the UA split. This question could not be resolved within this thesis and must be taken into account when assessing the absolute values; the comparisons *between* configurations in Chapter 5 are not affected, since all conditions use the same rubric texts.

3.5 Comparison of the Benchmarks and Rationale for Their Selection

Figure 3.1 compares the size and split structure of the three benchmarks; Table 3.4 adds the qualitative characteristics. The selection is not redundant but deliberately complementary in three respects.

Variability of the label scale. ASAP-SAS is the only one of the three benchmarks whose number of levels varies between individual tasks, and it is thus the direct empirical test case for the rubric-conditioned formulation (2.5) motivated in Section 2.1, under which a model with a fixed classification head is not applicable. Beetle and SciEntsBank, by contrast, mostly have a uniform three-level scale and thus serve as control cases in which any advantage of the variable formulation cannot stem from a different number of levels but must result solely from the semantic rubric conditioning itself.

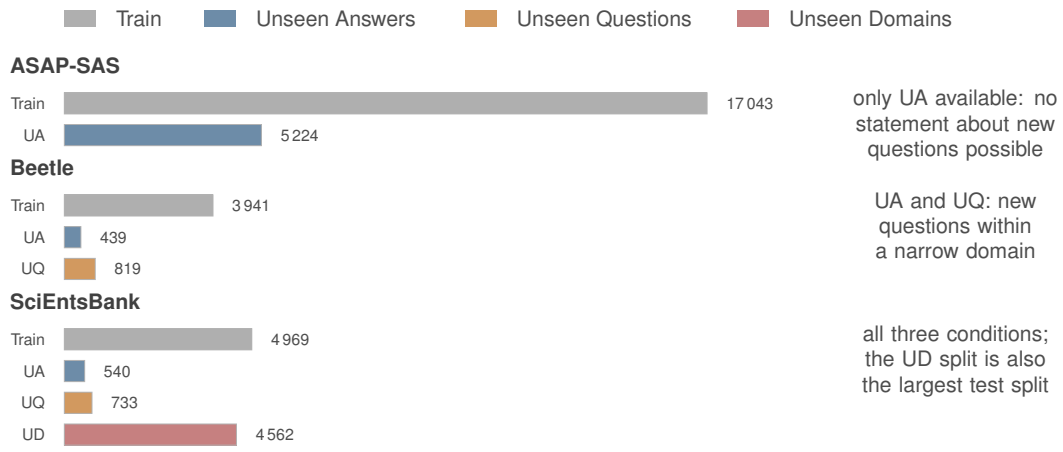


Figure 3.1: Size of the training and test splits of the three benchmarks; bar lengths are proportional to the number of instances. ASAP-SAS provides only the UA condition; the largest test split is SciEntsBank UD.

Table 3.4: Comparison of the three ASAG benchmarks used.

	ASAP-SAS	**Beetle**	**SciEntsBank**
Domain	science / language, 10 individual tasks	electricity (narrow, 1 domain)	broad science spectrum (15 topics)
Native rubrics	yes (task-specific)	no (generated)	no (generated)
Native reference answer	no (generated)	yes	yes
Label scale	variable, 3–4 levels, fixed per task	uniform, 3-level ordinal	uniform, mostly 3-level ordinal
Available splits	UA	UA, UQ	UA, UQ, UD
Answer length (median)	39 words	9 words	11 words
Training size	17 043	3 941	4 969

Availability of generalisation splits. Only SciEntsBank provides a genuine domain transfer condition with the UD split; Beetle at least permits testing on new questions within the same narrow domain with UQ, whereas ASAP-SAS allows only the UA condition. This gradation makes it possible to interpret the results reported in Chapter 5 in a staggered manner according to the generalisation requirement, rather than relying on a single test condition that may be favoured by lexical overlap.

Answer length and domain breadth. ASAP-SAS also differs structurally from Beetle and SciEntsBank: its answers are on average three to four times as long, its tasks come from different school subjects rather than from a single science domain, and its number of training instances is considerably larger. With its narrow focus on electricity, Beetle covers the opposite extreme – a single, internally consistent technical domain with short, precise answers –, while SciEntsBank, with 15 topic blocks, offers an intermediate domain breadth with likewise short answers. The three benchmarks thus differ systematically in answer length, domain breadth and subject, which makes it possible to check whether the architecture developed in Chapter 4 works robustly across these axes of variation or whether its effectiveness is tied to particular

structural properties of a benchmark.

Further benchmarks available in the RUSPAN software project – in particular a German-language dataset (ALICE), a Portuguese-language dataset (PT-ASAG) and iStudio – are not evaluated in this thesis, since the focus lies on the monolingual, English-language comparison of the fusion mechanisms and of bidirectional attention. Their evaluation is mentioned as a possible extension in Chapter 7.

4 Methodology: The RUSPAN System

This chapter develops the system investigated in this thesis. Section 4.1 describes the overall architecture, the input format and the extraction of the span representations. Section 4.2 specifies the four fusion mechanisms compared: CONDIFF, CONCAT, DIFF and GATE. Section 4.3 develops the partial removal of the causal attention mask. Section 4.4 presents the backbone models used, Section 4.5 specifies the variants required for the ablation experiments, and Section 4.6 documents the training configuration.

The notation introduced in Chapter 2 is used throughout: $H_i \in \mathbb{R}^{T \times d}$ denotes the hidden-state matrix of the last layer, z_i the sequence representation, $r_{i,k}$ the representation of the k-th rubric span and g the fusion function. The instance index i is omitted below where it is not needed for disambiguation.

4.1 System Architecture: Span-Alignment Model

Overview

The system implements the architecture family outlined in Section 2.5 in four processing steps:

1. **Serialisation.** The textual fields of an instance are assembled into a single string, keeping track of the character positions of the answer and of each rubric span.
2. **Encoding.** The sequence is processed by the language model in *one* forward pass; the result is the hidden-state matrix $H \in \mathbb{R}^{T \times d}$.
3. **Span pooling.** The sequence representation z and, for each scoring level k, the rubric representation r_k are extracted from H.
4. **Alignment and scoring.** Each pair (z, r_k) is combined via the fusion function g and mapped to a scalar s_k by a shared scoring head. A distribution is formed over the scalars and optimised listwise, i.e. jointly over all candidate levels of an instance (Equation (4.12)).

The decisive difference from a classifier with a fixed output head lies in step 4: the scoring head maps to *one* scalar per candidate level rather than to a vector of fixed length. The number of outputs is thus determined at run time by the number of rubric spans in the input rather than by the architecture. Figure 4.1 shows the complete processing pipeline.

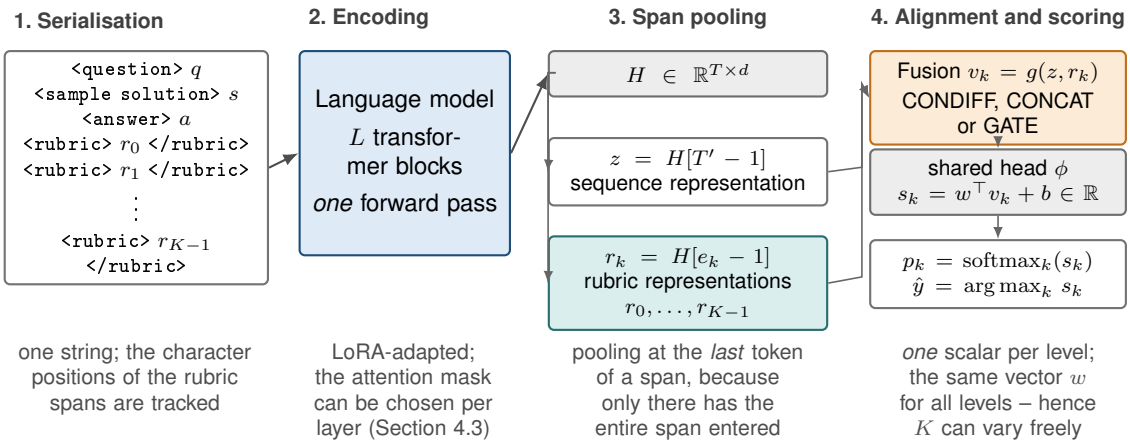


Figure 4.1: Processing pipeline of the span-alignment model: serialisation, encoding in one forward pass, pooling of z and r_k, fusion, and one scalar per candidate level.

4.1.1 Input Format

Serialisation with Field Markers

The fields of an instance are wrapped in XML-like tags and concatenated line by line. For an instance with K scoring levels, the following string is produced:

```
<question> q </question>
<sample solution> s </sample solution>
<answer> a </answer>
<rubric> r_0 </rubric>
<rubric> r_1 </rubric>
⋮
<rubric> r_{K-1} </rubric>
[instruction]
```

The tags serve two functions. First, they structure the input for the language model, which is familiar with XML-like markup from pre-training. Second – and essential for the architecture – the closing tags `</rubric>` provide unique end positions of the rubric spans that can be located in the character stream and at which pooling later takes place.

The order is not arbitrary: the rubrics come *after* all other fields and thus in particular after the student answer. Under causal attention (Section 2.6.2), it follows that each rubric representation has seen the preceding input including the student answer – the rubric representation is therefore not the embedding of an isolated text but a *contextualised* representation of the scoring level in view of the answer actually given. This property is the actual mechanism through which the architecture relates answer and criterion.

The candidate levels are serialised in ascending order of the score. The lowest level is thus always in the first position and the highest in the last.

Optional Components

Both the task context (question and reference answer) and the final instruction are configurable and can be omitted. For this thesis, this configuration must be stated explicitly: **all main runs were carried out without task context and without a final instruction** (`add_context` $=$ `False`, `add_suffix` $=$ `False`). The only exception is the context experiment described in Section 4.5.3, which deliberately lifts this restriction. The input actually processed thus consists exclusively of the student answer and the K rubric texts:

<answer> a </answer>
<rubric> r_0 </rubric>
$\vdots$
<rubric> r_{K-1} </rubric>

The influence of this decision is not a mere formality but essential for interpreting the results. Without question and reference answer, the sequence representation z must reconstruct the entire basis for assessment from the answer and the rubric texts; in particular, the model does not learn *what* the given answer is responding to. The rubric texts are thus the only task-related information in the input – a circumstance that makes the ablation reported in Section 5.3.2 particularly sharp: if they are removed as well, only the answer text remains.

Tokenisation

The serialised string is encoded with the tokenizer of the respective backbone. The maximum sequence length is 2048 tokens; longer inputs are truncated on the right. Shorter sequences within a batch are likewise padded on the right (*right padding*); for the decoder models, the EOS token serves as the padding token. This choice is relevant for pooling and is taken up again in Section 4.1.2.

4.1.2 Span Extraction and Pooling

Mapping Character to Token Positions

Serialisation initially yields the spans as character intervals. Since tokenisation does not map character positions to token positions in a length-preserving way, the intervals are translated using the *offset mapping* provided by the tokenizer. For a character interval $[c_{\text{start}}, c_{\text{end}})$, the token start position is determined as the token whose character interval contains c_{start}, and the token end position analogously as the first token whose interval ends at or extends beyond c_{end}. The resulting token spans follow the convention of half-open intervals

$$\text{Span}_k = [\, b_k,\ e_k\,), \qquad b_k, e_k \in \{0, \ldots, T\}, \tag{4.1}$$

i.e. e_k is the position *after* the last token of the span.

Rubric Representations

For each scoring level k, the representation is formed as the state of the *last* token of the corresponding span:

$$r_k = H\big[e_k - 1\big] \in \mathbb{R}^d. \tag{4.2}$$

The choice of the end position follows directly from causality: under the mask (2.27), $H[j]$ is a function of all positions $\leq j$; only at the end position of the span has the entire rubric text entered the representation. Concretely, this position falls on a token of the closing tag `</rubric>`, which has the additional advantage that the pooling anchor is structurally identical for all levels and all benchmarks.

Alternatively, the implementation supports averaging over all tokens of the span,

$$r_k^{\text{mean}} = \frac{1}{e_k - b_k} \sum_{j=b_k}^{e_k-1} H[j], \tag{4.3}$$

which, however, is not used in the experiments reported here.

Sequence Representation

The sequence representation is formed as the state of the last *valid* token of the sequence. If $m \in \{0,1\}^T$ denotes the padding mask and $T' = \sum_j m_j$ the actual sequence length – the symbol L being reserved for the number of transformer layers –, then with right padding and positions counted from 0,

$$z = H\big[T' - 1\big] \in \mathbb{R}^d. \tag{4.4}$$

Before pooling, the states at padding positions are set to zero and dropout with rate 0.1 is applied. Since the rubrics are at the end of the sequence, z has seen the entire input including *all* rubric texts under causal attention. z is thus not a pure answer representation but a summary of the entire scoring situation. Figure 4.2 shows serialisation, span boundaries and pooling anchors together.

Distinction from the Answer-Span Representation

The implementation additionally extracts a representation of the answer span using the same procedure as (4.2). For the variants investigated in this thesis, however, it is not used as the context: all reported fusion mechanisms (Section 4.2) combine the rubric representation with the *sequence* representation z, not with the answer representation. The alternative – aligning the answer span with the rubric spans – corresponds to the approach of Gombert et al. [23] and serves as a conceptual reference point in this thesis.

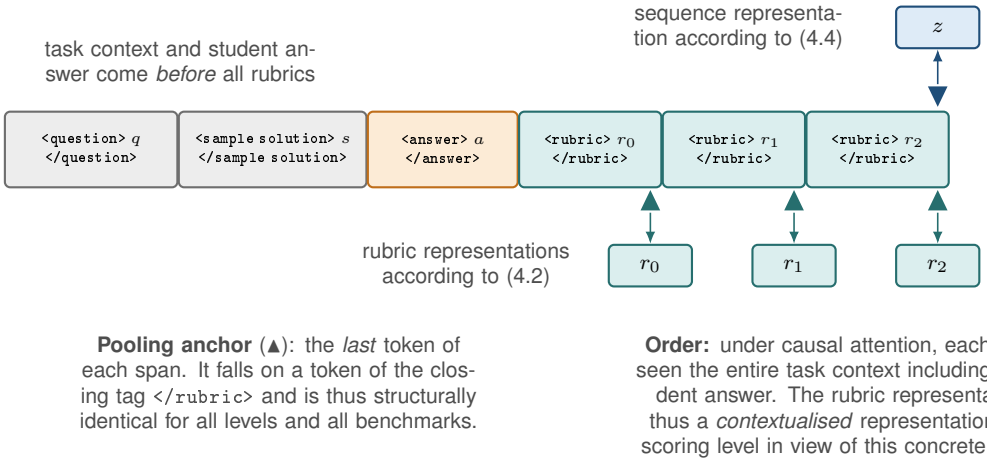


Figure 4.2: Serialisation of an instance with three scoring levels. Pooling takes place at the last token of each rubric span (r_k) and of the entire sequence (z); z has thus also seen all rubric texts.

Handling a Variable Number of Levels

Since the number of scoring levels K_i varies between instances (cf. Chapter 3), the instances of a batch are padded to the maximum number of levels occurring, $K_{\max}$. A validity mask $m^{\text{rub}} \in \{0,1\}^{B\times K_{\max}}$ marks the real levels. The scores of the padding positions are set to $-\infty$ before normalisation and thus contribute neither to the distribution nor to the gradient.

4.2 Fusion Mechanisms

The fusion function g from (2.18) combines the sequence representation $z \in \mathbb{R}^d$ with a rubric representation $r_k \in \mathbb{R}^d$ into a vector v_k, which a shared linear scoring head, identical for all levels, maps to a scalar:

$$s_k = w^\top \operatorname{Dropout}(v_k) + b, \qquad w \in \mathbb{R}^{d_g},\ b \in \mathbb{R}, \tag{4.5}$$

where d_g is the dimension produced by the respective fusion function and the dropout rate is 0.1. Using the *same* vector w for all levels is the architectural condition that allows the number of levels to vary freely: the head does not learn a level-specific pattern but a general assessment of the match between scoring situation and criterion description.

The four mechanisms compared differ exclusively in the construction of v_k and thus in the type of interaction modelled.

4.2.1 CONDIFF: Conditional Difference Fusion

CONDIFF concatenates both representations and adds their element-wise difference:

$$v_k^{\text{CONDIFF}} = \left[\, z \parallel r_k \parallel (z - r_k) \,\right] \in \mathbb{R}^{3d}, \qquad d_g = 3d. \tag{4.6}$$

The difference term is the actual subject of this variant. A linear head over the plain concatenation computes $w_z^\top z + w_r^\top r_k$, i.e. an additive combination of two independently weighted contributions. Such an expression can represent agreement between z and r_k only indirectly. The difference term, by contrast, provides the element-wise deviation explicitly: if the scoring situation is compatible with the rubric in those dimensions that describe the criterion, the corresponding components of $z - r_k$ are close to zero. The head can thus evaluate the match as proximity to the zero vector instead of having to reconstruct it from two separate terms. This construction transfers the heuristic matching known from sentence-pair modelling [41, 14, 11, 45] to the pair of sequence and rubric representation.

Notably, the difference term introduces no additional parameters in the fusion step itself – it is a parameter-free transformation. The additional cost arises exclusively in the scoring head, whose input dimension grows from $2d$ to $3d$, which means d additional parameters.

4.2.2 CONCAT: Plain Concatenation as a Baseline

CONCAT dispenses with the difference term and merely concatenates:

$$v_k^{\text{CONCAT}} = \left[\, z \parallel r_k \,\right] \in \mathbb{R}^{2d}, \qquad d_g = 2d. \tag{4.7}$$

CONCAT is designed as an *ablative* baseline: it is identical to CONDIFF in all components – same span extraction, same pooling, same head type, same loss function – and differs only by the absence of $(z - r_k)$. A performance difference between the two variants can therefore be attributed to the difference term and not to confounding architectural differences. Conversely, the absence of a difference indicates that the head can learn the comparison operation sufficiently well from the concatenation even without it being provided explicitly.

4.2.3 DIFF: The Difference as the Sole Input

DIFF is the counterpart of CONCAT: instead of the two representations, the head receives only their element-wise difference,

$$v_k^{\text{DIFF}} = z - r_k \in \mathbb{R}^d, \qquad d_g = d. \tag{4.8}$$

Together with CONCAT, DIFF decomposes the construction of CONDIFF into its two components: CONDIFF (4.6) contains exactly the components of CONCAT (4.7) and DIFF (4.8) at the same time. The three variants thus allow a complete attribution. An advantage of CONDIFF over CONCAT is attributable to the difference term; if DIFF alone reaches the same level as CONDIFF, the unchanged individual representations carry no additional usable information. Like CONCAT and CONDIFF, DIFF is parameter-free in the fusion step; with $d_g = d$ its head is even smaller than in both.

4.2.4 GATE: Gated Fusion

GATE replaces the concatenative combination with a multiplicative one. An element-wise gate is computed from the rubric representation that modulates the sequence representation; a residual connection and layer normalisation stabilise the result:

$$\gamma_k = \sigma\big(W_g\, r_k + b_g\big) \in (0,1)^d, \tag{4.9}$$

$$v_k^{\text{GATE}} = \mathrm{LayerNorm}\big(z + z \odot \gamma_k\big) \in \mathbb{R}^d, \qquad d_g = d, \tag{4.10}$$

with $W_g \in \mathbb{R}^{d\times d}$, σ the logistic function and $\odot$ the element-wise product.

The construction is to be read as follows. The gate γ_k depends *exclusively* on the rubric and acts as a criterion-specific feature filter: for each dimension of the sequence representation, the rubric decides how strongly this dimension enters the assessment of this level. Features that are irrelevant to the criterion under consideration can be suppressed. The residual term z in (4.10) ensures that the unfiltered sequence information is retained and the gate contributes only an additive amplification; since $\gamma_k \in (0,1)^d$, the expression $z + z \odot \gamma_k$ acts as an element-wise scaling of z by factors in $(1,2)$. Layer normalisation absorbs the resulting shift in scale.

Unlike CONDIFF, CONCAT and DIFF, GATE introduces additional parameters in the fusion step: the gate projection W_g with d^2 parameters and the normalisation parameters. For $d = 4096$ (Mistral-7B), W_g alone amounts to about $1.7 \cdot 10^7$ additional parameters that must be trained in full. Since the fusion modules are not part of the LoRA-adapted backbone but are trained from scratch (Section 4.6), GATE is the variant with the most parameters of the four and potentially more prone to overfitting – an aspect that must be taken into account when interpreting the results in Chapter 5.

Overview

Figure 4.3 contrasts three of the four mechanisms schematically – DIFF corresponds to the difference branch of CONDIFF and is not shown separately –, and Table 4.1 summarises the key figures of all four.

Table 4.1: The four fusion mechanisms compared. d denotes the model dimension, d_g the input dimension of the scoring head.

Variant	Construction of v_k	d_g	Additional parameters
CONCAT	$[z \,\|\, r_k]$	$2d$	—
DIFF	$z - r_k$	d	—
CONDIFF	$[z \,\|\, r_k \,\|\, (z - r_k)]$	$3d$	—
GATE	$\mathrm{LayerNorm}(z + z \odot \sigma(W_g r_k + b_g))$	d	$W_g \in \mathbb{R}^{d\times d}$, LayerNorm

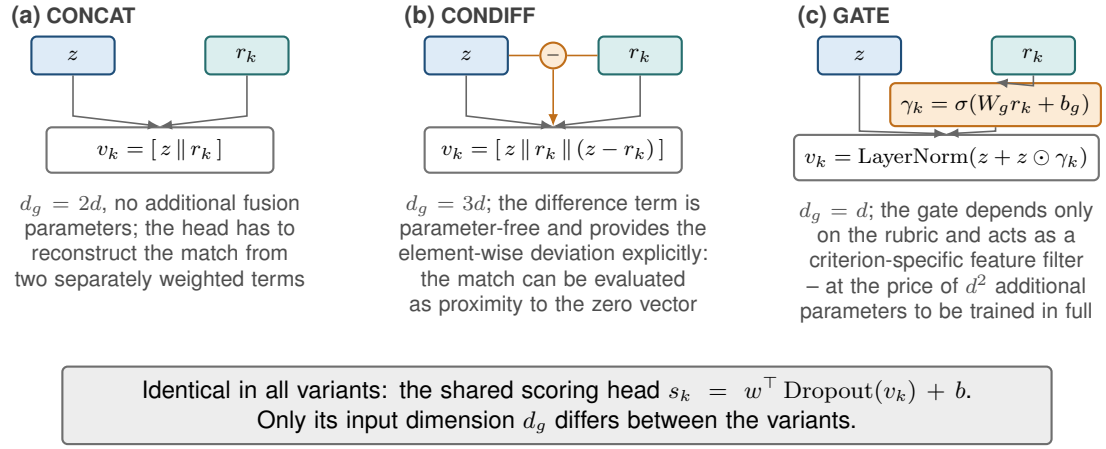


Figure 4.3: Construction of v_k in CONCAT **(a)**, CONDIFF **(b)** and GATE **(c)**. DIFF (4.8) corresponds to the orange branch in **(b)**.

Training Objective

Irrespective of the fusion mechanism, the scalars s_k are normalised over the valid candidate levels,

$$p_k = \frac{\exp(s_k)}{\sum_{k' \in \mathcal{Y}_i} \exp(s_{k'})}, \tag{4.11}$$

and the cross-entropy loss with respect to the reference level y_i is minimised over the training set:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \log p_{i,y_i}. \tag{4.12}$$

This formulation is *listwise*: it assesses all candidate levels of an instance jointly and in competition with one another, instead of treating each answer–rubric pairing in isolation as a binary decision. The gradient pushes the score of the correct level up and the scores of the other levels *of the same instance* down; the model thus learns a relative ranking among the criteria of a concrete task. Formally, (4.12) is identical to the cross-entropy loss of a classifier, but over an instance-dependent candidate set instead of a fixed label space. The prediction is obtained as the argmax over the candidate levels according to (2.5).

4.3 Bidirectional Attention in Decoder-Only Models

4.3.1 Motivation

The observation developed in Section 2.6.2 is applied here to the concrete architecture. Under the causal mask, the representations from (4.2) and (4.4) are subject to an asymmetry that results from the serialisation order:

- The tokens of the **student answer** precede all rubrics. They can therefore see *no* rubric. The answer is thus encoded without the model knowing against which criteria it is to be checked.
- The representation r_k of the k**-th rubric** sees the context, the answer and all *preceding* rubrics $r_0, \ldots, r_{k-1}$, but not the subsequent ones. The representations of the levels are thus conditioned asymmetrically on one another: the first level is encoded without knowledge of the alternatives, the last with complete knowledge.

Neither asymmetry has a substantive motivation. For the scoring task, the temporal order of the fields is an artefact of serialisation, not a substantive feature: a human rater reads criteria and answer in any order and switches between the two. Removing causality establishes exactly this mutual visibility – the answer tokens can take the criteria into account, and all levels are encoded under the same visibility conditions.

This is countered by the objection discussed in Section 2.6.2: the backbone was pre-trained exclusively under a causal mask. A complete removal represents a deviation from the pre-training distribution and can disrupt established mechanisms such as the attention sink [60, 35]. Instead of deciding between the two extremes, this thesis treats the degree of removal as a continuously controllable parameter.

4.3.2 Implementation: Partial Removal of the Causal Mask

Layer-Wise Choice of Mask

Let L be the number of transformer layers of the backbone and $n_{\text{bidir}} \in \{0, \ldots, L\}$ the number of layers computing bidirectionally. The bidirectional layers are always chosen as the *top* contiguous group; in contrast to (2.28), the layer indices are counted from 0 here, in line with the implementation:

$$\mathcal{B} = \{\, \ell \;:\; L - n_{\text{bidir}} \leq \ell < L \,\}, \tag{4.13}$$

and the mask used in layer ℓ is

$$M^{(\ell)} = \begin{cases} M^{\text{bidir}}, & \ell \in \mathcal{B}, \\ M^{\text{causal}}, & \text{otherwise}, \end{cases} \tag{4.14}$$

with M^{bidir} and M^{causal} according to (2.26) and (2.27). The choice of the top layers follows the consideration that lower layers predominantly extract local lexical and syntactic features and should therefore remain closer to the pre-training conditions, whereas task-specific aggregation takes place in the upper layers, where it can benefit from complete visibility.

Depth of the Intervention

The intervention is limited to the additive mask term in (2.25). In particular, the following remain unchanged:

- all weights of the backbone, including the projections W_Q, W_K, W_V, W_O,

- the positional encoding: RoPE is applied unchanged and the position indices remain the original ones. This is possible because RoPE encodes only relative positions and contains no directional assumption (Section 2.6.1),
- the computation rule of scaled dot-product attention itself.

Concretely, only the mask matrix is exchanged for the affected layers: instead of the lower triangular structure, a pure padding mask is passed that makes all valid positions mutually visible. In addition, the causality flag of the affected attention modules is deactivated so that efficient attention implementations do not add a causal mask of their own. The intervention is thus minimal and allows a clean attribution of observed effects.

Parametrisation via a Proportion

Since the three backbones have different numbers of layers (Section 4.4), n_{bidir} is specified not in absolute terms but via a proportion ρ. The conversion follows

$$n_{\text{bidir}}(\rho) = \begin{cases} 0, & \rho = 0, \\ \max\big(1,\ \lfloor \rho \cdot L \rfloor\big), & 0 < \rho < 1, \\ \rho, & \rho \geq 1 \text{ and } \rho \in \mathbb{N}, \end{cases} \tag{4.15}$$

i.e. values in the open interval $(0, 1)$ are interpreted as a proportion of the number of layers and rounded down, while values of 1 and above are read as an absolute number of layers. Table 4.2 gives the resulting numbers of layers for the proportions and backbones investigated; Figure 2.8 illustrates the split using Llama-3.2-1B as an example.

Table 4.2: Number of bidirectional layers n_{bidir} resulting from the proportion ρ per backbone according to rule (4.15). The rows are ordered by the actual number of bidirectional layers, not by the nominal value of ρ. The fully bidirectional condition $\rho = L$ is available for Llama-3.2-1B and -3B, but not for Mistral-7B; for both Llama sizes it is available with three random seeds.

ρ	**Llama-3.2-1B** ($L = 16$)	**Llama-3.2-3B** ($L = 28$)	**Mistral-7B** ($L = 32$)
1.0	1	1	1
0.2	3	5	6
0.4	6	11	12
0.6	9	16	19
0.8	12	22	25
$\rho = L$	16	28	32

The edge cases deserve explicit mention. For $\rho = 0$, the model is identical to the unchanged decoder; this condition corresponds to the runs without a bidir label. The lower bound $\max(1, \cdot)$ in (4.15) ensures that a non-zero proportion always covers at least one layer. According to (4.15), the value $\rho = 1$ already falls into the branch of the absolute specification and therefore denotes *one* layer, not the fully bidirectional model; for the latter, $\rho = L$ must be set.

Relation to the Fusion Mechanism

The variation of ρ is orthogonal to the choice of fusion mechanism. To be able to assess both effects separately, the fusion mechanism is fixed to CONDIFF in all runs with varied ρ. Differences between these runs can thus be attributed exclusively to the attention mask.

4.4 Backbone Models

The architecture is backbone-agnostic: it only requires that the model provides a hidden-state matrix $H \in \mathbb{R}^{T \times d}$ and that token spans can be located in this matrix. Three decoder-only models of different sizes and two encoder models as baselines are investigated.

4.4.1 Llama-3.2-1B/3B

The two smaller backbones come from the Llama 3.2 family. They follow the architecture described for Llama 3 [25]; their model configuration shows the same characteristics, including eight key-value heads and the RoPE base $500\,000$. They are decoder-only transformers with a pre-norm structure, RMS normalisation [62], SwiGLU feed-forward layers [49], grouped-query attention and RoPE as positional encoding [53]. The 1B variant has 16 layers with a model dimension of $d = 2048$, the 3B variant 28 layers with $d = 3072$.

The size difference between the two variants is of particular interest for this thesis: it makes it possible to examine whether the effectiveness of the fusion mechanisms and of bidirectional attention depends on the capacity of the backbone. In particular, a fusion mechanism with many parameters such as GATE can be expected to carry relatively more weight in smaller backbones, since the ratio between newly trained fusion parameters and adapted backbone parameters is more favourable.

4.4.2 Mistral-7B

The largest backbone is Mistral-7B-v0.1 [29] with 32 layers and a model dimension of $d = 4096$. The model likewise uses pre-norm, RMS normalisation, SwiGLU and RoPE, supplemented by grouped-query attention and sliding-window attention for efficiency on long sequences.

It is integrated via the same extension as Llama: the layer-wise choice of mask according to (4.14) is implemented identically for both model families, so that the bidir conditions are comparable across all three decoder backbones.

4.4.3 BERT/RoBERTa as Encoder Baselines

BERT-base-uncased [15] and RoBERTa-base [37] serve as baselines. Both are encoder-only models with 12 layers and $d = 768$ that already compute fully bidirectionally during pre-training.

They thus make it possible to answer an obvious question: if bidirectional visibility is advantageous for the task, how does a model perform that possesses this property from the outset – albeit with lower capacity and without the world knowledge of large language models?

The span-alignment architecture is adopted unchanged for the encoder baselines; only the serialisation is adapted. Since the encoder tokenizers have not been pre-trained with XML-like tags as structural separators, the fields are instead separated by the model's own `[SEP]` token, and the rubric span is defined so that it includes the following `[SEP]` token. The pooling anchor according to (4.2) thus falls on this separator, which takes over the structural role of the closing tag `</rubric>`. The fusion mechanisms, the scoring head and the loss function remain identical, so that the comparison between encoder and decoder backbones keeps the architecture constant.

4.5 Variants for the Ablation Experiments

In addition to the four fusion mechanisms and the variation of the attention mask, five further configurations are evaluated in Chapter 5. Four of them do not serve to improve the system but to attribute observed performance to individual components; the fifth tests the transferability of a trained model between benchmarks. This section specifies them.

4.5.1 Fixed Classification Head as a Baseline

The most basic baseline condition dispenses with span alignment entirely. The input is serialised and encoded unchanged, and the sequence representation z is also formed according to (4.4); however, the rubric representations r_k are not included in the scoring. Instead, a single linear map

$$\big(s_0, \ldots, s_{C-1}\big)^\top = W_{\text{cls}}\,\mathrm{Dropout}(z) + b_{\text{cls}}, \qquad W_{\text{cls}} \in \mathbb{R}^{C \times d}, \tag{4.16}$$

maps the sequence representation directly to C logits, and the ordinary cross-entropy loss over the fixed label space is optimised. This variant thus corresponds exactly to the classical formulation (2.4) from Section 2.1.

Two properties are essential for the evaluation. First, the model still *sees* the rubric texts – they are part of the serialised input and enter z via causal attention; they are merely not read out as spans. A comparison of this variant with span alignment therefore measures the contribution of explicit span extraction, not that of the rubric information (Section 5.3.1). Second, (4.16) presupposes a number of levels C that is constant across all instances. The variant is therefore not applicable to ASAP-SAS, whose number of levels varies between three and four depending on the task (Chapter 3) – direct practical evidence for the justification of the rubric-conditioned formulation given in Section 2.1.

4.5.2 Ablation of the Rubric Texts

For the ablation, all rubric texts are removed from the instance before serialisation. The input then consists only of those fields that remain according to the task-context configuration. Since

the rubric spans thus disappear, this condition can only be combined with the fixed classification head from Section 4.5.1; the implementation enforces this. All other components – backbone, LoRA configuration, pooling, head, loss function and all hyperparameters – remain unchanged, so that an observed difference can be attributed solely to the presence of the rubric texts.

4.5.3 Variation of the Task Context

As documented in Section 4.1.1, the main runs of this thesis contain neither question nor reference answer. To separate the contribution of the rubric texts from that of the task description, this restriction is lifted in a dedicated experiment: question and reference answer are prepended to the serialisation, and everything else remains unchanged. In combination with the ablation from Section 4.5.2, this yields a full 2×2 design over the availability of both information sources, which is evaluated in Section 5.3.2.

4.5.4 Answer Span as the Anchor of the Fusion

The mechanisms specified in Section 4.2 consistently use the sequence representation z as the anchor. To test this design decision, a variant is investigated that uses the representation of the answer span instead. It is formed using the same procedure as the rubric representations, i.e. as the state of the last token of the answer span,

$$t = H\big[e_{\text{ans}} - 1\big] \in \mathbb{R}^d, \tag{4.17}$$

and replaces z in the fusion:

$$v_k^{\text{t-concat}} = \big[\, t \parallel r_k \,\big] \in \mathbb{R}^{2d}. \tag{4.18}$$

The variant is thus the exact counterpart of CONCAT (4.7) with the anchor exchanged; any difference can be attributed solely to this exchange. In terms of content, the two quantities differ in the context they have seen: under the causal mask, t precedes all rubric texts and does not know them, whereas z is at the end of the sequence and has seen all levels. This variant also corresponds to the approach of Gombert et al. [23], who anchor the alignment at the answer span; it is evaluated in Section 5.4.2.

4.5.5 Transfer Between Benchmarks

In all configurations described so far, training and testing take place on the same benchmark. The transfer variant separates the two: training and model selection take place on ASAP-SAS only, and testing is carried out without further training on all test splits of Beetle and SciEntsBank. Architecture, fusion (CONDIFF), mask (causal) and all hyperparameters correspond to the main runs. At test time, each instance contains the rubric texts of the target benchmark; thanks to the variable candidate set (Section 4.1.2), the model scores it against their levels without adapting the scoring head. This variant cannot be carried out with the fixed classification head from Section 4.5.1, since training on ASAP-SAS already rules out a constant number of levels; the implementation rejects the combination. The evaluation follows in Section 5.3.4.

4.6 Training Configuration

Parameter-Efficient Fine-Tuning

The backbones are not fine-tuned in full but adapted using LoRA [28] (cf. Section 2.7). The configuration uses rank $r = 64$ and scaling factor $\alpha = 64$, which according to (2.29) yields an effective scaling factor of $\alpha/r = 1$. The LoRA dropout rate is 0.1; bias terms are not adapted. Adapters are applied to *all* linear modules of the transformer blocks, i.e. for the Llama and Mistral backbones to the four attention projections W_Q, W_K, W_V, W_O and to the three projections of the SwiGLU feed-forward layer.

The architecture-specific components – the gate projection and layer normalisation of the GATE variant and the scoring head (4.5) – are excluded from the LoRA adaptation. They are not part of the pre-trained backbone, are randomly initialised and are trained in full.

Optimisation

Table 4.3 summarises the training configuration. Optimisation uses AdamW; for the decoder backbones, a memory-efficient paged variant in 32-bit precision is used, for the encoder baselines the standard implementation. The learning rate follows a cosine schedule with a short warm-up phase. The decoder backbones compute in `bfloat16` during training and inference and additionally use gradient checkpointing to reduce the memory required for activations; the encoder baselines compute in 32-bit floating-point precision.

Deviating Configuration of the Encoder Baselines

The encoder backbones are not trained in the same configuration as the decoder models. Since BERT and RoBERTa have a maximum context length of 512 tokens, the limit of $2\,048$ tokens used for the decoders is not applicable to them; inputs are therefore truncated to 512 tokens. Furthermore, the encoders are trained for ten instead of five epochs and undergo full fine-tuning without LoRA, since their number of parameters readily allows this. They compute in 32-bit floating-point precision, with a batch size of 16 and without gradient accumulation.

For the encoder main runs, however, this configuration is not documented by files: they stem from an early training wave of which only the predictions have been preserved; in particular, their learning rate cannot be verified. Since BERT reaches only chance-level values on all three SciEntsBank splits (Table 5.9), a learning-rate variation with fully documented configuration was additionally trained for both encoders. Their configuration files show ten epochs, full fine-tuning without LoRA, 32-bit floating-point precision and a batch size of 16 without gradient accumulation, and a batch size of 32 for one BERT run. That the main runs were configured in the same way is plausible but not documented. Table 4.4 reports the results.

The variation serves solely to safeguard the encoder baselines and does not constitute a separate axis of investigation. Two results are decisive for the evaluation. First, BERT remains at chance level on SciEntsBank at every learning rate tested (-0.026 to 0.009 QWK). The failure is

Table 4.3: Training configuration of the decoder backbones. The values correspond to the configuration used in the evaluated decoder runs; the deviations of the encoder baselines are described in the following subsection.

Parameter	Value
LoRA rank r / scaling α	64 / 64
LoRA dropout	0.1
LoRA target modules	all linear modules
Batch size (per device)	4
Gradient accumulation	4
Effective batch size	16
Maximum number of epochs	5
Learning rate	$5 \cdot 10^{-5}$
Learning rate schedule	cosine
Warm-up ratio	0.01
Weight decay	0.01
Adam-ε	10^{-8}
Gradient clipping (norm)	1.0
Dropout (fusion / scoring head)	0.1
Numerical format	`bfloat16`
Maximum sequence length	2048 tokens
Task context (question, reference answer)	not included
Final instruction	not included
Random seeds	$114\,514$ (reference), 1, 2

thus not merely a learning-rate problem; its cause cannot be determined with the available data, and the BERT row for SciEntsBank in Table 5.9 is excluded from any substantive interpretation. Second, the RoBERTa performance of the main runs is not substantially underestimated: on SciEntsBank UA, the best rate reaches 0.634, practically the value of the main run (0.633), and on Beetle UQ RoBERTa remains between 0.288 and 0.362 QWK at all three rates. The drop on the transfer split reported in Section 5.3.3 is consequently not tied to a particular learning rate.

At the same time, the variation shows that the encoders are sensitive to the learning rate. RoBERTa fails on SciEntsBank at $2 \cdot 10^{-5}$ and $5 \cdot 10^{-5}$ (-0.054 and 0.049 QWK), and on Beetle UA the values vary between 0.665 and 0.768. Since each cell is based on a single run and this range lies below the largest seed range observed in Section 5.2, the influence of the learning rate and seed noise cannot be separated there. For the decoder backbones, the learning rate is addressed in Section 5.4.4, where it confounds the comparison of model sizes.

These differences must be taken into account when interpreting the encoder–decoder comparison: the comparison keeps the *architecture* constant – serialisation, span pooling, fusion mechanism, scoring head and loss function are identical –, but not all optimisation conditions.

Table 4.4: Learning-rate variation of the encoder baselines with CONDIFF fusion, QWK. Single runs, random seed $114\,514$. The BERT runs use batch size 16 at $1 \cdot 10^{-5}$ and $2 \cdot 10^{-5}$ and 32 at $3 \cdot 10^{-5}$, the RoBERTa runs 16 throughout. The last column reproduces the main run from Table 5.9, whose learning rate is not documented.

Model	Benchmark	Split	$1 \cdot 10^{-5}$	$2 \cdot 10^{-5}$	$3 \cdot 10^{-5}$	$5 \cdot 10^{-5}$	Main run
RoBERTa-base	Beetle	UA	0.665	**0.768**	—	0.688	0.694
		UQ	**0.362**	0.337	—	0.288	0.289
	SciEntsBank	UA	**0.634**	-0.054	—	0.049	0.633
BERT-base	SciEntsBank	UA	0.006	0.009	-0.026	—	0.052

Model Selection and Validation

Since the three benchmarks do not provide an official development split, $10\,\%$ of the training split is randomly held out as a validation set; the split uses a fixed random seed and is therefore identical across all runs. Evaluation on this set takes place after each epoch. Accuracy serves as the selection criterion; the checkpoint with the highest value is loaded at the end of training and used for test evaluation. In addition, an early-stopping criterion ends training if the validation performance does not improve for three consecutive epochs.

That model selection uses accuracy rather than the primarily reported metrics QWK or Macro-F_1 is a property of the configuration used. It must be taken into account when interpreting the results, since under imbalanced label distributions (cf. Chapter 3) accuracy can favour checkpoints that perform well on frequent levels and poorly on rare ones.

Evaluation Protocol

Test evaluation takes place on the splits described in Chapter 3. The metrics defined in Section 2.2 are reported: Quadratic Weighted Kappa with the quadratic weight matrix (2.12), Macro-F_1 (2.9) and accuracy (2.7). They are computed jointly over all test instances of a split. In addition, the metrics are computed and logged separately for each task, which allows a differentiated evaluation in particular for ASAP-SAS with its ten heterogeneous tasks (cf. Chapter 5).

Data collection took place in two stages. In the first stage, each configuration was trained once with the reference seed $114\,514$; these runs cover the entire space of the investigation and serve as an overview. In the second stage, those configurations on which the substantive statements of this thesis rest were additionally repeated with seeds 1 and 2. Across all benchmarks, backbones, variants and splits, this yields 93 cells with three runs each, excluding the transfer runs. One group was extended to eight seeds to check the noise threshold (Section 5.2).

This two-stage design carries through to the evaluation: single runs are reported in Chapter 5 for documentation only, whereas every substantive statement rests on means over three seeds and is checked against the noise threshold determined in Section 5.2. Where possible, the three runs of a group come from the same software environment, since a change of environ-

ment represents a separate source of disturbance that is independent of the random seed (Section 5.2).

5 Experiments and Results

This chapter reports the empirical results along the two research questions. Section 5.1 describes the experimental design, and Section 5.2 determines the noise threshold against which every reported difference is subsequently measured. Section 5.3 answers RQ1 and compares span alignment with a fixed classification head, with conditions without rubric text or task information, and with encoder models, and tests the transfer of a trained model to other benchmarks. Section 5.4 answers RQ2 and examines the design of the alignment, the attention mask and the backbone size. Section 5.5 safeguards the findings through a task-level evaluation, a diagnosis of unstable runs and a check under Macro-F_1, and Section 5.6 summarises them.

Unless stated otherwise, all figures are rounded half-up to three decimal places; differences are computed from the unrounded values. The raw values are contained in the evaluation files of the respective runs.

5.1 Experimental Design

Conditions Investigated

Five axes of variation are investigated. The first four concern the architecture, the fifth the backbone:

1. **Availability of the rubric text.** The contrast between a model to which the rubric texts are available as input and an otherwise identical model from which they are withheld (Section 5.3.2).
2. **Fusion mechanism** with unchanged causal attention: CONDIFF, CONCAT, DIFF and GATE (Section 4.2); DIFF was investigated on Llama-3.2-3B only.
3. **Anchor of the fusion**: the sequence representation z versus the representation of the answer span (Section 5.4.2).
4. **Attention mask** with the fusion mechanism fixed to CONDIFF: the proportions $\rho \in \{0.2;\ 0.4;\ 0.6;\ 0.8;\ 1.0\}$ and the fully bidirectional condition $n_{\text{bidir}} = L$ (Section 4.3.2).
5. **Backbone**: Llama-3.2-1B, Llama-3.2-3B and Mistral-7B as decoder models and BERT-base-uncased and RoBERTa-base as encoder baselines (Section 4.4).

The first axis, together with the comparison against the fixed classification head (Section 4.5.1), the encoder models and a transfer experiment in which training takes place on ASAP-SAS only and testing on Beetle and SciEntsBank (Section 4.5.5), serves to answer RQ1; axes two to four and the comparison of the decoder backbones serve to answer RQ2.

Since the runs with a varied mask consistently use CONDIFF, the run with CONDIFF and unchanged causal mask is at the same time the reference condition $\rho = 0$ for the fourth axis. Fusion and mask effects can thus be separated from each other.

Pairwise methods in the sense of rubric retrieval (Section 2.5) and generative scoring via prompting (Section 2.4) were not trained as separate conditions. The research question concerns variants within the span-alignment architecture; the fixed classification head (Section 4.5.1) and the encoder models serve as baselines outside this architecture.

Interpretation of the Proportion Parameter

For the fourth axis, the conversion rule (4.15) described in Section 4.3.2 is of central importance. Since values of 1 and above are read as an absolute number of layers, the condition $\rho = 1.0$ corresponds to *a single* bidirectional layer and not to the fully bidirectional model; the latter requires the specification $\rho = L$. Figure 2.8 and Table 4.2 in Chapter 4 give the resulting numbers of layers.

The conditions therefore do *not* form a monotonically increasing series in their nominal labels. Ordered by the actual degree of bidirectionality,

$$\rho = 0 \;\prec\; \rho = 1.0 \;\prec\; \rho = 0.2 \;\prec\; \rho = 0.4 \;\prec\; \rho = 0.6 \;\prec\; \rho = 0.8 \;\prec\; \rho = L, \tag{5.1}$$

i.e. the condition $\rho = 1.0$, which nominally appears “complete”, is closest to the causal starting model. All result tables in this chapter are therefore sorted by the actual number of layers n_{bidir} and state it explicitly.

Completeness of the Evaluation

The evaluation is not complete in all cells. For Mistral-7B, the fusion variants CONDIFF, CONCAT and GATE are available on Beetle and SciEntsBank with three random seeds, whereas the variation of the attention mask is available only with the proportions $\rho = 0.2$, $\rho = 0.6$ and $\rho = 1.0$ as single runs; the proportions $\rho = 0.4$ and $\rho = 0.8$ and the fully bidirectional condition were not run for this backbone on these two benchmarks. On ASAP-SAS, all mask conditions including the fully bidirectional one are available for Mistral-7B as single runs, as is one transfer run (Section 5.3.4). For BERT, only the UA split is available on Beetle. The fully bidirectional condition and its causal CONDIFF reference are available for both Llama sizes on all three benchmarks with three random seeds. CONDIFF with Llama-3.2-1B on Beetle was additionally repeated with five further seeds in order to check the reliability of the noise threshold (Section 5.2). The variation of the anchor and the fusion variant DIFF were investigated with three seeds only on Llama-3.2-3B, in each case on Beetle and SciEntsBank; the ablation of the rubric text was repeated on both Llama sizes. The learning-rate variation of the encoders covers RoBERTa on

Beetle and SciEntsBank and BERT on SciEntsBank. Missing cells are marked with "—" in the tables.

The single runs of the anchor variant on Llama-3.2-1B are not listed in separate tables, since with one random seed they support no interpretation. The same applies to t-concat on ASAP-SAS (0.814 and 0.837 QWK for Llama-3.2-1B and -3B compared with 0.828 and 0.842 for CONCAT).

Where several random seeds are available, the mean over the seeds is reported together with the range. Where only one run is available, this is noted in the table caption; such values serve documentation, not interpretation.

5.2 Noise Threshold: Variation Between Runs

Before individual configurations are compared, the noise threshold introduced in Section 2.2 must be determined empirically. Otherwise it would not be possible to decide whether an observed difference is due to the variation under investigation or to the stochastic components of training.

Reproducibility with Identical Random Seed

A first finding concerns repeatability itself, and it turns out more clearly than expected. Two configurations were run a second time with an identical random seed, identical hyperparameters and an identical software environment. The resulting prediction files are *byte-identical* to those of the first run; all four measured values (two configurations $\times$ two splits) match exactly. Within a fixed environment, training in this set-up is thus fully deterministic: the non-associative floating-point addition on the graphics processor, named as a possible source of disturbance in Section 2.2, has no effect here.

This finding corrects a tempting misinterpretation. The data contain a further pair of runs with the same random seed that does *not* match: for CONDIFF with Llama-3.2-1B on Beetle, the repetition yielded 0.718 instead of 0.696 QWK on UA and 0.472 instead of 0.532 on UQ. This pair, however, stems from two different software environments and thus measures not repeatability but the influence of the environment, which is treated in the next subsection.

The practical consequence is favourable: the range over several random seeds determined below is a pure seed effect and is not additionally overlaid by repetition noise.

Size of the Noise Threshold

To determine the variation, configurations were repeated with three random seeds each. Across all benchmarks, backbones, variants and splits, this yields 93 cells from 39 groups with three runs. Not included are the transfer runs (Section 5.3.4), whose predictions are almost constant and whose range therefore says nothing about the variation of a functioning training run. Table 5.1 summarises the distribution of the ranges (2.14).

Table 5.1: Range Δ of QWK over three random seeds, aggregated over 93 cells.

Statistic	Range Δ (QWK)
Median	0.036
Mean	0.037
1st quartile	0.020
3rd quartile	0.049
Minimum	0.004
Maximum	0.118
Share of cells with $\Delta \leq 0.05$	$76\,\%$

The typical range is about 0.036 QWK, and just over three quarters of all cells lie at 0.05 or below. The maximum of 0.118 is attained by GATE with Llama-3.2-1B on the Beetle UQ split. Remarkable is the breadth of the distribution itself: it ranges from 0.004 to 0.118, so a blanket threshold is insufficient. A two-stage check is therefore applied below – against the median of 0.036 as orientation and, where three seeds are available, against the actual range of the cells concerned.

Influence of the Software Environment

Runs from different software environments, by contrast, are not directly comparable; the CONDIFF pair mentioned in the previous subsection deviates by up to 0.060 QWK with an identical seed. Not all groups with three random seeds could be formed within a single environment: in 21 of the 39 groups, the run with seed $114\,514$ comes from a different environment than the two other runs. In these groups, the contribution of the environment is contained in the reported range. For comparisons without a demonstrable difference this is conservative, since a larger range makes the evaluation rule stricter. For the comparisons with a finding, it was additionally checked whether the direction also holds within a single environment; this is the case throughout. Since only one run of a condition enters each of these checks, they confirm the direction, not the size, of the effects.

Evaluation Rule

The preceding findings yield the rule applied throughout this chapter. A difference between two configurations is only called an effect if the difference of the means exceeds the larger of the two ranges. If it lies below that but the ranges do not overlap, this is noted as a weak indication and not claimed as a result. If the ranges overlap, the difference is considered not demonstrable. Figure 2.4 depicts this rule schematically.

Reliability of Three Seeds

Three runs allow only a rough estimate of the variation, and the range of a small sample systematically underestimates the spread of the underlying distribution. To estimate by how much, one group – CONDIFF with Llama-3.2-1B on Beetle – was extended by five further seeds (3 to 7) to eight runs in total. The additional runs stem from the same environment as the runs with seeds 1 and 2. Table 5.2 contrasts the variation over all eight runs with the variation obtained for the 56 possible groups of three formed from these eight runs.

Table 5.2: Beetle, Llama-3.2-1B, CONDIFF: variation of QWK over eight random seeds and over the 56 groups of three that can be formed from them. *Original* denotes the group of seeds $114\,514$, 1 and 2 on which the other tables of this chapter are based.

	eight runs			**range over three runs**			
Split	Mean	Range	Std. dev.	original	Median	Min.	Max.
UA	0.732	0.079	0.025	0.023	0.039	0.013	0.079
UQ	0.525	0.073	0.024	0.073	0.034	0.003	0.073

The extension leads to two findings. First, it confirms the level of the noise threshold as a typical value: the range of the groups of three has a median of 0.039 (UA) and 0.034 QWK (UQ) and thus lies close to the global median of 0.036. Second, it shows how far a single group of three can deviate from it. On UA, the range of the groups of three extends from 0.013 to 0.079. The original group lay in the lower part with 0.023, while the range over all eight runs is more than three times as large. The run with seed 6 alone reaches 0.781 QWK and thus lies 0.040 above the best of the three original runs. On UQ, by contrast, the original group was already the widest possible.

A narrow range over three seeds is therefore no reliable evidence of narrow variation. Findings that pass the evaluation rule only because the cells concerned vary unusually little must be read with corresponding caution. For the pairwise comparisons involving this group, the evaluation rule was therefore repeated with all eight runs (Table 5.3).

Under QWK, one verdict changes: the gain of the smaller backbone from fully bidirectional attention on Beetle UA, which narrowly passed the rule with three seeds, is no longer demonstrable with eight seeds (Section 5.4.3). The advantage of span alignment over the fixed head on Beetle UA, by contrast, persists and also exceeds the range over eight runs (Section 5.3.1). Under Macro-F_1, two further verdicts disappear with eight seeds, the gain from bidirectional attention on UA and a weak indication in favour of CONDIFF over CONCAT on UA; the remaining Macro-F_1 verdicts of these comparisons are unchanged. The result tables continue to report the three original seeds for this group, so that all groups rest on the same number of runs; where the verdict differs with eight seeds, this is noted at the relevant place.

Table 5.3: Pairwise comparisons involving CONDIFF, Llama-3.2-1B, Beetle: verdict with the three original and with all eight seeds of the CONDIFF condition, QWK. The other condition is based on three seeds in each case. Δ as in the result tables of the respective sections; "—" means not demonstrable.

		three seeds		eight seeds	
Comparison	**Split**	Δ	Verdict	Δ	Verdict
Span alignment vs. fixed head	UA	$+0.088$	effect	$+0.092$	effect
	UQ	$+0.054$	—	$+0.069$	—
bidirectional vs. causal	UA	$+0.036$	effect	$+0.033$	—
	UQ	-0.016	—	-0.032	—
CONDIFF vs. CONCAT	UA	$+0.005$	—	$+0.008$	—
	UQ	$+0.006$	—	$+0.022$	—
CONDIFF vs. GATE	UA	$+0.033$	—	$+0.036$	—
	UQ	$+0.038$	—	$+0.054$	—

5.3 RQ1: Span Alignment Versus the Baselines

The first research question asks whether span alignment outperforms the baselines. This is examined against three baselines: a classifier with a fixed output head that receives the same input (Section 5.3.1), models from which the rubric texts or any task-related information are withheld (Section 5.3.2), and the encoder models BERT and RoBERTa (Section 5.3.3). Finally, Section 5.3.4 tests a property that belongs to span alignment alone: applying a trained model to a different benchmark with different rubrics.

5.3.1 Span Alignment Versus a Fixed Classification Head

The most obvious baseline is a model that sees the rubric texts in its input but does not extract them as spans and instead derives the score from the sequence representation via a fixed head with C outputs (Section 4.5.1). The difference from span alignment thus lies solely in the scoring mechanism. Like all main runs, both conditions contain no task context; the fixed head corresponds to the condition "without context, with rubric text" from Section 5.3.2.

Table 5.4 shows a picture that differs by benchmark and backbone. The clearest finding concerns Beetle: with Llama-3.2-1B, span alignment is ahead of the fixed head by 0.088 QWK on UA. The ranges of the two conditions are disjoint, and the difference exceeds not only both ranges but also, clearly, the global median of 0.036 QWK. Both groups also stem from the same software environment (Section 5.2). The finding also persists when the span condition is extended to eight seeds: the difference is then $+0.092$ and also exceeds the range over all eight runs (0.079). On Beetle UQ, span alignment is nominally ahead by 0.054 QWK; since the ranges overlap, the difference is not demonstrable under QWK. Under Macro-F_1, by contrast, both Beetle advantages hold as effects (Section 5.5.3).

Table 5.4: Span alignment (CONDIFF) versus fixed classification head, both with rubric texts and without task context. Means over three random seeds, ranges in parentheses. *Above range* means that the difference exceeds the larger of the two cell-specific ranges. Bold: higher value per row.

Benchmark	Backbone	Split	Span	Fixed head	Δ	Verdict
Beetle	Llama-3.2-1B	UA	**0.729** (0.023)	0.641 (0.055)	$+0.088$	above range
		UQ	**0.509** (0.073)	0.456 (0.068)	$+0.054$	within noise
	Llama-3.2-3B	UA	**0.730** (0.041)	0.722 (0.081)	$+0.008$	within noise
		UQ	**0.544** (0.044)	0.517 (0.089)	$+0.027$	within noise
SciEntsBank	Llama-3.2-1B	UA	**0.702** (0.007)	0.682 (0.007)	$+0.021$	above range
		UQ	0.544 (0.058)	**0.555** (0.051)	-0.011	within noise
		UD	**0.560** (0.026)	0.534 (0.016)	$+0.026$	above range (narrowly)
	Llama-3.2-3B	UA	0.734 (0.047)	**0.746** (0.028)	-0.011	within noise
		UQ	0.576 (0.009)	**0.580** (0.076)	-0.004	within noise
		UD	0.617 (0.028)	**0.641** (0.026)	-0.024	within noise

On SciEntsBank, the differences are smaller. With the smaller backbone, span alignment is ahead on UA and UD; the advantages amount to 0.021 and 0.026 QWK. Both lie *below* the global median of the seed range and are demonstrable only because the cells concerned vary unusually little (0.007 on UA). On UD, the decision is moreover extremely narrow: the difference exceeds the relevant range by only 0.0003 QWK. With the larger backbone, all three differences lie within the noise, and all three are nominally negative.

With the larger backbone, the advantage also disappears on Beetle. Span alignment is nominally ahead there by 0.008 (UA) and 0.027 QWK (UQ); both differences lie within the noise. Notable is the high variation of the fixed head, with ranges of 0.081 and 0.089, more than twice the global median. A difference of the size found for the smaller backbone would nevertheless have been demonstrable. Under Macro-F_1, the verdict is the same.

The contrast between the two model sizes must be read with caution. Both Llama variants were trained at $5 \cdot 10^{-5}$ instead of their intended rate of $2 \cdot 10^{-4}$ (Section 5.4.4); whether they are affected to the same extent cannot be settled with one run each at the reference rate. The comparison *within* each model size is unaffected by this, the statement about the relation between the two sizes is not.

For the interpretation it is decisive that both conditions contain the rubric texts in the input. The comparison therefore does not measure the value of the rubric information – that is the subject of Section 5.3.2 and amounts to a multiple –, but merely the value of its explicit extraction as spans. This additional benefit is clear for the smaller backbone on Beetle, small on SciEntsBank and not demonstrable for the larger backbone on either benchmark. The latter suggests the reading that a sufficiently large language model already condenses the relevant rubric information into the sequence representation, so that the targeted read-out of the individual levels makes no further contribution. That the advantage disappears on both benchmarks at once supports this reading; with two model sizes and the learning-rate limitation mentioned above, it nevertheless cannot be confirmed.

The value of the span architecture thus lies, on the one hand, in its structural ability to handle

a number of levels that varies at run time (Section 4.1) – a property that the fixed head fundamentally lacks and that makes the joint modelling of all tasks possible in the first place on benchmarks such as ASAP-SAS with varying scale lengths. Whether this property also enables transfer between benchmarks is examined in Section 5.3.4. On the other hand, a clear accuracy advantage also emerges under certain conditions, namely on Beetle with the smaller backbone; an advantage that is reliable across benchmarks and backbones, however, is not demonstrated.

5.3.2 The Contribution of Task-Related Information

The second baseline concerns not the architecture but the input: what do the rubric texts contribute, and what happens when the model lacks any task-related information? To answer this, an ablation in the sense of Section 2.2 is carried out: one information source is deliberately removed from a working configuration, while serialisation, backbone, pooling, scoring head, loss function and all hyperparameters remain unchanged.

The architecture is the same in all conditions of this section, including those without rubric text: the fixed classification head from Section 4.5.1. The decoder backbone (Llama-3.2-1B or -3B with LoRA) encodes the serialised input, the sequence representation z is formed according to (4.4) as the state of the last valid token, and a single linear map projects z onto the logits of the fixed label space. Rubric representations r_k and fusion function are used in none of the conditions. The conditions differ solely in which fields remain in the input before serialisation (Sections 4.5.2 and 4.5.3); in the condition *with* rubric text, the rubric texts are part of the input and act exclusively through z. The comparison is thus not confounded by the fusion architecture.

Decisive for the design of the experiment is the specification documented in Section 4.1.1 that the main runs of this thesis contain *neither question nor reference answer*. The rubric texts are the only task-related information there; withdrawing them leaves only the answer text. An ablation under this condition therefore cannot distinguish between the specific contribution of the criterion description and the general contribution of *any* task information. To separate the two, the ablation is carried out here in a full 2×2 design that additionally varies the availability of question and reference answer.

The Full Design

Table 5.5 leads to a clearly structured finding that corrects the obvious interpretation of the simple ablation.

Task-related information is indispensable. If it is missing entirely, performance collapses: on UQ from 0.555 to 0.035 QWK, on UD from 0.534 to 0.172. The effect amounts to $+0.235$ / $+0.520$ / $+0.361$ and exceeds the noise threshold six- to fourteenfold; the ranges of the two conditions do not overlap in any of the three cells.

The rubric text, however, contributes nothing once question and reference answer are available. The difference between the two upper rows is -0.012 / $+0.015$ / $+0.006$ QWK. All

Table 5.5: SciEntsBank, Llama-3.2-1B, fixed classification head: availability of task context (question and reference answer) and rubric text in the full 2×2 design. Means over three random seeds, ranges in parentheses.

Context	Rubric text	UA	UQ	UD
yes	yes	0.769 (0.024)	**0.630** (0.026)	**0.617** (0.016)
yes	no	**0.782** (0.041)	0.615 (0.022)	0.611 (0.038)
no	yes	0.682 (0.007)	0.555 (0.051)	0.534 (0.016)
no	no	0.447 (0.055)	0.035 (0.023)	0.172 (0.038)

three values lie below the noise threshold, and in all three cases the ranges overlap. According to the rule formulated in Section 5.2, a contribution of the rubric text is not demonstrable under this condition. Under Macro-F_1 this holds only with qualifications: there, the rubric text contributes demonstrably on UD when context is present (Section 5.5.3).

Question and reference answer are even superior to the rubric text. Comparing the second with the third row – context without rubric versus rubric without context – yields $+0.100$ / $+0.060$ / $+0.077$ QWK in favour of the context. All three values exceed the noise threshold. Conversely, adding the context to an existing rubric yields $+0.088$ / $+0.075$ / $+0.083$ QWK, likewise consistently above the noise threshold. On UA, both findings also hold under Macro-F_1. On the transfer splits, by contrast, the advantage of the context rests mainly on gross misjudgements becoming rarer; under Macro-F_1 it is not demonstrable there or only as a weak indication (Section 5.5.3).

Finally, it is noteworthy that the best single configuration of this design – context without rubric text – reaches the highest SciEntsBank value of the entire thesis on the UA split, with 0.782 QWK on average and 0.799 in the best run, above all runs with Llama-3.2-3B and Mistral-7B. A 1.2-billion-parameter model with question, reference answer and answer is thus ahead of all larger backbones. These, however, were never trained with task context, so the comparison permits no statement about model size.

Consequence for the Interpretation

The effect of up to 0.52 QWK measured in the simple ablation is therefore *not* to be read as the contribution of the criterion description but as the contribution of task-related information as such. Rubric text on the one hand and question and reference answer on the other prove to be largely interchangeable carriers of the same information – with an advantage for the latter that shows on all splits under QWK but only on UA under Macro-F_1. This finding is consistent with the origin of the rubric texts: no original scoring guidelines exist for Beetle and SciEntsBank; the texts used were generated with the help of a language model from question and reference answer (Chapter 3) and visibly incorporate their content. For the SciEntsBank question *What is the difference between a rock and a mineral?*, for example, the description of the highest level states that correct answers accurately distinguish between rock and mineral and explain that a rock consists of several minerals, whereas a mineral consists of a single substance – which is exactly the content of the corresponding reference answer.

The premise of the rubric-conditioned formulation (2.5) is thus empirically qualified. According to these data, its benefit does not lie in criterion descriptions being an information source superior to the task description. It lies in the structural ability to handle a number of levels that varies at run time (Section 4.1) and to transfer to unseen rubric sets – properties that a fixed classification head fundamentally lacks and that cannot show up in an accuracy comparison on fixed label spaces.

Two Clarifications on the Collapse

The collapse without any task information is not a degenerate classifier. It would be conceivable that the model collapses onto predicting a single level and that the low QWK merely reflects this collapse. An inspection of the prediction files refutes this for all six runs without any task information, the three in the bottom table row and the three corresponding runs with Llama-3.2-3B (Table 5.6): each of them predicts all three levels, and the share of the most frequently predicted level never exceeds $54\,\%$.

On the transfer splits, performance then lies at or below chance level. Without any task information, the models reach an accuracy between 0.381 and 0.416 on the UQ split, compared with a majority baseline of 0.411; in five of the six runs it lies *below* this baseline. On UD, the same picture emerges with 0.353 to 0.440 compared with a baseline of 0.420. On UA, by contrast, accuracy remains clearly above the local baseline of 0.431, at 0.574 to 0.611. The contrast is revealing: without task information, the model can still partly reproduce the scoring convention of known questions – apparently via linguistic regularities of the answers –, whereas it loses any basis on new questions and domains. This provides empirical evidence for what was introduced as a methodological argument in Section 2.1: UA and UQ measure different abilities.

Model Size and the Ablation

The simple ablation without context was additionally carried out on Llama-3.2-3B (Table 5.6). The effect remains equally large there; on UD it is nominally larger, at 0.440 compared with 0.361 QWK. Whether this increase is more than noise cannot be decided: both model sizes were trained at the same learning rate, which is not the one intended for them (Section 5.4.4). Only the sign of the comparison *within* each size is certain: additional model capacity does not compensate for missing task information; without it, both model sizes are equally at the bottom.

An Exception on Beetle

On Beetle, the ablation without context turns out differently than on SciEntsBank (Table 5.7). This comparison, too, rests on three random seeds from the same environment each.

On UA, all three runs with rubric text lie above all three runs without; the difference of 0.054 QWK, however, remains just below the range of the condition with rubric and therefore counts as an indication, not as a confirmed effect, according to the evaluation rule. Under Macro-F_1, the same comparison passes the rule as an effect ($+0.040$). On UQ, the difference is exactly

Table 5.6: SciEntsBank: ablation of the rubric texts without task context, across two model sizes. Means over three random seeds, ranges in parentheses. *Gap* denotes the distance between the worst run with and the best run without rubric text.

Backbone	Split	With rubric text	Without	Effect	Gap
Llama-3.2-1B	UA	0.682 (0.007)	0.447 (0.055)	$+0.235$	0.206
	UQ	0.555 (0.051)	0.035 (0.023)	$+0.520$	0.485
	UD	0.534 (0.016)	0.172 (0.038)	$+0.361$	0.338
Llama-3.2-3B	UA	0.746 (0.028)	0.491 (0.025)	$+0.255$	0.236
	UQ	0.580 (0.076)	0.052 (0.063)	$+0.528$	0.454
	UD	0.641 (0.026)	0.201 (0.029)	$+0.440$	0.406

Table 5.7: Beetle, Llama-3.2-1B, fixed classification head, without task context: ablation of the rubric texts. Means over three random seeds, ranges in parentheses.

Split	With rubric text	Without	Effect	Verdict
UA	0.641 (0.055)	0.587 (0.011)	$+0.054$	ranges disjoint
UQ	0.456 (0.068)	0.457 (0.037)	-0.001	within noise

zero at -0.001 QWK, and both conditions lie clearly above the majority baseline of 0.420, with an accuracy of 0.581 to 0.611 and 0.586 to 0.602, respectively.

On Beetle, the model thus generalises to unseen questions even without any task information, whereas it fails completely on SciEntsBank. A plausible explanation lies in the domain structure: Beetle comprises 47 training questions from a single, narrowly bounded domain (electricity), whose nine UQ questions are very close in content to the training questions. SciEntsBank distributes 135 training questions over twelve topic blocks (Chapter 3). The narrower the domain, the more readily a scoring convention can be transferred from one question to the next on the basis of the answer text alone.

The Design on the Second Benchmark

To check whether the finding on the interchangeability of the two information sources is limited to SciEntsBank, the 2×2 design was repeated on Beetle. The two conditions with task context are available there with one run each and should therefore be assessed more cautiously than the SciEntsBank values.

Table 5.8 confirms the pattern from SciEntsBank and sharpens it in one respect. The task context is clearly effective here too: it yields $+0.095$ / $+0.111$ QWK when the rubric is present and $+0.211$ / $+0.157$ without it. Context alone outperforms rubric alone by $+0.131$ / $+0.158$, and thus even more clearly than on SciEntsBank.

The rubric text, by contrast, not only adds nothing when context is present but costs performance: -0.036 on UA and -0.047 on UQ. Both values are of the order of the noise threshold and rest on one run each, so they are not to be read as confirmed harm. It should be noted,

Table 5.8: Beetle, Llama-3.2-1B, fixed classification head: availability of task context and rubric text. All values at random seed $114\,514$ to keep the cells comparable with one another. For the two lower rows, three seeds are additionally available (means 0.641 / 0.456 and 0.587 / 0.457, respectively).

Context	Rubric text	UA	UQ
yes	yes	0.764	0.554
yes	no	**0.799**	**0.602**
no	yes	0.669	0.443
no	no	0.588	0.445

however, that on both benchmarks the sign speaks against an additional benefit of the rubric texts once the task itself is described.

Figure 5.1 shows both designs side by side. Under QWK, the pattern is the same on both benchmarks. The two conditions with task context cannot be reliably distinguished on any split; on Beetle they rest on single runs whose distances are of the order of the noise threshold. The condition with rubric text alone lies consistently below, and without any task-related information performance collapses on the transfer splits.

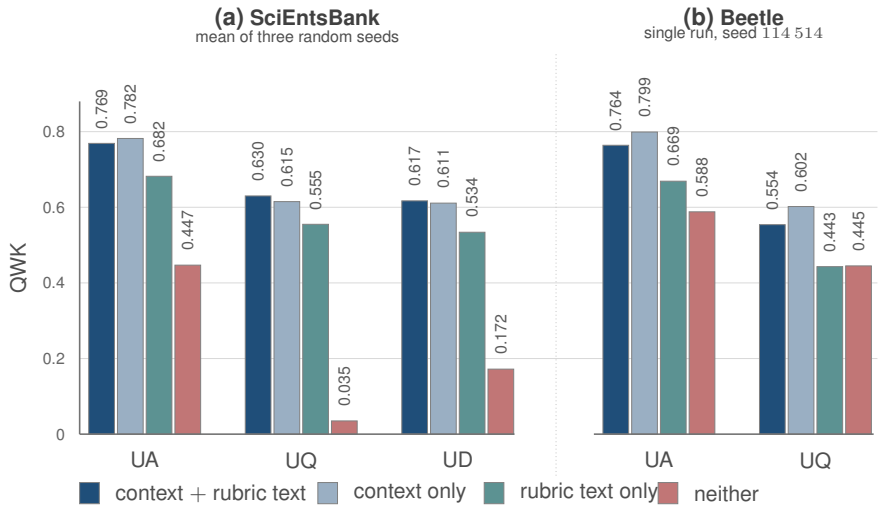


Figure 5.1: Availability of task context and rubric text, Llama-3.2-1B with a fixed classification head (values from Tables 5.5 and 5.8).

On Beetle too, as on SciEntsBank, the strongest of the small configurations investigated is thus the one *without* rubric texts. A Llama-3.2-1B with question, reference answer and answer reaches 0.799 QWK on UA and thus outperforms all runs with Llama-3.2-3B, which, however, were never trained with task context. On UQ, the same configuration reaches 0.602 QWK and is thus practically on a par with Mistral-7B, which achieves a mean of 0.605 over three seeds there with rubric texts and without context – with about six times as many parameters. Among the two small backbones, only the run with a deviating learning rate reported in Section 5.4.4 lies higher on UQ (0.605).

One limitation remains for both benchmarks: in the implementation used, question and reference answer are switched on or off together. Whether the observed context effect is due to the task description, the reference answer or both cannot be decided with the available runs and remains open (Chapter 7).

Assessment

This section delivers by far the largest effect of this thesis – and at the same time a qualification of its initial assumption. What matters is *that* task-related information is available to the model, not in which form. How this information is subsequently extracted and processed is the subject of RQ2 (Section 5.4).

5.3.3 Encoder- Versus LLM-Based Approaches

Table 5.9: Encoder baselines versus Llama-3.2-1B (CONDIFF, causal in each case), QWK. Encoder values are single runs from an early training wave of which only the predictions have been preserved; Llama values are means over three random seeds. ‡Run with chance-level result, excluded from interpretation (Section 4.6).

Benchmark	Split	BERT-base	RoBERTa-base	Llama-3.2-1B
ASAP-SAS	UA	0.788	0.821	0.827
Beetle	UA	0.478	0.694	0.729
	UQ	—	0.289	**0.509**
SciEntsBank	UA	0.052‡	0.633	**0.702**
	UQ	0.061‡	0.451	**0.544**
	UD	-0.004‡	0.472	**0.560**

The comparison in Table 5.9 once again shows the choice of evaluation split to be decisive.

On UA splits, encoders are competitive. On ASAP-SAS, RoBERTa nearly reaches the level of Llama-3.2-1B (0.827) with 0.821 QWK – with a fraction of the parameters and without LoRA overhead. On Beetle UA, RoBERTa is 0.035 behind the Llama mean with 0.694 and thus within the range of the noise threshold. On SciEntsBank UA this does not hold: there the gap amounts to 0.069 QWK and clearly exceeds the noise threshold. Had this thesis evaluated only the UA splits of ASAP-SAS and Beetle, it would have been tempting to conclude that decoder-only LLMs offer no added value for this task.

On transfer splits, the encoder advantage collapses. On Beetle UQ, RoBERTa drops to 0.289 QWK, while Llama-3.2-1B holds 0.509 – a gap of 0.221, i.e. a multiple of the noise threshold. The drop from UA to UQ amounts to 0.406 for RoBERTa but only 0.220 for Llama-3.2-1B. On SciEntsBank, Llama-3.2-1B is ahead on all three splits, by margins of 0.069 to 0.093.

The pattern is consistent with the motivation reviewed in Section 2.4: if a model can learn the scoring convention of a known question, the representational capacity of a medium-sized encoder appears to suffice; if the question is new, the knowledge acquired by the larger language

model during pre-training could make the difference. This mechanism is not demonstrated by the available runs, since encoder and decoder differ simultaneously in architecture, number of parameters, context length and training configuration (Section 4.6).

The BERT values on SciEntsBank lie at chance level on all three splits. A supplementary learning-rate variation (Section 4.6, Table 4.4) does not remedy this failure and at the same time confirms that the RoBERTa values in Table 5.9 are not substantially underestimated; the drop of RoBERTa on Beetle UQ occurs at every learning rate tested.

5.3.4 Transfer Between Benchmarks

The comparisons so far measure performance on fixed label spaces. A property that is structurally reserved for span alignment cannot show there: since the levels are present in the input as rubric spans, a trained model can be applied to tasks with other rubrics and another number of levels without adapting the output head (Section 4.1). Whether this structural transferability leads to usable scoring is tested by the transfer experiment from Section 4.5.5. Llama-3.2-1B and -3B were trained and validated with CONDIFF and causal attention on ASAP-SAS only and then evaluated without further training on the test splits of Beetle and SciEntsBank, in each case with the rubric texts of the target benchmark. A fixed classification head is not applicable in this set-up, since ASAP-SAS mixes three- and four-level tasks. The CONDIFF runs of the same backbone trained on the respective target benchmark serve as the reference.

Table 5.10: Transfer from ASAP-SAS to Beetle and SciEntsBank (CONDIFF, causal), QWK. *Transfer*: trained on ASAP-SAS only; *in-domain*: trained on the target benchmark. Means over three random seeds, ranges in parentheses.

		Llama-3.2-1B		**Llama-3.2-3B**	
Benchmark	**Split**	Transfer	In-domain	Transfer	In-domain
Beetle	UA	-0.001 (0.003)	0.729 (0.023)	-0.004 (0.011)	0.730 (0.041)
	UQ	0.000 (0.001)	0.509 (0.073)	-0.017 (0.060)	0.544 (0.044)
SciEntsBank	UA	0.008 (0.012)	0.702 (0.007)	0.022 (0.014)	0.734 (0.047)
	UQ	0.005 (0.014)	0.544 (0.058)	0.008 (0.005)	0.576 (0.009)
	UD	0.003 (0.008)	0.560 (0.026)	0.017 (0.015)	0.617 (0.028)

The transfer fails completely. On all five splits and both backbones, QWK lies between -0.017 and 0.022 and thus at chance level (Table 5.10). The distance to the in-domain runs amounts to 0.510 to 0.734 QWK and exceeds the noise threshold in every cell many times over; under Macro-F_1, it likewise holds as an effect in all ten cells.

The cause is a collapse onto the lowest level. In each of the 15 evaluations (three seeds, five splits), Llama-3.2-1B assigns between 97.2 and $100\,\%$ of the answers to the level *incorrect*, Llama-3.2-3B between 80.2 and $97.1\,\%$. The accuracy of the single runs, 0.303 to 0.374, thus roughly corresponds to the share of this level in the reference data and lies throughout below the majority baseline of 0.401 to 0.431, which falls to the level *correct*. Unlike the models without any task information in Section 5.3.2, which predict all three levels, this is a degenerate classifier.

The largest backbone collapses as well. A single transfer run with Mistral-7B (seed $114\,514$) shows the same picture: QWK lies between 0.000 and 0.017 on the five splits, and between 97.2 and $100\,\%$ of the answers are assigned to the level *incorrect* – a collapse as complete as for Llama-3.2-1B. Additional model capacity thus does not remedy the failure of transfer; a collapse that diminishes with size is not discernible across the three backbones.

Answer length offers an obvious explanation. In the training split of ASAP-SAS, the level depends strongly on the length of the answer: of the answers with at most ten words, $89.8\,\%$ receive level 0, of those with 11 to 20 words $66.9\,\%$, and of those with more than 40 words only $19.3\,\%$. The answers of the target benchmarks, by contrast, are short, with a median length of 9 (Beetle) and 11 words (SciEntsBank) compared with 39 on ASAP-SAS (Table 3.4); 95 to $96\,\%$ of the Beetle and 86 to $89\,\%$ of the SciEntsBank test answers comprise at most 20 words. A model that has learned on ASAP-SAS to associate short answers with low levels assigns almost every answer of the target benchmarks to the lowest level. Consistent with this, the few answers to which the transfer models assign a higher level are longer on average: 24.0 compared with 11.1 words for Llama-3.2-1B, 15.2 compared with 10.9 words for Llama-3.2-3B and 22.6 compared with 11.0 words for Mistral-7B. This association is favoured by the scoring guidelines of ASAP-SAS, which for four of the ten tasks (items 1, 2, 5 and 6) define the level via the number of content elements mentioned, for item 5 for instance “Four key elements” for the highest level. Length and level are thus correlated there by the guideline itself.

The length explanation is a hypothesis. It is consistent with the data but was not tested causally, for instance through an evaluation with controlled answer length. Differences in domain and scoring convention may equally contribute to the failure, as may the origin of the rubric texts, which on ASAP-SAS stem from the task authors and on the target benchmarks were generated with a language model (Chapter 3).

Structural transferability is not substantive transferability. For RQ1, the experiment separates two properties that coincide in the motivation of the architecture. Structurally, the transfer succeeds: the same model scores tasks with three, four and, on the target benchmarks, again three levels without any adaptation; with a fixed head, even joint training on ASAP-SAS is ruled out. Substantively, however, no transfer takes place. The rubric texts of the target benchmark are not used in a way that overrides the heuristic learned on ASAP-SAS. Structural transferability is thus a necessary but not a sufficient condition for scoring across benchmarks.

5.4 RQ2: Adapting Decoder-Only Models

The second research question asks whether and how decoder-only language models can be adapted to span alignment. This covers the design of the alignment, i.e. the fusion mechanism (Section 5.4.1) and the anchor of the fusion (Section 5.4.2), as well as the attention mask (Section 5.4.3) and the size of the backbone (Section 5.4.4).

5.4.1 Fusion Mechanisms

Results per Benchmark

Tables 5.11 and 5.12 document the results of the fusion variants, as single runs on ASAP-SAS and as means over three random seeds on Beetle and SciEntsBank, there for all three decoder backbones. DIFF was investigated on Llama-3.2-3B only. The tables serve completeness; the interpretation that follows is based on the means over three random seeds.

Table 5.11: ASAP-SAS (UA): fusion mechanisms with causal attention. Single runs, random seed 114 514. Bold: highest value per column within a backbone; the marking makes no statement about whether a difference is demonstrable (cf. Table 5.13).

	Llama-3.2-1B			**Llama-3.2-3B**			**Mistral-7B**		
	QWK	F_1	Acc	QWK	F_1	Acc	QWK	F_1	Acc
CONDIFF	0.818	0.729	0.767	0.838	0.762	0.786	**0.860**	**0.803**	**0.809**
CONCAT	0.828	**0.750**	**0.781**	**0.842**	**0.769**	**0.796**	0.855	0.788	0.800
GATE	**0.829**	**0.750**	0.777	0.841	0.765	0.789	0.858	0.790	0.801

Table 5.12: Beetle and SciEntsBank: fusion mechanisms with causal attention, QWK per split. Means over three random seeds; DIFF was investigated on Llama-3.2-3B only. Bold: highest value per column within a backbone; the marking makes no statement about whether a difference is demonstrable (cf. Table 5.13).

		Beetle		**SciEntsBank**		
Backbone	**Fusion**	UA	UQ	UA	UQ	UD
Llama-3.2-1B	CONDIFF	**0.729**	**0.509**	0.702	0.544	0.560
	CONCAT	0.725	0.503	0.698	**0.569**	**0.577**
	GATE	0.696	0.471	**0.718**	0.564	0.559
Llama-3.2-3B	CONDIFF	0.730	0.544	0.734	0.576	0.617
	CONCAT	0.735	0.546	0.720	0.582	0.626
	GATE	**0.757**	0.550	**0.737**	**0.624**	**0.630**
	DIFF	0.740	**0.572**	0.722	0.579	0.623
Mistral-7B	CONDIFF	0.779	**0.605**	0.755	0.639	0.641
	CONCAT	0.772	0.604	0.750	**0.652**	**0.647**
	GATE	**0.781**	0.567	**0.758**	0.632	0.638

The Difference Between the Variants Lies Below the Noise Threshold

Table 5.13 contrasts the difference *between* the fusion mechanisms with the variation *within* a fusion variant. In all fifteen configurations of benchmark, backbone and split, the variation over the seeds exceeds the distance between the variants, in the extreme case by a factor

Table 5.13: Difference between the fusion mechanisms (range of the means of CONDIFF, CONCAT and GATE, for Llama-3.2-3B additionally DIFF) compared with the largest seed range of the same cells.

Benchmark	Backbone	Split	Between variants	Max. seed Δ	Ratio
Beetle	Llama-1B	UA	0.033	0.051	$1.5\times$
Beetle	Llama-1B	UQ	0.038	0.118	$3.1\times$
Beetle	Llama-3B	UA	0.027	0.042	$1.6\times$
Beetle	Llama-3B	UQ	0.028	0.048	$1.7\times$
Beetle	Mistral-7B	UA	0.008	0.046	$5.6\times$
Beetle	Mistral-7B	UQ	0.038	0.058	$1.5\times$
SciEntsBank	Llama-1B	UA	0.020	0.039	$1.9\times$
SciEntsBank	Llama-1B	UQ	0.025	0.058	$2.3\times$
SciEntsBank	Llama-1B	UD	0.017	0.047	$2.7\times$
SciEntsBank	Llama-3B	UA	0.016	0.047	$2.8\times$
SciEntsBank	Llama-3B	UQ	0.049	0.060	$1.2\times$
SciEntsBank	Llama-3B	UD	0.013	0.035	$2.7\times$
SciEntsBank	Mistral-7B	UA	0.009	0.050	$5.9\times$
SciEntsBank	Mistral-7B	UQ	0.020	0.063	$3.2\times$
SciEntsBank	Mistral-7B	UD	0.009	0.037	$3.9\times$

of almost six. This is most pronounced for Mistral-7B: on the UA splits of both benchmarks, the three variants lie only 0.008 and 0.009 QWK apart on average, with seed ranges of up to 0.050. According to the rule formulated in Section 5.2, a difference between the fusion variants is consequently *not demonstrable as an effect* on this data basis.

Four pairwise comparisons do, however, meet the weaker criterion: GATE versus CONDIFF on Beetle UA and on SciEntsBank UQ, and GATE versus DIFF on Beetle UA, all with Llama-3.2-3B, and CONDIFF versus GATE on Beetle UQ with Mistral-7B. There, the ranges of the two conditions do not overlap, while the difference of the means (0.027, 0.049, 0.017 and 0.038 QWK) remains below the larger of the two ranges in each case. According to the evaluation rule, this is a weak indication, not a result. The indication for Mistral-7B points in the opposite direction to the other three: there, GATE is behind. Under Macro-F_1, nine pairwise verdicts shift between these categories; the overall picture remains unaffected (Section 5.5.3).

The consequence reaches further than it first appears. Table 5.14 shows which variant attains the highest value in which cell.

With the smaller backbone, the ranking changes between benchmarks and splits without a discernible pattern – exactly the picture to be expected if the underlying differences are in fact zero and the order is determined by the random seed alone. With Llama-3.2-3B, by contrast, GATE leads in four of five cells; in the fifth, Beetle UQ, DIFF leads. This picture is consistent with the three disjoint ranges mentioned above for this backbone, all of which concern GATE, and it must be assessed cautiously: none of the differences exceeds the range of the conditions involved, and the five cells are not independent of each other, since they rest on the same runs and are merely evaluated separately by split.

Table 5.14: Best fusion variant per benchmark, split and backbone according to the mean over three random seeds. According to Table 5.13 the ranking is not robust and is given here solely to illustrate its arbitrariness.

Benchmark	Split	Llama-1B	Llama-3B	Mistral-7B
Beetle	UA	CONDIFF	GATE	GATE
	UQ	CONDIFF	DIFF	CONDIFF
SciEntsBank	UA	GATE	GATE	GATE
	UQ	CONCAT	GATE	CONCAT
	UD	CONCAT	GATE	CONCAT

It remains to be noted that this indication runs *counter* to the expectation formulated in Section 4.4.1: there it had been assumed that a mechanism with many parameters such as GATE would carry relatively more weight with smaller backbones. Mistral-7B, however, does not confirm the pattern of Llama-3.2-3B. There, GATE leads in two cells, CONCAT in two and CONDIFF in one – again the picture of a random alternation –, and the only indication for this backbone goes against GATE. Across the three backbones, GATE leads in one, four and two of five cells; an advantage of either mechanism that grows with the size of the backbone is not discernible. An interpretation of individual rankings – for instance that the difference term of CONDIFF benefits generalisation – is equally unjustifiable on this basis. Such interpretations would have been quite plausible on single runs; they do not survive repetition with several random seeds.

For the research question of this thesis, this is an informative negative result: the parameter-free difference term of CONDIFF (4.6) brings no demonstrable advantage over plain concatenation on any of the three backbones, and conversely the unchanged individual representations add nothing over the difference term alone: on Llama-3.2-3B, DIFF (4.8) is nominally ahead of CONDIFF in four of five cells, throughout within the noise. The d^2 additional parameters of the gate projection (4.9) likewise do not justify their cost according to the evaluation rule. Since CONCAT and DIFF are the simplest constructions – DIFF with the smallest scoring head –, a practical recommendation in favour of a simple construction follows; for DIFF, however, it rests on Llama-3.2-3B alone.

5.4.2 Anchor of the Fusion

The second question regarding the design of the fusion concerns the choice of context representation in (2.18). This thesis consistently uses the sequence representation z; Gombert et al. [23] instead align the representation of the *answer* span with the rubric spans. To test this, the variant "t-concat" was investigated, which differs from CONCAT solely in the anchor: instead of $[\, z \,\|\, r_k \,]$ it forms $[\, t \,\|\, r_k \,]$, where t denotes the representation of the answer span pooled by the same procedure (Section 4.1.2).

Table 5.15 shows completely overlapping ranges in four of the five cells, with differences between 0.002 and 0.013 QWK. The only exception is Beetle UA: there the ranges of t-concat and CONCAT do not overlap, but the gap is only 0.007 QWK, and the difference of the means (0.026) remains below the range of CONCAT (0.042). According to the evaluation rule, this is a weak

Table 5.15: Anchor of the fusion on Llama-3.2-3B: answer span (t-concat) versus sequence representation (CONCAT, CONDIFF). Means over three random seeds, ranges in parentheses.

Benchmark	Split	t-concat	CONCAT	CONDIFF	Δ to CONCAT
Beetle	UA	0.761 (0.006)	0.735 (0.042)	0.730 (0.041)	$+0.026$
	UQ	0.552 (0.019)	0.546 (0.046)	0.544 (0.044)	$+0.006$
SciEntsBank	UA	0.734 (0.066)	0.720 (0.042)	0.734 (0.047)	$+0.013$
	UQ	0.584 (0.059)	0.582 (0.049)	0.576 (0.009)	$+0.002$
	UD	0.615 (0.019)	0.626 (0.035)	0.617 (0.028)	-0.011

indication, not an effect. Under Macro-F_1, the same comparison narrowly passes the rule as an effect ($+0.014$); the difference, however, remains small (Section 5.5.3).

This result deserves a methodological remark, because it illustrates the need for repetition particularly vividly. If only the run with random seed $114\,514$ were available, a markedly different picture would emerge: t-concat reached 0.762 on SciEntsBank UA there, compared with a CONCAT range of 0.701 to 0.743, and 0.601 on SciEntsBank UQ, compared with 0.550 to 0.598 – in both cases above all comparison runs. The run with seed 1, at 0.744 and 0.604, also lies just above these ranges. The run with seed 2, by contrast, drops to 0.696 and 0.545, respectively, and thus below the weakest CONCAT run in each case. On average, only a lead of 0.013 and 0.002 QWK remains, which lies entirely within the ranges. Two of three draws would have suggested an advantage as a single run, which the third cancels out.

For the thesis it follows that whether the answer span or the sequence representation serves as the anchor makes no difference above the noise threshold. This is remarkable insofar as the two representations carry different information – t ends before the rubric texts and has not seen them under the causal mask, while z is at the end of the sequence and has seen them all (Section 4.1.2). That this difference has no consequences indicates that the information decisive for scoring is already contained in the answer representation and that the additional knowledge of the other levels makes no usable contribution.

5.4.3 Attention Mask

The attention mask is the only variation within the span-alignment architecture whose effect clearly exceeds the noise threshold. It is evaluated in two steps: first the partial removal of causality on single runs, then the complete removal with three random seeds on both Llama sizes.

Partial Removal

Across the single runs, there is *no* monotonic relationship between the number of bidirectional layers and performance (Figure 5.2). Depending on the cell, the strongest condition is the causal reference, a single bidirectional layer or an intermediate number of layers. Since all values in these two tables rest on single runs and the observed distances lie predominantly

Table 5.16: ASAP-SAS (UA): attention mask with CONDIFF fusion, sorted by the actual number of bidirectional layers n_{bidir}. Single runs, random seed $114\,514$; for the causal and the fully bidirectional condition of the two Llama sizes, three seeds are additionally available (Tables 5.18 and 5.19). For Mistral-7B, the fully bidirectional condition stems from a different software environment than the other runs in this column. Bold: highest value per column; the marking makes no statement about whether a difference is demonstrable.

	Llama-3.2-1B ($L = 16$)			**Llama-3.2-3B** ($L = 28$)			**Mistral-7B** ($L = 32$)		
ρ	n_{bidir}	QWK	F_1	n_{bidir}	QWK	F_1	n_{bidir}	QWK	F_1
0 (causal)	0	0.818	0.729	0	0.838	0.762	0	**0.860**	**0.803**
1.0	1	0.821	0.743	1	**0.843**	**0.773**	1	0.849	0.782
0.2	3	0.822	0.745	5	0.836	0.764	6	0.851	0.784
0.4	6	0.822	0.738	11	0.834	0.765	12	0.845	0.783
0.6	9	0.819	0.744	16	0.834	0.765	19	0.855	0.792
0.8	12	0.823	0.748	22	0.839	0.767	25	0.847	0.787
L (full)	16	**0.825**	**0.754**	28	0.840	0.770	32	0.853	0.782

below the noise threshold of 0.036 QWK, they cannot be interpreted individually. The fully bidirectional condition on ASAP-SAS also deviates only slightly from the causal reference; with three seeds, the difference amounts to $+0.003$ (Llama-3.2-1B) and $+0.001$ QWK (Llama-3.2-3B), and to -0.007 in the single run with Mistral-7B.

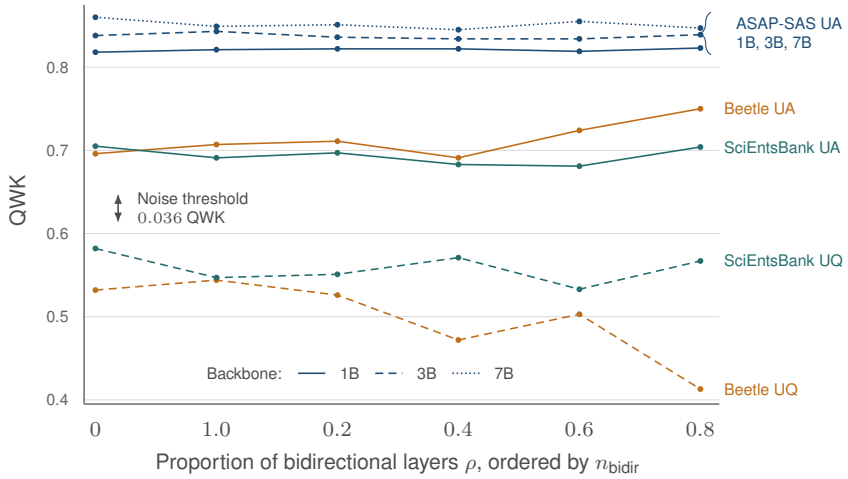


Figure 5.2: QWK as a function of the number of bidirectional layers n_{bidir}, CONDIFF, single runs, with the noise threshold indicated (values from Tables 5.16 and 5.17).

One observation exceeds this threshold: for Llama-3.2-1B on Beetle, the UA value rises by 0.053 QWK over the range from zero to twelve bidirectional layers, while the UQ value falls by 0.119 over the same range. This opposing development is clear enough to be noteworthy, but it rests on one run each and would require confirmation through repetition.

Table 5.17: Beetle and SciEntsBank: attention mask with CONDIFF fusion, QWK per split, sorted by n_{bidir}. Single runs, random seed $114\,514$. For Llama-3.2-3B and Mistral-7B, the same conditions are likewise available as single runs; they are not listed here, since they contribute no statement that does not already follow from the complete removal. Bold: highest value per column; the marking makes no statement about whether a difference is demonstrable.

	Beetle, Llama-3.2-1B			SciEntsBank, Llama-3.2-1B			
ρ	n_{bidir}	UA	UQ	n_{bidir}	UA	UQ	UD
0 (causal)	0	0.696	0.532	0	**0.705**	**0.582**	0.572
1.0	1	0.707	**0.544**	1	0.691	0.547	0.558
0.2	3	0.711	0.526	3	0.697	0.551	0.555
0.4	6	0.691	0.472	6	0.683	0.571	0.561
0.6	9	0.724	0.503	9	0.681	0.533	0.541
0.8	12	**0.750**	0.413	12	0.704	0.567	**0.589**

Complete Removal

The more informative condition is the complete removal of causality in all layers. It was run for Llama-3.2-3B (28 layers) and Llama-3.2-1B (16 layers) with three random seeds each and compared with the causal CONDIFF reference of the same backbone. Table 5.18 first reports the larger backbone.

Table 5.18: Fully bidirectional attention ($n_{\text{bidir}} = L = 28$) versus causal attention, Llama-3.2-3B, CONDIFF fusion. Means over three random seeds, ranges in parentheses.

Benchmark	Split	Bidirectional	Causal	Δ	Verdict
ASAP-SAS	UA	0.836 (0.008)	0.835 (0.004)	$+0.001$	within noise
Beetle	UA	0.657 (0.061)	0.730 (0.041)	-0.073	above noise threshold
	UQ	0.358 (0.094)	0.544 (0.044)	-0.186	far above noise threshold
SciEntsBank	UA	0.767 (0.024)	0.734 (0.047)	$+0.032$	ranges disjoint
	UQ	0.636 (0.026)	0.576 (0.009)	$+0.060$	above noise threshold
	UD	0.625 (0.036)	0.617 (0.028)	$+0.008$	within noise

For Llama-3.2-3B, Table 5.18 contains the clearest architectural effect of this thesis – and at the same time the only one that changes sign between benchmarks.

On Beetle, complete removal causes considerable harm. The loss amounts to 0.073 QWK on UA and 0.186 on UQ. On both splits the ranges do not overlap; on UQ, the best bidirectional run (0.419) is still 0.109 below the worst causal one (0.528). The increased variation is also remarkable: the range rises from 0.044 to 0.094, so training becomes not only worse but also less stable.

On SciEntsBank, it helps. The gain amounts to 0.060 QWK on UQ – with disjoint ranges and a gap of 0.043 – and 0.032 on UA, where the ranges likewise do not overlap but the difference of the means remains below the range of the causal reference. On UD, no difference is

demonstrable.

On ASAP-SAS, it has no effect. At 0.836 versus 0.835 QWK, both conditions are practically level, with ranges of at most 0.008.

Under Macro-F_1, the two UQ findings, the loss on Beetle and the gain on SciEntsBank, hold unchanged as effects. The loss on Beetle UA and the indication on SciEntsBank UA, by contrast, are not demonstrable there (Section 5.5.3).

On the smaller backbone, a different picture emerges. Table 5.19 presents the same condition for Llama-3.2-1B, likewise with three random seeds.

Table 5.19: Fully bidirectional attention ($n_{\text{bidir}} = L = 16$) versus causal attention, Llama-3.2-1B, CONDIFF fusion. Means over three random seeds, ranges in parentheses. *Above range* as in Table 5.4. With eight seeds of the causal condition, the difference on Beetle UA is not demonstrable (Table 5.3).

Benchmark	Split	Bidirectional	Causal	Δ	Verdict
ASAP-SAS	UA	0.830 (0.010)	0.827 (0.016)	$+0.003$	within noise
Beetle	UA	0.765 (0.017)	0.729 (0.023)	$+0.036$	above range
	UQ	0.493 (0.058)	0.509 (0.073)	-0.016	within noise
SciEntsBank	UA	0.692 (0.049)	0.702 (0.007)	-0.010	within noise
	UQ	0.553 (0.087)	0.544 (0.058)	$+0.009$	within noise
	UD	0.588 (0.046)	0.560 (0.026)	$+0.028$	within noise

On ASAP-SAS and SciEntsBank, no difference is demonstrable for the smaller backbone on any split; all ranges overlap. The role of the individual seeds is revealing here: the run with seed $114\,514$ is the weakest of the three on all three SciEntsBank splits and, as a single run, would have suggested a loss of up to 0.084 QWK compared with the causal reference of the same seed.

On Beetle, by contrast, the sign nominally reverses. On UA, the smaller backbone gains 0.036 QWK with three seeds; the ranges are disjoint, and the difference exceeds the larger of them (0.023), so that there would be an effect according to the evaluation rule. It passes the rule, however, only because both conditions vary unusually little, and it does not survive the extension of the causal reference to eight seeds: against their range of 0.079, the difference of then $+0.033$ is no longer demonstrable (Section 5.2). On UQ, where the larger backbone loses 0.186 QWK, no difference is demonstrable for the smaller one under QWK. Under Macro-F_1, by contrast, the smaller backbone loses on Beetle UQ (-0.041), and this loss passes the evaluation rule with eight seeds as well (-0.038).

The contrast between the two benchmarks for the larger backbone requires explanation, since they share the same task structure, the same three-level scale and the same backbone. An obvious hypothesis relates to the data structure (Chapter 3): Beetle comprises $3\,941$ training instances for 47 questions from a single narrowly bounded domain, SciEntsBank $4\,969$ instances for 135 questions from twelve topic blocks. Complete removal of causality removes the condition under which the backbone was pre-trained (Section 2.6.2); the model first has to learn to use the two-sided context during fine-tuning. A narrow and homogeneous data basis would

offer too little material for this, a broader one would suffice. The runs on ASAP-SAS speak against this explanation in its simple form: with $17\,043$ training instances, ASAP-SAS offers by far the broadest data basis, and yet no difference is demonstrable there on either backbone. The amount of training data alone thus does not determine the effect of the mask. The direction of the Beetle finding is, by contrast, compatible with the smaller backbone: once the gain on UA disappears with eight seeds, Llama-3.2-1B shows no advantage on Beetle, and under Macro-F_1 even a loss on UQ – the same direction as the larger backbone, albeit much weaker. Which property of the benchmarks is decisive cannot be isolated with three benchmarks; varying the amount of training data with the mask held fixed could at least delimit the contribution of data quantity.

In any case, the methodological core remains: the effect of the attention mask cannot be generalised across benchmarks or across backbones. A recommendation to operate decoder models for ASAG bidirectionally as a matter of principle can be derived from these data just as little as the opposite.

5.4.4 Effect of Model Size

Table 5.20: Effect of backbone size with CONDIFF and causal attention (QWK). Means over three random seeds; for Mistral-7B on ASAP-SAS, a single run.

Benchmark	Split	Llama-1B	Llama-3B	Mistral-7B
ASAP-SAS	UA	0.827	0.835	0.860
Beetle	UA	0.729	0.730	0.779
	UQ	0.509	0.544	0.605
SciEntsBank	UA	0.702	0.734	0.755
	UQ	0.544	0.576	0.639
	UD	0.560	0.617	0.641

At first glance, the size effect is the most stable finding of this thesis (Table 5.20). From Llama-3.2-1B to Mistral-7B, the gain on Beetle amounts to 0.050 QWK on UA and 0.095 on UQ, on SciEntsBank to 0.053 / 0.095 / 0.082 and on ASAP-SAS to 0.033; five of the six cells would thus lie above the noise threshold, ASAP-SAS just below it. A supplementary check shows, however, that this impression does not hold.

The Comparison Is Confounded with the Learning Rate

All decoder main runs of this thesis use the learning rate $5 \cdot 10^{-5}$ (Section 4.6). The reference implementation, however, provides this value only for models with seven billion parameters or more; for the two Llama backbones it specifies $2 \cdot 10^{-4}$. The largest backbone was thus trained at its intended rate, the two smaller ones at a quarter of theirs.

To estimate the consequences, CONDIFF on both Llama sizes was additionally trained at $2 \cdot 10^{-4}$ and compared with the three existing runs at $5 \cdot 10^{-5}$ (Table 5.21).

Table 5.21: CONDIFF at the reference learning rate $2 \cdot 10^{-4}$ compared with the runs at $5 \cdot 10^{-5}$ (mean and range over three random seeds), QWK. The runs at $2 \cdot 10^{-4}$ rest on a single random seed each.

Backbone	Benchmark	Split	$2 \cdot 10^{-4}$	$5 \cdot 10^{-5}$	Position rel. to range
Llama-3.2-1B	Beetle	UA	**0.779**	0.729 (0.718–0.741)	above ($+0.038$)
		UQ	**0.605**	0.509 (0.472–0.545)	above ($+0.060$)
	SciEntsBank	UA	0.679	0.702 (0.698–0.705)	below (-0.019)
		UQ	0.537	0.544 (0.524–0.582)	within
		UD	0.496	0.560 (0.546–0.572)	below (-0.050)
Llama-3.2-3B	Beetle	UA	**0.775**	0.730 (0.707–0.748)	above ($+0.027$)
		UQ	0.564	0.544 (0.528–0.573)	within
	SciEntsBank	UA	0.749	0.734 (0.703–0.750)	within
		UQ	**0.609**	0.576 (0.571–0.580)	above ($+0.030$)
		UD	0.564	0.617 (0.605–0.633)	below (-0.041)

The finding is twofold and decisive for the evaluation of this axis.

On Beetle, the smallest backbone was considerably undertrained. At the reference learning rate it reaches 0.779 QWK on UA and 0.605 on UQ – in both cases above all runs at $5 \cdot 10^{-5}$. The gain over the mean amounts to 0.050 and 0.096 QWK, respectively. The gap between Llama-3.2-1B and Mistral-7B on Beetle reported in Table 5.20 thus disappears completely: it now amounts to less than 0.001 QWK on UA and -0.001 on UQ, i.e. the smallest backbone is on a par with the mean over three Mistral runs. The extension to eight seeds (Section 5.2) qualifies the finding on UA: one of the five additional runs at $5 \cdot 10^{-5}$ reaches 0.781 there and thus lies just above the run at $2 \cdot 10^{-4}$. The gain on UA is consequently not secured. On UQ, the run at $2 \cdot 10^{-4}$ also lies above all eight runs, by 0.060.

On SciEntsBank, the same change acts in the opposite direction. There the run at $2 \cdot 10^{-4}$ lies below the entire range at $5 \cdot 10^{-5}$ on two of three splits, by 0.050 QWK on UD.

For Llama-3.2-3B, the picture is mixed. On Beetle, the run at $2 \cdot 10^{-4}$ lies 0.027 QWK above the range at $5 \cdot 10^{-5}$ on UA and within it on UQ. On SciEntsBank, it lies above on UQ ($+0.030$), below on UD (-0.041) and within on UA. A gain as clear as for the smallest backbone on Beetle does not emerge. Consequently, for neither of the two Llama backbones is there a learning rate that is more favourable across both benchmarks; it interacts with dataset and backbone.

Consequence

Both observations together imply that Table 5.20 permits no statement about the influence of model size. In the available runs, the three backbones differ not only in their size but also in how far their learning rate deviates from the one intended for them – and on Beetle the extent of this deviation is large enough to explain the entire observed difference. A robust investigation of this axis would require training each backbone at its respectively appropriate learning rate; this was not feasible within this thesis.

This thesis therefore makes *no* statement about the effect of backbone size. Table 5.20 is

retained as documentation of the runs carried out and is explicitly not to be read as a size comparison.

One limitation also concerns this finding itself: for both Llama sizes, it rests on one run each at $2 \cdot 10^{-4}$, whose distance from the range at $5 \cdot 10^{-5}$ is of the order of the noise threshold in several cells.

All comparisons *within* a backbone, in which the learning rate is identical throughout, remain unaffected – and with them the findings on fusion mechanism, anchor and attention mask. Affected, by contrast, are the few places at which this chapter compares effect sizes *between* two model sizes: the increase of the ablation effect from 1B to 3B (Section 5.3.2), the disappearance of the span advantage with the larger backbone (Section 5.3.1) and the interpretation of the task-level differences between Llama-3.2-1B and Mistral-7B (Section 5.5.1). All three are qualified accordingly there. The finding does, however, fit into a pattern that this chapter shows repeatedly: several effects that initially appear clear turn out, on closer inspection, to be properties of the configuration rather than of the object under investigation.

5.5 Supplementary Analyses

The following analyses do not answer a research question of their own but safeguard the findings on RQ1 and RQ2: the evaluation per task on ASAP-SAS (Section 5.5.1), the diagnosis of unstable runs (Section 5.5.2) and the check of which findings hold under Macro-F_1 (Section 5.5.3).

5.5.1 Results per Task on ASAP-SAS

Since ASAP-SAS consists of ten tasks that are heterogeneous in content and structure (Chapter 3), the value aggregated over all instances conceals considerable differences. Table 5.22 breaks down QWK by task and compares the smallest with the largest backbone.

The variation between tasks exceeds all architectural effects investigated. With the 1B backbone, QWK ranges from 0.625 (item 8) to 0.834 (item 9), with the 7B backbone from 0.653 to 0.898. This range of about 0.21 and 0.25, respectively, is many times larger than the differences between fusion mechanisms, anchors or attention conditions on the same benchmark, which reach at most 0.015 QWK, and of the same order of magnitude as the effect of the rubric text. It also exceeds the largest architectural effect of this thesis, the loss of 0.186 QWK due to full bidirectionality on Beetle UQ, albeit only narrowly. Which tasks a benchmark contains thus has a stronger effect than any of the architectural variants investigated here – a circumstance to be taken into account when generalising aggregated benchmark values.

Figure 5.3 makes this relation visible: the range over the ten tasks is contrasted there with the noise threshold against which all architectural comparisons in this chapter are to be measured.

The gap between the two backbones concentrates on the four-level science tasks. On the four items with scale $\{0, \ldots, 3\}$ – all Science or Biology – Mistral-7B is ahead by 0.071 QWK on average, on the six three-level items by only 0.024. On the two English Language Arts

Table 5.22: ASAP-SAS (UA): QWK per task for CONDIFF with causal attention. Single runs. Bold: higher value per row.

Item	Subject	Scale	Llama-1B	Mistral-7B	Difference
1	Science	$\{0..3\}$	0.829	**0.881**	$+0.052$
2	Science	$\{0..3\}$	0.776	**0.898**	$+0.122$
5	Biology	$\{0..3\}$	0.832	**0.852**	$+0.020$
6	Biology	$\{0..3\}$	0.801	**0.892**	$+0.091$
3	English Language Arts	$\{0..2\}$	**0.663**	0.653	-0.011
4	English Language Arts	$\{0..2\}$	0.728	**0.734**	$+0.007$
7	English	$\{0..2\}$	0.702	**0.738**	$+0.035$
8	English	$\{0..2\}$	0.625	**0.697**	$+0.072$
9	English	$\{0..2\}$	**0.834**	0.832	-0.002
10	Science	$\{0..2\}$	0.731	**0.772**	$+0.041$
Mean of four-level items			0.810	**0.881**	$+0.071$
Mean of three-level items			0.714	**0.738**	$+0.024$

tasks, the gap is practically zero: -0.011 on item 3 and $+0.007$ on item 4. The pattern could be linked to the description of these items in Chapter 3: the four-level science tasks require the identification of several content aspects, for which subject knowledge is needed that a larger language model is more likely to possess from pre-training. It is, however, not to be read as a size effect. Both columns rest on single runs, and Llama-3.2-1B was trained at a quarter of its reference learning rate, as explained in Section 5.4.4, Mistral-7B at its own. Whether the concentration on the four-level tasks would persist with an adjusted learning rate is open.

5.5.2 Unstable Runs

Three runs lie clearly below their comparison conditions (Table 5.23). The corresponding prediction files were evaluated separately to distinguish between a substantive effect and training instability; the two Mistral runs were also repeated. The reference points are the accuracy of a trivial model that always predicts the most frequent level and the QWK under random permutation of the predictions – i.e. with an unchanged marginal distribution but the assignment removed.

The two original Mistral runs are functional but weak. Both predict all levels with an approximately correct marginal distribution – for the SciEntsBank run with $\rho = 0.6$, the predicted distribution on the UD split, $(1733, 1001, 1828)$, matches the reference distribution $(1659, 986, 1917)$ almost exactly. Their accuracy, at 0.674 and 0.578, lies clearly above the majority baseline, and their confusion matrices show a sensible diagonal structure. These are thus not aborted training runs or models that merely reproduce the class distribution, but models that have learned the task – only considerably worse. An evaluation per question further shows that the deterioration is widely distributed: on Beetle, the run with $\rho = 0.2$ drops by more than 0.1 QWK relative to the causal reference on 28 of the 47 questions and improves on only four. A single problematic item or a data artefact can thus be ruled out as an explanation.

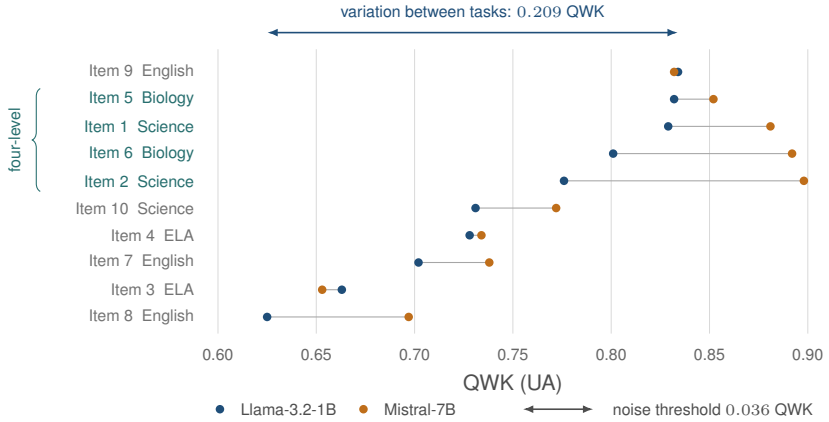


Figure 5.3: ASAP-SAS (UA): QWK per task for CONDIFF with causal attention, single runs (values from Table 5.22).

The gaps of up to 0.22 QWK on UA and 0.40 on UQ lie far above the noise threshold determined in Section 5.2 and also above the largest range observed there (0.118). At the same time, a substantive interpretation as a mask effect is contradicted by the fact that $\rho = 0.6$, with 19 bidirectional layers, reaches 0.801 on the same benchmark – more than three times as many bidirectional layers with almost unchanged performance.

The repetition rules out a mask effect. Both configurations were trained again with the same random seed. The Beetle run with $\rho = 0.2$ now reaches 0.778 instead of 0.587 QWK on UA and 0.593 instead of 0.210 on UQ, the SciEntsBank run with $\rho = 0.6$ reaches 0.657 instead of 0.378 on UQ; on UA and UD it likewise lies within the range of the other Mistral runs at 0.751 and 0.651. Both repetitions thus lie close to the causal reference of the same seed (0.803 / 0.606 on Beetle, 0.621 on SciEntsBank UQ), at most 0.025 below it on Beetle and 0.036 above it on SciEntsBank UQ. Since training within one environment is deterministic (Section 5.2), the repetition necessarily ran in a different software environment than the original run; whether the deviation is due to the environment or to an instability that occurred by chance in the original run therefore cannot be separated. In any case, it is not an effect of the mask. The two original runs remain excluded from interpretation.

The BERT run on SciEntsBank, by contrast, is a genuine failure. Its accuracy is 0.431 and thus matches the majority baseline exactly to three decimal places. The confusion matrix shows that 148 of 194 instances of level 0 and 76 of 113 of level 1 are assigned to level 2; overall, $77\,\%$ of all predictions fall on the most frequent level, compared with $43\,\%$ in the reference. The model has therefore learned nothing beyond the class distribution. The learning-rate variation reported in Section 5.3.3 further shows that this failure cannot be remedied by any of the three additionally tested learning rates. The run is thus to be classified as a failure whose cause remains open; it is not to be interpreted substantively in any case.

5.5.3 Robustness to the Choice of Metric

All verdicts on RQ1 and RQ2 are based on QWK. Since Section 2.2 introduces Macro-F_1 as a second metric, every pairwise comparison with three random seeds reported there was re-

Table 5.23: Diagnosis of the conspicuous runs. *Acc*$_{majority}$ is the accuracy when constantly predicting the most frequent level, *QWK*$_{perm}$ the QWK under a single random permutation of the predictions. *Rep.*: repetition with the same random seed in the software environment with transformers 4.57.6.

Run	Split	QWK	Acc	$\text{Acc}_{\text{majority}}$	QWK_{perm}
Beetle, Mistral, $\rho = 0.2$	UA	0.587	0.674	0.401	-0.018
Beetle, Mistral, $\rho = 0.2$, rep.	UA	0.778	0.770	0.401	0.044
Beetle, Mistral, causal	UA	0.803	0.788	0.401	0.000
Beetle, Mistral, $\rho = 0.2$	UQ	0.210	0.480	0.420	-0.001
Beetle, Mistral, $\rho = 0.2$, rep.	UQ	0.593	0.652	0.420	-0.055
Beetle, Mistral, causal	UQ	0.606	0.661	0.420	-0.015
SciEntsBank, Mistral, $\rho = 0.6$	UQ	0.378	0.578	0.411	-0.004
SciEntsBank, Mistral, $\rho = 0.6$, rep.	UQ	0.657	0.700	0.411	0.030
SciEntsBank, Mistral, $\rho = 0.2$	UQ	0.627	0.690	0.411	-0.005
SciEntsBank, BERT	UA	0.052	**0.431**	**0.431**	-0.012
SciEntsBank, RoBERTa	UA	0.633	0.680	0.431	0.005

peated under Macro-F_1, applying the same evaluation rule to the means and ranges of Macro-F_1. 114 comparisons are included, with each split of a pairwise comparison counted separately: the ablation and context comparisons from Section 5.3.2, span alignment versus fixed head on both Llama sizes, the transfer runs versus the in-domain runs, all pairs of fusion variants per backbone including Mistral-7B, t-concat versus CONCAT, and the fully bidirectional versus the causal condition on all three benchmarks.

In 94 of the 114 comparisons, the verdict is the same under both metrics. Table 5.24 lists the 20 deviating ones.

Task Information

The core finding is independent of the metric. The collapse without any task information holds as an effect under Macro-F_1 on both Llama sizes and all three SciEntsBank splits, just as under QWK. The same applies on UA to the advantage of context alone over rubric alone and to the gain from context when the rubric is present; the latter also holds on UD.

The relation between the two information sources on the transfer splits, by contrast, depends on the metric. Under QWK, context alone outperforms rubric alone as an effect on UQ and UD as well. Under Macro-F_1, the two are level on UQ (-0.003), and on UD a weak indication remains ($+0.014$). Conversely, the rubric text contributes something under Macro-F_1 when context is present: as an effect on UD ($+0.029$) and as an indication on UQ ($+0.019$), while QWK demonstrates no difference in either case.

The discrepancy can be traced to the type of errors. With three levels, QWK weights confusing *incorrect* and *correct* four times as heavily as confusing adjacent levels; Macro-F_1 does not

Table 5.24: Pairwise comparisons with three random seeds whose verdict differs between QWK and Macro-F_1. Δ is the difference of the means (first minus second condition); the verdict follows the evaluation rule from Section 5.2, applied to the respective metric. "—" means not demonstrable. Context denotes question and reference answer.

Sect.	Comparison	Split	Δ QWK	Verdict	Δ F_1	Verdict
5.3.1	Beetle 1B: span alignment vs. fixed head	UQ	$+0.054$	—	$+0.044$	effect
5.3.2	SciEntsBank 1B: context with vs. without rubric	UQ	$+0.015$	—	$+0.019$	indication
5.3.2	SciEntsBank 1B: context with vs. without rubric	UD	$+0.006$	—	$+0.029$	effect
5.3.2	SciEntsBank 1B: context only vs. rubric only	UQ	$+0.060$	effect	-0.003	—
5.3.2	SciEntsBank 1B: context only vs. rubric only	UD	$+0.077$	effect	$+0.014$	indication
5.3.2	SciEntsBank 1B: context and rubric vs. rubric only	UQ	$+0.075$	effect	$+0.017$	—
5.3.2	Beetle 1B: rubric vs. neither	UA	$+0.054$	indication	$+0.040$	effect
5.4.1	Beetle 1B: CONDIFF vs. CONCAT	UA	$+0.005$	—	$+0.006$	indication
5.4.1	Beetle 1B: CONDIFF vs. GATE	UQ	$+0.038$	—	$+0.041$	indication
5.4.1	Beetle 1B: CONCAT vs. GATE	UQ	$+0.032$	—	$+0.036$	indication
5.4.1	SciEntsBank 1B: CONDIFF vs. CONCAT	UD	-0.017	—	-0.011	indication
5.4.1	SciEntsBank 1B: CONCAT vs. GATE	UD	$+0.017$	—	$+0.019$	indication
5.4.1	Beetle 3B: DIFF vs. GATE	UA	-0.017	indication	-0.009	effect
5.4.1	SciEntsBank 3B: CONDIFF vs. GATE	UQ	-0.049	indication	-0.019	—
5.4.1	SciEntsBank 3B: DIFF vs. GATE	UA	-0.014	—	-0.012	indication
5.4.1	Beetle 7B: CONDIFF vs. GATE	UQ	$+0.038$	indication	$+0.013$	—
5.4.2	Beetle 3B: t-concat vs. CONCAT	UA	$+0.026$	indication	$+0.014$	effect
5.4.3	Beetle 1B: bidirectional vs. causal	UQ	-0.016	—	-0.041	effect
5.4.3	Beetle 3B: bidirectional vs. causal	UA	-0.073	effect	-0.028	—
5.4.3	SciEntsBank 3B: bidirectional vs. causal	UA	$+0.032$	indication	$+0.011$	—

distinguish between the two types of error. Table 5.25 shows that the task context reduces precisely the wide errors.

Without context, prediction and reference are two levels apart for just over $11\,\%$ of the instances on both transfer splits, with context for about $7\,\%$; the ranges over the seeds do not overlap in any case. Accuracy, by contrast, differs little, and on UQ rubric alone is even nominally ahead. The context thus does not make the model correct substantially more often, but grossly wrong less often – a difference that QWK rewards and Macro-F_1 does not capture. Conversely, when context is present, the rubric text increases accuracy on UD without changing the share of wide errors; this explains why Macro-F_1 indicates a contribution there and QWK does not. On the transfer splits, the statement that question and reference answer are superior to the rubric text thus concerns the severity of the errors, not their frequency.

Table 5.25: SciEntsBank, Llama-3.2-1B, fixed classification head: share of wide errors (prediction and reference two levels apart) and accuracy. Means over three random seeds, for the share of wide errors with the range over the seeds.

	UQ		UD	
Condition	Wide errors	Acc	Wide errors	Acc
Context only	0.068 (0.064–0.074)	0.652	0.074 (0.062–0.082)	0.641
Rubric only	0.117 (0.109–0.128)	0.666	0.114 (0.108–0.122)	0.627
Context and rubric	0.072 (0.065–0.079)	0.675	0.078 (0.072–0.084)	0.665

Architecture

Fusion and anchor. Among the fusion comparisons, the verdict shifts in nine cases, eight of them between "not demonstrable" and "indication". The only shift to an effect concerns GATE versus DIFF on Beetle UA with Llama-3.2-3B, with a difference of only 0.009 Macro-F_1 and correspondingly small variation. The pattern described in Section 5.4.1 is broadly retained under Macro-F_1: with the smaller backbone, GATE lies behind CONDIFF and CONCAT on Beetle UQ with disjoint ranges, with the larger ahead of DIFF; the indication of an advantage of GATE over CONDIFF on SciEntsBank UQ, by contrast, disappears, as does the indication of an advantage of CONDIFF over GATE on Beetle UQ with Mistral-7B. All other fusion comparisons with Mistral-7B are not demonstrable under either metric. The indication of an advantage of t-concat on Beetle UA becomes an effect under Macro-F_1, with a difference of 0.014.

Span alignment and fixed head. The comparisons turn out the same under both metrics, with one exception: on Beetle UQ, the advantage of span alignment with the smaller backbone holds as an effect under Macro-F_1 ($+0.044$), whereas it is not demonstrable under QWK because the ranges overlap. The advantage on Beetle UA holds under both metrics ($+0.088$ QWK, $+0.058$ Macro-F_1). With the larger backbone, no difference is demonstrable on Beetle under either metric.

Transfer. The collapse when transferring from ASAP-SAS to Beetle and SciEntsBank holds as an effect under both metrics in all ten cells.

Attention mask. The loss of the larger backbone on Beetle UQ and its gain on SciEntsBank UQ hold as effects under both metrics. The gain of the smaller backbone on Beetle UA holds under both metrics with three seeds, but under neither with eight (Section 5.2). Not robust to the metric are the loss of the larger backbone on Beetle UA and the indication on SciEntsBank UA. In addition, under Macro-F_1 there is a loss of the smaller backbone on Beetle UQ (-0.041), which also holds with eight seeds. On Beetle UQ, complete removal of causality thus harms both backbones under this metric. On ASAP-SAS, no difference is demonstrable under either metric.

Conclusion

Robust to the choice of metric are the advantage of span alignment over the fixed head on Beetle UA with the smaller backbone, the collapse without task information, the advantage of the

context on UA, the failure of transfer between benchmarks, the absence of fusion differences above the noise threshold on all three backbones and the mask effects on the UQ splits of the larger backbone. Dependent on the metric are the superiority of the context on the transfer splits and the missing contribution of the rubric text when context is present; both statements hold only under QWK, which weights gross misjudgements more heavily.

5.6 Summary of Findings

The results of this chapter can be organised along an overarching distinction: what matters above all is *which information* is available to the model, less *how* it is processed architecturally.

RQ1: Span Alignment Versus the Baselines

1. **Compared with the fixed classification head, span alignment shows an advantage only with the smaller backbone** (Section 5.3.1). On Beetle, it is ahead by 0.088 QWK on UA with Llama-3.2-1B, as an effect under both metrics and also with eight seeds. On SciEntsBank, the advantages are small with the smaller backbone; with Llama-3.2-3B, no difference is demonstrable on either benchmark.
2. **Task-related information is the determining factor – not its form** (Section 5.3.2). Withdrawing all task information from the model costs between 0.235 and 0.528 QWK, and on the transfer splits performance drops to or below chance level; additional model capacity does not compensate for this. The rubric text, however, is *interchangeable*: if question and reference answer are available, it contributes nothing further under QWK (-0.012 to $+0.015$ QWK, all within the noise), and as the sole information source it is inferior to them by 0.060 to 0.100 QWK. Under Macro-F_1 both statements hold only with qualifications: the inferiority shows only on UA there, and when context is present the rubric text contributes demonstrably on UD (Section 5.5.3).
3. **Compared with the encoder models, the choice of evaluation split is decisive** (Section 5.3.3). On UA, encoder models are competitive; on Beetle UQ, Llama-3.2-1B is ahead of RoBERTa by 0.221 QWK – and a clear deficit of the encoder exists at every learning rate tested.
4. **Structural transferability does not lead to a transfer of scoring** (Section 5.3.4). Trained on ASAP-SAS only, both Llama sizes drop to chance level on Beetle and SciEntsBank (-0.017 to 0.022 QWK) and assign almost every answer to the lowest level; a single run with Mistral-7B collapses in the same way. Consistent with the data is the hypothesis that the models learned on ASAP-SAS to associate short answers with low levels.

RQ2: Adapting Decoder-Only Models

5. **The design of the alignment has no demonstrable effect** (Sections 5.4.1 and 5.4.2). Neither the choice between CONDIFF, CONCAT, DIFF and GATE nor the anchor of the

fusion produces differences above the noise threshold on any of the three backbones. The weak indication in favour of GATE on Llama-3.2-3B, where the variant leads in four of five cells, is not confirmed on Mistral-7B.

6. **The attention mask has an effect, but its sign depends on the benchmark** (Section 5.4.3). Complete removal of causality costs Llama-3.2-3B 0.186 QWK on Beetle UQ and gains it 0.060 on SciEntsBank; on ASAP-SAS, no difference is demonstrable for either Llama size. For the smaller backbone, no effect under QWK survives the extension of the reference to eight seeds; under Macro-F_1, it loses on Beetle UQ.
7. **No statement can be made about backbone size** (Section 5.4.4). The seemingly clear size effect is confounded with the learning rate: at its reference learning rate, Llama-3.2-1B reaches 0.779 and 0.605 QWK on Beetle and is thus on a par with the mean over three Mistral-7B runs.

Overarching

8. **The variation between tasks exceeds all architectural effects** (Section 5.5.1) and reaches a range of 0.21 to 0.25 QWK between the ten items of the same benchmark on ASAP-SAS.
9. **Three seeds can clearly underestimate the variation** (Section 5.2). Extended to eight seeds, the range of one group grows from 0.023 to 0.079 QWK; the median of the groups of three, by contrast, confirms the noise threshold of 0.036 as a typical value.

The methodologically most far-reaching consequence concerns the relation between finding and measurement precision. Several patterns that emerged clearly on single runs – a reversal of the fusion ranking between splits, an advantage of answer-span alignment, a loss due to full bidirectionality with the smaller backbone on SciEntsBank – do not survive repetition with three random seeds. A gain from full bidirectionality with the smaller backbone on Beetle passed the evaluation rule with three seeds and disappeared only with eight. Without the determination of the noise threshold carried out in Section 5.2 and its verification, these patterns would have been reported as results.

6 Discussion

This chapter answers the research questions formulated in Section 1.3 on the basis of the results from Chapter 5, relates them to the literature and identifies the limitations of the study. The yardstick for all answers is the evaluation rule determined in Section 5.2: only a difference that exceeds the variation over three random seeds counts as an effect.

6.1 Answers to the Research Questions

RQ1: Does the Span-Based Classifier Outperform the Baselines?

Compared with the fixed classification head, span alignment shows an advantage only with the smaller backbone (Section 5.3.1). On Beetle, it is ahead by 0.088 QWK on UA with Llama-3.2-1B. The lead clearly exceeds the ranges of both conditions, is more than twice as large as the global median of the seed range, also holds under Macro-F_1 and persists when the span condition is extended to eight seeds; under Macro-F_1, the lead on UQ is additionally demonstrable as an effect. On SciEntsBank, the advantages with the smaller backbone are small at 0.021 and 0.026 QWK. With Llama-3.2-3B, no difference is demonstrable on either benchmark; on SciEntsBank the differences are nominally negative, on Beetle small at $+0.008$ and $+0.027$ QWK.

Since both conditions contain the rubric texts in the input, this comparison does not measure the value of the rubric information but only the value of its explicit extraction as spans. Why this value is considerably larger with the smaller backbone on Beetle than on SciEntsBank cannot be clarified with the available runs. That it disappears with the larger backbone on both benchmarks is compatible with the reading that a larger model already condenses the rubric information into the sequence representation; because of the unadjusted learning rate, however, this reading is not secured.

In transfer between benchmarks, span alignment fails (Section 5.3.4). Trained on ASAP-SAS only, it can be applied to Beetle and SciEntsBank without any adaptation, but reaches only chance level there and assigns almost all answers to the lowest level, on both Llama sizes as well as in a single run with Mistral-7B. Consistent with the data is the explanation that the models learned on ASAP-SAS to treat the length of an answer as an indication of its level and therefore score the consistently short answers of the target benchmarks low. The rubric texts of the target benchmark do not correct this heuristic.

Compared with conditions without rubric text or task information, a twofold picture emerges (Section 5.3.2). Task-related information is indispensable: if it is withdrawn from the model entirely, the model loses between 0.235 and 0.528 QWK on SciEntsBank, and on the transfer splits performance drops to or below chance level. The rubric text as such, by contrast, is interchangeable. If question and reference answer are available, it contributes nothing further under QWK; the differences of -0.012 to $+0.015$ QWK all lie within the noise. As the sole information source, it is inferior to question and reference answer by 0.060 to 0.100 QWK. On Beetle, the same sign appears, though there on single runs.

The obvious explanation lies in the origin of the rubric texts. For Beetle and SciEntsBank, they were generated with the help of a language model from question and reference answer and visibly incorporate their content (Section 3.4). They are thus not an independent information source but a paraphrase of the same information. The model consequently uses the rubric texts as carriers of task-related information and not as descriptions of scoring criteria that go beyond the task description. As a qualification, this finding depends on the metric. Under Macro-F_1, the rubric text contributes demonstrably on SciEntsBank UD when context is present, and the superiority of question and reference answer is limited to UA (Section 5.5.3). The advantage of the context on the transfer splits thus consists mainly in gross misjudgements becoming rarer.

Compared with the encoder models, the evaluation split is decisive (Section 5.3.3). On UA splits RoBERTa is competitive, whereas on Beetle UQ Llama-3.2-1B is ahead by 0.221 QWK, and a clear deficit of the encoder exists at every learning rate tested. Since encoder and decoder differ simultaneously in architecture, number of parameters, context length and training configuration, this lead cannot be attributed to span alignment alone.

The answer to RQ1 is thus differentiated. The span-based classifier clearly outperforms the fixed head under certain conditions, namely on Beetle with the smaller backbone, without this advantage being reliable across benchmarks and backbones. Its strength also lies in a property that cannot show up in an accuracy comparison on fixed label spaces: it processes a number of scoring levels that varies at run time (Section 4.1) and thereby makes the joint modelling of all tasks possible in the first place on ASAP-SAS with its varying scale lengths. The transfer experiment, however, shows the limit of this property: it makes transfer to new rubrics technically possible but does not ensure that the new rubrics are also used substantively. The premise that the rubric texts, as criterion descriptions, form an independent information source, by contrast, is not confirmed for the rubrics used here, which are derived from the task description.

RQ2: Can Decoder-Only Models Be Adapted to the Span-Based Classifier?

In principle, yes: with Llama-3.2-1B, Llama-3.2-3B and Mistral-7B, span alignment works on all three benchmarks, and on the transfer splits the decoder models are clearly ahead of the encoder models. For the individual design decisions, the following picture emerges.

The design of the alignment has no demonstrable effect. In all fifteen configurations of benchmark, backbone and split investigated, the variation over the random seeds exceeds the distance between the fusion variants, for Mistral-7B by up to a factor of six (Section 5.4.1). Neither the parameter-free difference term of CONDIFF nor the additional parameters of the gate projection bring an advantage above the noise threshold over plain concatenation. The weak indication in favour of GATE on Llama-3.2-3B, where the variant is nominally ahead in four

of five cells, is not confirmed on Mistral-7B; there, as with the smallest backbone, the ranking alternates without a discernible pattern. The same applies to the anchor of the fusion, i.e. to the choice between aligning the rubric spans with the sequence representation or with the answer span. In four of five cells it makes a difference of at most 0.013 QWK with overlapping ranges; on Beetle UA, a weak indication in favour of the answer span remains (Section 5.4.2). This is remarkable because the two representations carry different information under causal attention: the answer span has not seen the rubrics, the sequence representation has seen them all. That this difference has no consequences suggests that the information decisive for scoring is already contained in the representation of the answer. For practice, this implies a recommendation in favour of the simplest construction, i.e. CONCAT or DIFF.

The attention mask is effective but not reliable. It is the only variation within the span-alignment architecture whose effect clearly exceeds the noise threshold, but its sign depends on benchmark and backbone (Section 5.4.3). Complete removal of causality costs Llama-3.2-3B 0.073 QWK on Beetle UA and 0.186 on UQ and at the same time increases the variation between runs. On SciEntsBank, the same backbone gains 0.060 QWK on UQ; on ASAP-SAS, the mask has no effect for either Llama size. For the smaller backbone, no effect is secured under QWK: a gain on Beetle UA that narrowly passed the evaluation rule with three seeds disappeared with eight; under Macro-F_1, Llama-3.2-1B loses on Beetle UQ, in the same direction as the larger backbone. Across the single runs, the partial removal shows no monotonic relationship between the number of bidirectional layers and performance. A recommendation to operate decoder-only models for ASAG bidirectionally as a matter of principle can be derived from the data just as little as the opposite; the mask should rather be treated as a dataset- and model-dependent hyperparameter.

No statement about backbone size is possible. The seemingly clear lead of Mistral-7B is confounded with the learning rate: at its reference learning rate, Llama-3.2-1B is on a par on Beetle with the mean over three Mistral runs (Section 5.4.4).

The answer to RQ2 is therefore: decoder-only models can be adapted to the span-based classifier without changing their weight architecture, and the design of the alignment is uncritical in doing so. The only effective intervention, the removal of the causal mask, does not improve the adaptation reliably, however, but has to be validated for each application.

6.2 Relation to Prior Work

Rubric-Guided Scoring

Work on exploiting label semantics shows its effectiveness mainly in zero- and few-shot scenarios, in which the model cannot infer the meaning of a label from numerous training examples [48, 61]. This thesis, by contrast, investigates a fully supervised scenario with several thousand training instances per benchmark. That the rubric text makes no additional contribution there compared with question and reference answer does not contradict these findings but limits their transferability: the benefit of label semantics appears to be greatest where little other information about the task is available. The transfer experiment also shows that a model trained in a supervised manner does not readily use the label semantics of a new benchmark:

trained on ASAP-SAS, it follows a heuristic learned there instead of the rubric texts of the target benchmark. A zero-shot scenario in the sense of the works cited thus presupposes that training itself encourages the use of the rubric texts, for instance through a variety of training tasks.

With respect to the approach of Gombert et al. [23], which aligns the representation of the answer span with the rubric spans via a bilinear network, this thesis provides a finding on the anchor: on a decoder-only backbone and with concatenative fusion, it makes no difference whether the answer span or the sequence representation is used. The bilinear fusion itself was not investigated, so no statement about the approach as a whole can be derived from this.

Language Models as Encoders

Work on repurposing decoder-only models as encoders regards the removal of the causal mask as a shift relative to the pre-training condition, which is countered by additional masked further pre-training [3] or by measures against the attention sink [35]. In this thesis, the mask was removed without such an intermediate step, solely within LoRA fine-tuning on a few thousand instances. The inconsistent results are compatible with the assumption that the model does not reliably learn to use the two-sided context under these conditions. Whether preceding further pre-training stabilises the effect of the mask is an open question.

The lead of the decoder models on the transfer splits fits the motivation of this line of research. It is compatible with the assumption that the knowledge a larger language model acquires during pre-training comes into play mainly for unseen questions; this mechanism is not demonstrated, since the two model classes differ simultaneously in several properties.

Variance Between Runs

The finding that the choice of random seed substantially influences the outcome of fine-tuning has been established for other tasks [44, 16]. This thesis confirms it for ASAG with decoder-only models: the range over three seeds amounts to 0.036 QWK at the median and 0.118 at the maximum and thus exceeds most of the architectural differences investigated. Several patterns that emerged clearly on single runs did not survive repetition; in the case of the anchor, two of three random seeds would have suggested an advantage as a single run, which the third cancels out (Section 5.4.2). Three seeds, however, can also underestimate the variation: in a group extended to eight seeds, the range grew from 0.023 to 0.079 QWK, and a narrowly passed effect disappeared (Section 5.2).

Choice of Metric

The criticism of QWK as the sole measure [17] finds a concrete counterpart in the results. QWK and Macro-F_1 reach different verdicts in 20 of 114 pairwise comparisons (Section 5.5.3). The discrepancies are not random but are explained by the type of errors: the task context mainly reduces wide misjudgements, which QWK weights heavily and Macro-F_1 does not distinguish from adjacent confusions. The two metrics thus measure different properties of a scoring system and should be reported together.

6.3 Methodological Implications

Beyond the individual research questions, three methodological conclusions follow for the evaluation of ASAG systems.

First, comparisons between architectural variants cannot be interpreted without repetition with several random seeds as soon as the differences are of the order of a few hundredths of QWK. Findings that pass an evaluation rule only because of unusually narrow variation should be checked with additional seeds.

Second, the choice of evaluation split determines the conclusion. Had the evaluation been limited to the UA splits of ASAP-SAS and Beetle, it would have been tempting to conclude that decoder-only models offer no added value over encoders. Likewise, without task information performance drops to chance level only on the transfer splits, whereas it remains clearly above it on UA. Statements about the transferability of a system therefore require UQ and UD splits.

Third, the variation between the tasks of a benchmark exceeds all architectural effects investigated; on ASAP-SAS, QWK per task spans a range of 0.21 and 0.25, respectively (Section 5.5.1). Aggregated benchmark values thus depend strongly on the composition of the tasks and should be generalised with caution to other task collections.

6.4 Limitations

Learning rate of the Llama backbones. All decoder runs use the learning rate $5 \cdot 10^{-5}$, which the reference implementation intends for models with seven billion parameters or more; for the Llama backbones, $2 \cdot 10^{-4}$ would be intended. At this rate, Llama-3.2-1B gains 0.096 QWK on Beetle UQ, while a mixed picture emerges for Llama-3.2-3B (Section 5.4.4). Both findings rest on one run each. Comparisons *within* a backbone are not affected by this, but comparisons *between* model sizes are; the thesis therefore makes no statement about the influence of backbone size.

Incomplete evaluation for Mistral-7B. For the largest backbone, the fusion variants CONDIFF, CONCAT and GATE are available on Beetle and SciEntsBank with three random seeds, whereas the variation of the attention mask is available only as single runs with three proportions and without the fully bidirectional condition. Two of these runs were conspicuously weak; their repetition lies within the range of the other runs (Section 5.5.2). On ASAP-SAS, all mask conditions are available for Mistral-7B, but only as single runs, and transfer was tested with a single seed. The findings on the mask therefore rest on the two Llama sizes.

Mixed software environments. In 21 of the 39 groups with three random seeds, the reference run stems from a different software environment than the two repetitions. For the comparisons with a finding, only the direction, not the size, of the effects is thus safeguarded within one environment (Section 5.2).

Origin of the rubric texts. No original scoring guidelines exist for Beetle and SciEntsBank; the rubric texts used were generated with a language model from question and reference answer. The statements on the contribution of the rubric texts (RQ1) therefore apply in the first

instance to such derived rubrics. Whether rubrics written by teachers, which formulate criteria beyond the reference answer, make an independent contribution remains open; ASAP-SAS does contain original scoring guidelines but was not investigated in the 2×2 design.

Task context in the main runs. The main runs on which the comparison with the fixed head and all findings on RQ2 are based contain neither question nor reference answer (Section 4.1.1). It is precisely this information that proves to be the most effective factor in Section 5.3.2. Whether the findings on fixed head, fusion, anchor and mask persist in the stronger configuration with task context has not been tested. Moreover, question and reference answer are switched on or off together, so their respective shares in the context effect cannot be separated.

Scope of the data basis. All three benchmarks are in English. Only SciEntsBank provides a UD split, so all statements on domain transfer rest on one benchmark. ASAP-SAS is available only as a UA split; there, only the causal and the fully bidirectional condition of the two Llama sizes were evaluated with three seeds, all other conditions as single runs. Transfer between benchmarks was tested in one direction only, from ASAP-SAS to Beetle and SciEntsBank, and the length explanation for its failure has not been tested causally. Finally, the noise threshold rests on groups with three seeds; its check with eight seeds covers a single group.

Further limitations. Model selection uses accuracy on the validation set and not the primarily reported metrics (Section 4.6). The configuration of the encoder main runs is not documented by files, and BERT fails on SciEntsBank regardless of the learning rate. Finally, pairwise methods and generative scoring were not trained as separate baseline conditions (Section 5.1); the comparison is limited to variants within the span-alignment architecture, the fixed classification head and the encoder models.

6.5 Practical Implications

Four recommendations for the use of rubric-guided ASAG systems can be derived from the results. The most effective lever is the input: question and reference answer should be available to the model, since their absence is compensated neither by any of the architectural variants investigated nor by additional model capacity. As for the architecture, a simple fusion suffices; more elaborate variants bring no demonstrable advantage. The attention mask can change performance considerably, but in both directions, and should therefore be validated per dataset on a transfer split. Finally, the structural ability of span alignment to handle varying scoring scales speaks for its use wherever tasks with different numbers of levels are to be scored in one model. Such a model should not, however, be transferred to a new set of tasks without training on suitable data: a model trained on ASAP-SAS fails completely on Beetle and SciEntsBank (Section 5.3.4).

For the requirement of local operability, one finding is revealing: with question, reference answer and answer, Llama-3.2-1B reaches 0.602 QWK on Beetle UQ and is thus practically on a par with Mistral-7B, which is about six times larger and reaches a mean of 0.605 over three seeds there without task context (Section 5.3.2). The value for Llama-3.2-1B rests on a single run, and the two conditions differ in their input; they do, however, indicate that a small, locally operable model with complete task information is not fundamentally at a disadvantage for this task.

7 Conclusion and Outlook

List of Tables

3.1 ASAP-SAS: overview of tasks with subject, grade, label scale and instance counts (computed from `train.csv` / `test_ua.csv`). 26

3.2 Beetle: instance and question counts per split (computed from the provided CSV files). 27

3.3 SciEntsBank: instance and question counts per split (computed from the provided CSV files). 28

3.4 Comparison of the three ASAG benchmarks used. 30

4.1 The four fusion mechanisms compared. d denotes the model dimension, d_g the input dimension of the scoring head. 39

4.2 Number of bidirectional layers n_{bidir} resulting from the proportion ρ per backbone according to rule (4.15). The rows are ordered by the actual number of bidirectional layers, not by the nominal value of ρ. The fully bidirectional condition $\rho = L$ is available for Llama-3.2-1B and -3B, but not for Mistral-7B; for both Llama sizes it is available with three random seeds. 42

4.3 Training configuration of the decoder backbones. The values correspond to the configuration used in the evaluated decoder runs; the deviations of the encoder baselines are described in the following subsection. 47

4.4 Learning-rate variation of the encoder baselines with CONDIFF fusion, QWK. Single runs, random seed $114\,514$. The BERT runs use batch size 16 at $1 \cdot 10^{-5}$ and $2 \cdot 10^{-5}$ and 32 at $3 \cdot 10^{-5}$, the RoBERTa runs 16 throughout. The last column reproduces the main run from Table 5.9, whose learning rate is not documented. 48

5.1 Range Δ of QWK over three random seeds, aggregated over 93 cells. 54

5.2 Beetle, Llama-3.2-1B, CONDIFF: variation of QWK over eight random seeds and over the 56 groups of three that can be formed from them. *Original* denotes the group of seeds $114\,514$, 1 and 2 on which the other tables of this chapter are based. 55

5.3 Pairwise comparisons involving CONDIFF, Llama-3.2-1B, Beetle: verdict with the three original and with all eight seeds of the CONDIFF condition, QWK. The other condition is based on three seeds in each case. Δ as in the result tables of the respective sections; "—" means not demonstrable. 56

5.4 Span alignment (CONDIFF) versus fixed classification head, both with rubric texts and without task context. Means over three random seeds, ranges in parentheses. *Above range* means that the difference exceeds the larger of the two cell-specific ranges. Bold: higher value per row. 57
5.5 SciEntsBank, Llama-3.2-1B, fixed classification head: availability of task context (question and reference answer) and rubric text in the full 2×2 design. Means over three random seeds, ranges in parentheses. 59
5.6 SciEntsBank: ablation of the rubric texts without task context, across two model sizes. Means over three random seeds, ranges in parentheses. *Gap* denotes the distance between the worst run with and the best run without rubric text. . . 61
5.7 Beetle, Llama-3.2-1B, fixed classification head, without task context: ablation of the rubric texts. Means over three random seeds, ranges in parentheses. . . . 61
5.8 Beetle, Llama-3.2-1B, fixed classification head: availability of task context and rubric text. All values at random seed $114\,514$ to keep the cells comparable with one another. For the two lower rows, three seeds are additionally available (means 0.641 / 0.456 and 0.587 / 0.457, respectively). 62
5.9 Encoder baselines versus Llama-3.2-1B (CONDIFF, causal in each case), QWK. Encoder values are single runs from an early training wave of which only the predictions have been preserved; Llama values are means over three random seeds. ‡Run with chance-level result, excluded from interpretation (Section 4.6). 63
5.10 Transfer from ASAP-SAS to Beetle and SciEntsBank (CONDIFF, causal), QWK. *Transfer*: trained on ASAP-SAS only; *in-domain*: trained on the target benchmark. Means over three random seeds, ranges in parentheses. 64
5.11 ASAP-SAS (UA): fusion mechanisms with causal attention. Single runs, random seed $114\,514$. Bold: highest value per column within a backbone; the marking makes no statement about whether a difference is demonstrable (cf. Table 5.13). 66
5.12 Beetle and SciEntsBank: fusion mechanisms with causal attention, QWK per split. Means over three random seeds; DIFF was investigated on Llama-3.2-3B only. Bold: highest value per column within a backbone; the marking makes no statement about whether a difference is demonstrable (cf. Table 5.13). 66
5.13 Difference between the fusion mechanisms (range of the means of CONDIFF, CONCAT and GATE, for Llama-3.2-3B additionally DIFF) compared with the largest seed range of the same cells. 67
5.14 Best fusion variant per benchmark, split and backbone according to the mean over three random seeds. According to Table 5.13 the ranking is not robust and is given here solely to illustrate its arbitrariness. 68
5.15 Anchor of the fusion on Llama-3.2-3B: answer span (t-concat) versus sequence representation (CONCAT, CONDIFF). Means over three random seeds, ranges in parentheses. 69
5.16 ASAP-SAS (UA): attention mask with CONDIFF fusion, sorted by the actual number of bidirectional layers n_{bidir}. Single runs, random seed $114\,514$; for the causal and the fully bidirectional condition of the two Llama sizes, three seeds are additionally available (Tables 5.18 and 5.19). For Mistral-7B, the fully bidirectional condition stems from a different software environment than the other runs in this column. Bold: highest value per column; the marking makes no statement about whether a difference is demonstrable. 70

5.17 Beetle and SciEntsBank: attention mask with CONDIFF fusion, QWK per split, sorted by n_{bidir}. Single runs, random seed $114\,514$. For Llama-3.2-3B and Mistral-7B, the same conditions are likewise available as single runs; they are not listed here, since they contribute no statement that does not already follow from the complete removal. Bold: highest value per column; the marking makes no statement about whether a difference is demonstrable. 71
5.18 Fully bidirectional attention ($n_{\text{bidir}} = L = 28$) versus causal attention, Llama-3.2-3B, CONDIFF fusion. Means over three random seeds, ranges in parentheses. 71
5.19 Fully bidirectional attention ($n_{\text{bidir}} = L = 16$) versus causal attention, Llama-3.2-1B, CONDIFF fusion. Means over three random seeds, ranges in parentheses. *Above range* as in Table 5.4. With eight seeds of the causal condition, the difference on Beetle UA is not demonstrable (Table 5.3). 72
5.20 Effect of backbone size with CONDIFF and causal attention (QWK). Means over three random seeds; for Mistral-7B on ASAP-SAS, a single run. 73
5.21 CONDIFF at the reference learning rate $2 \cdot 10^{-4}$ compared with the runs at $5 \cdot 10^{-5}$ (mean and range over three random seeds), QWK. The runs at $2 \cdot 10^{-4}$ rest on a single random seed each. 74
5.22 ASAP-SAS (UA): QWK per task for CONDIFF with causal attention. Single runs. Bold: higher value per row. 76
5.23 Diagnosis of the conspicuous runs. $Acc_{majority}$ is the accuracy when constantly predicting the most frequent level, QWK_{perm} the QWK under a single random permutation of the predictions. *Rep.*: repetition with the same random seed in the software environment with transformers 4.57.6. 78
5.24 Pairwise comparisons with three random seeds whose verdict differs between QWK and Macro-F_1. Δ is the difference of the means (first minus second condition); the verdict follows the evaluation rule from Section 5.2, applied to the respective metric. "—" means not demonstrable. Context denotes question and reference answer. 79
5.25 SciEntsBank, Llama-3.2-1B, fixed classification head: share of wide errors (prediction and reference two levels apart) and accuracy. Means over three random seeds, for the share of wide errors with the range over the seeds. 80

List of Figures

2.1 Classical and rubric-conditioned task formulation 7
2.2 Generalisation dimensions UA, UQ and UD 8
2.3 Weight matrices of accuracy and QWK 10
2.4 Noise threshold and interpretability of differences 13
2.5 Three architecture families for rubric-guided scoring 16
2.6 Structure of the transformer architecture 18
2.7 Causal and bidirectional attention mask 21
2.8 Layer-wise removal of the causal mask 22

3.1 Size and split structure of the three benchmarks 30

4.1 Overall architecture of the RUSPAN system 34
4.2 Serialisation and span pooling 37
4.3 Constructions of the fusion mechanisms 40

5.1 Context and rubric text in the 2×2 design 62
5.2 Attention mask over the proportion of bidirectional layers 70
5.3 QWK per task on ASAP-SAS 77

Bibliography

[1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. *Layer Normalization*. arXiv:1607.06450. 2016. arXiv: 1607.06450.

[2] Xiaoyu Bai and Manfred Stede. "A Survey of Current Machine Learning Approaches to Student Free-Text Evaluation for Intelligent Tutoring". In: *International Journal of Artificial Intelligence in Education* 33.4 (2023), pp. 994–1032. DOI: 10.1007/s40593-022-00323-0.

[3] Parishad BehnamGhader et al. "LLM2Vec: Large Language Models Are Secretly Powerful Text Encoders". In: *First Conference on Language Modeling (COLM)*. 2024. URL: https://openreview.net/forum?id=IW1PR7vEBf.

[4] MV Berry et al. "Can apparent superluminal neutrino speeds be explained as a quantum weak measurement?" In: *arXiv preprint arXiv:1110.2832* (2011).

[5] Marie Bexte, Andrea Horbach, and Torsten Zesch. "Similarity-Based Content Scoring – How to Make S-BERT Keep Up with BERT". In: *Proceedings of the 17th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2022)*. Seattle, Washington: Association for Computational Linguistics, 2022, pp. 118–123.

[6] Gary Blake and Robert W Bly. *The elements of technical writing*. Macmillan New York, NY, 1993.

[7] Steven Burrows, Iryna Gurevych, and Benno Stein. "The Eras and Trends of Automatic Short Answer Grading". In: *International Journal of Artificial Intelligence in Education* 25.1 (2015), pp. 60–117.

[8] Leon Camus and Anna Filighera. "Investigating Transformers for Automatic Short Answer Grading". In: *Artificial Intelligence in Education: 21st International Conference, AIED 2020, Proceedings, Part II*. Springer, 2020, pp. 43–48.

[9] Imran Chamieh, Torsten Zesch, and Klaus Giebermann. "LLMs in Short Answer Scoring: Limitations and Promise of Zero-Shot and Few-Shot Approaches". In: *Proceedings of the 19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2024)*. Mexico City, Mexico: Association for Computational Linguistics, 2024, pp. 309–315. URL: https://aclanthology.org/2024.bea-1.25/.

[10] Li-Hsin Chang and Filip Ginter. "Automatic Short Answer Grading for Finnish with ChatGPT". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 38. 21. 2024, pp. 23173–23181. DOI: 10.1609/aaai.v38i21.30363.

[11] Qian Chen et al. “Enhanced LSTM for Natural Language Inference”. In: *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Vancouver, Canada: Association for Computational Linguistics, 2017, pp. 1657–1668.

[12] Jacob Cohen. “A Coefficient of Agreement for Nominal Scales”. In: *Educational and Psychological Measurement* 20.1 (1960), pp. 37–46.

[13] Jacob Cohen. “Weighted Kappa: Nominal Scale Agreement Provision for Scaled Disagreement or Partial Credit”. In: *Psychological Bulletin* 70.4 (1968), pp. 213–220.

[14] Alexis Conneau et al. “Supervised Learning of Universal Sentence Representations from Natural Language Inference Data”. In: *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Copenhagen, Denmark: Association for Computational Linguistics, 2017, pp. 670–680.

[15] Jacob Devlin et al. “BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding”. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*. Minneapolis, Minnesota: Association for Computational Linguistics, 2019, pp. 4171–4186.

[16] Jesse Dodge et al. *Fine-Tuning Pretrained Language Models: Weight Initializations, Data Orders, and Early Stopping*. arXiv:2002.06305. 2020. arXiv: 2002.06305.

[17] Afrizal Doewes, Nughthoh Arfawi Kurdhi, and Akrati Saxena. “Evaluating Quadratic Weighted Kappa as the Standard Performance Metric for Automated Essay Scoring”. In: *Proceedings of the 16th International Conference on Educational Data Mining (EDM 2023)*. 2023. DOI: 10.5281/zenodo.8115784.

[18] Myroslava Dzikovska et al. “SemEval-2013 Task 7: The Joint Student Response Analysis and 8th Recognizing Textual Entailment Challenge”. In: *Second Joint Conference on Lexical and Computational Semantics (*SEM), Volume 2: Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013)*. Atlanta, Georgia, USA: Association for Computational Linguistics, 2013, pp. 263–274.

[19] Ahmed Elshabrawy et al. “Enabling Natural Zero-Shot Prompting on Encoder Models via Statement-Tuning”. In: *Findings of the Association for Computational Linguistics: NAACL 2025*. Albuquerque, New Mexico: Association for Computational Linguistics, 2025, pp. 8302–8321.

[20] Europäisches Parlament und Rat der Europäischen Union. *Verordnung (EU) 2016/679 des Europäischen Parlaments und des Rates vom 27. April 2016 zum Schutz natürlicher Personen bei der Verarbeitung personenbezogener Daten, zum freien Datenverkehr und zur Aufhebung der Richtlinie 95/46/EG (Datenschutz-Grundverordnung)*. Amtsblatt der Europäischen Union, L 119 vom 4.5.2016, S. 1, https://eur-lex.europa.eu/legal-content/DE/TXT/?uri=CELEX:32016R0679. 2016.

[21] Rafael Ferreira Mello et al. “Automatic Short Answer Grading in the LLM Era: Does GPT-4 with Prompt Engineering Beat Traditional Models?” In: *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. ACM, 2025, pp. 93–103.

[22] Zorik Gekhman et al. “Inside-Out: Hidden Factual Knowledge in LLMs”. In: *Second Conference on Language Modeling (COLM)*. 2025. URL: https://openreview.net/forum?id=f7GG1MbsSM.

[23] Sebastian Gombert et al. “Are Rubrics All You Need? Towards Rubric-Based Automatic Short Answer Scoring via Guided Rubric-Answer Alignment”. In: *Proceedings of LAK26: 16th International Learning Analytics and Knowledge Conference*. 2026.

[24] Sebastian Gombert et al. “Report on the BEA 2026 Shared Task on Rubric-Based Short Answer Scoring for German”. In: *Proceedings of the 21st Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2026)*. Association for Computational Linguistics, 2026.

[25] Aaron Grattafiori et al. *The Llama 3 Herd of Models*. arXiv:2407.21783. 2024. arXiv: 2407.21783.

[26] Momchil Hardalov et al. “Few-Shot Cross-Lingual Stance Detection with Sentiment-Based Pre-Training”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 36. 10. 2022, pp. 10729–10737. DOI: 10.1609/aaai.v36i10.21318.

[27] Neil Houlsby et al. “Parameter-Efficient Transfer Learning for NLP”. In: *Proceedings of the 36th International Conference on Machine Learning (ICML)*. 2019, pp. 2790–2799.

[28] Edward J. Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models”. In: *The Tenth International Conference on Learning Representations (ICLR)*. 2022.

[29] Albert Q. Jiang et al. *Mistral 7B*. arXiv:2310.06825. 2023. arXiv: 2310.06825.

[30] Gerd Kortemeyer. “Performance of the Pre-Trained Large Language Model GPT-4 on Automated Short Answer Grading”. In: *Discover Artificial Intelligence* 4.1 (2024).

[31] Yaman Kumar et al. “Get IT Scored Using AutoSAS – An Automated System for Scoring Short Answers”. In: *Proceedings of the Thirty-Third AAAI Conference on Artificial Intelligence*. 2019, pp. 9662–9669.

[32] Chankyu Lee et al. “NV-Embed: Improved Techniques for Training LLMs as Generalist Embedding Models”. In: *The Thirteenth International Conference on Learning Representations (ICLR)*. 2025. URL: https://openreview.net/forum?id=DtFbrhsXfU.

[33] Xiang Lisa Li and Percy Liang. “Prefix-Tuning: Optimizing Continuous Prompts for Generation”. In: *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Association for Computational Linguistics, 2021, pp. 4582–4597.

[34] Zhaohui Li et al. “Answer-State Recurrent Relational Network (AsRRN) for Constructed Response Assessment and Feedback Grouping”. In: *Findings of the Association for Computational Linguistics: EMNLP 2023*. Singapore: Association for Computational Linguistics, 2023, pp. 3879–3891.

[35] Ziyong Lin et al. “Look Both Ways and No Sink: Converting LLMs into Text Encoders Without Training”. In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Vienna, Austria: Association for Computational Linguistics, 2025, pp. 22839–22853.

[36] Chun Liu et al. “LLMEmbed: Rethinking Lightweight LLM’s Genuine Function in Text Classification”. In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Bangkok, Thailand: Association for Computational Linguistics, 2024, pp. 7994–8004.

[37] Yinhan Liu et al. *RoBERTa: A Robustly Optimized BERT Pretraining Approach*. arXiv:1907.11692. 2019. arXiv: 1907.11692.

[38] Kun Luo et al. “Large Language Models as Foundations for Next-Gen Dense Retrieval: A Comprehensive Empirical Assessment”. In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Miami, Florida, USA: Association for Computational Linguistics, 2024, pp. 1354–1365.

[39] Tom Mitchell et al. “Towards Robust Computerised Marking of Free-Text Responses”. In: *Proceedings of the 6th International Computer Assisted Assessment (CAA) Conference*. Loughborough: Loughborough University, 2002. URL: https://hdl.handle.net/2134/1884.

[40] Michael Mohler and Rada Mihalcea. “Text-to-Text Semantic Similarity for Automatic Short Answer Grading”. In: *Proceedings of the 12th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*. Association for Computational Linguistics, 2009, pp. 567–575.

[41] Lili Mou et al. “Natural Language Inference by Tree-Based Convolution and Heuristic Matching”. In: *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Berlin, Germany: Association for Computational Linguistics, 2016, pp. 130–136.

[42] Christopher Ormerod. *Short-Answer Scoring with Ensembles of Pretrained Language Models*. arXiv:2202.11558. 2022. arXiv: 2202.11558.

[43] Dan Qiao et al. “Decoder-Only LLMs Can Be Masked Auto-Encoders”. In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Vienna, Austria: Association for Computational Linguistics, 2025, pp. 713–723.

[44] Nils Reimers and Iryna Gurevych. “Reporting Score Distributions Makes a Difference: Performance Study of LSTM-Networks for Sequence Tagging”. In: *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Copenhagen, Denmark: Association for Computational Linguistics, 2017, pp. 338–348.

[45] Nils Reimers and Iryna Gurevych. “Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks”. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 3982–3992.

[46] Brian Riordan et al. “Investigating Neural Architectures for Short Answer Scoring”. In: *Proceedings of the 12th Workshop on Innovative Use of NLP for Building Educational Applications*. Copenhagen, Denmark: Association for Computational Linguistics, 2017, pp. 159–168.

[47] Qian Ruan, Ilia Kuznetsov, and Iryna Gurevych. “Are Large Language Models Good Classifiers? A Study on Edit Intent Classification in Scientific Document Revisions”. In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Miami, Florida, USA: Association for Computational Linguistics, 2024, pp. 15049–15067. URL: https://aclanthology.org/2024.emnlp-main.839/.

[48] Timo Schick and Hinrich Schütze. “It’s Not Just Size That Matters: Small Language Models Are Also Few-Shot Learners”. In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Online: Association for Computational Linguistics, 2021, pp. 2339–2352. URL: https://aclanthology.org/2021.naacl-main.185/.

[49] Noam Shazeer. *GLU Variants Improve Transformer*. arXiv:2002.05202. 2020. arXiv: 2002.05202.

[50] Mark D. Shermis. “Contrasting State-of-the-Art in the Machine Scoring of Short-Form Constructed Responses”. In: *Educational Assessment* 20.1 (2015), pp. 46–65. DOI: 10.1080/10627197.2015.997617.

[51] Shashank Sonkar et al. “Automated Long Answer Grading with RiceChem Dataset”. In: *Artificial Intelligence in Education (AIED 2024)*. Lecture Notes in Computer Science. Springer, 2024, pp. 163–176. DOI: 10.1007/978-3-031-64302-6_12.

[52] Wolfgang G Stock and Mechtild Stock. *Wissensrepräsentation: Informationen auswerten und bereitstellen*. Vol. 2. Oldenbourg Verlag, 2008, p. 382. ISBN: 978-3-486-58439-4.

[53] Jianlin Su et al. “RoFormer: Enhanced Transformer with Rotary Position Embedding”. In: *Neurocomputing* 568 (2024), p. 127063.

[54] Md Arafat Sultan, Cristobal Salazar, and Tamara Sumner. “Fast and Easy Short Answer Grading with High Accuracy”. In: *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*. Association for Computational Linguistics, 2016, pp. 1070–1075.

[55] Chul Sung et al. “Pre-Training BERT on Domain Resources for Short Answer Grading”. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 6071–6075.

[56] The Hewlett Foundation. *The Hewlett Foundation: Short Answer Scoring*. Kaggle-Wettbewerb (ASAP-SAS), https://www.kaggle.com/c/asap-sas. Abgerufen am 05.09.2026. 2012.

[57] Hugo Touvron, Louis Martin, Kevin Stone, et al. *Llama 2: Open Foundation and Fine-Tuned Chat Models*. arXiv:2307.09288. 2023. arXiv: 2307.09288.

[58] UK Department for Education. *Generative Artificial Intelligence (AI) in Education*. Policy paper, https://www.gov.uk/government/publications/generative-artificial-intelligence-in-education/generative-artificial-intelligence-ai-in-education. Zuletzt aktualisiert am 12.08.2025, abgerufen am 05.09.2026. 2025.

[59] Ashish Vaswani et al. “Attention Is All You Need”. In: *Advances in Neural Information Processing Systems 30 (NIPS 2017)*. 2017, pp. 5998–6008.

[60] Guangxuan Xiao et al. “Efficient Streaming Language Models with Attention Sinks”. In: *The Twelfth International Conference on Learning Representations (ICLR)*. 2024.

[61] Haike Xu et al. “A Universal Discriminator for Zero-Shot Generalization”. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada: Association for Computational Linguistics, 2023, pp. 10559–10575.

[62] Biao Zhang and Rico Sennrich. “Root Mean Square Layer Normalization”. In: *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*. 2019.

Appendices